Memorix Anatomy

9 Heart and blood vessels

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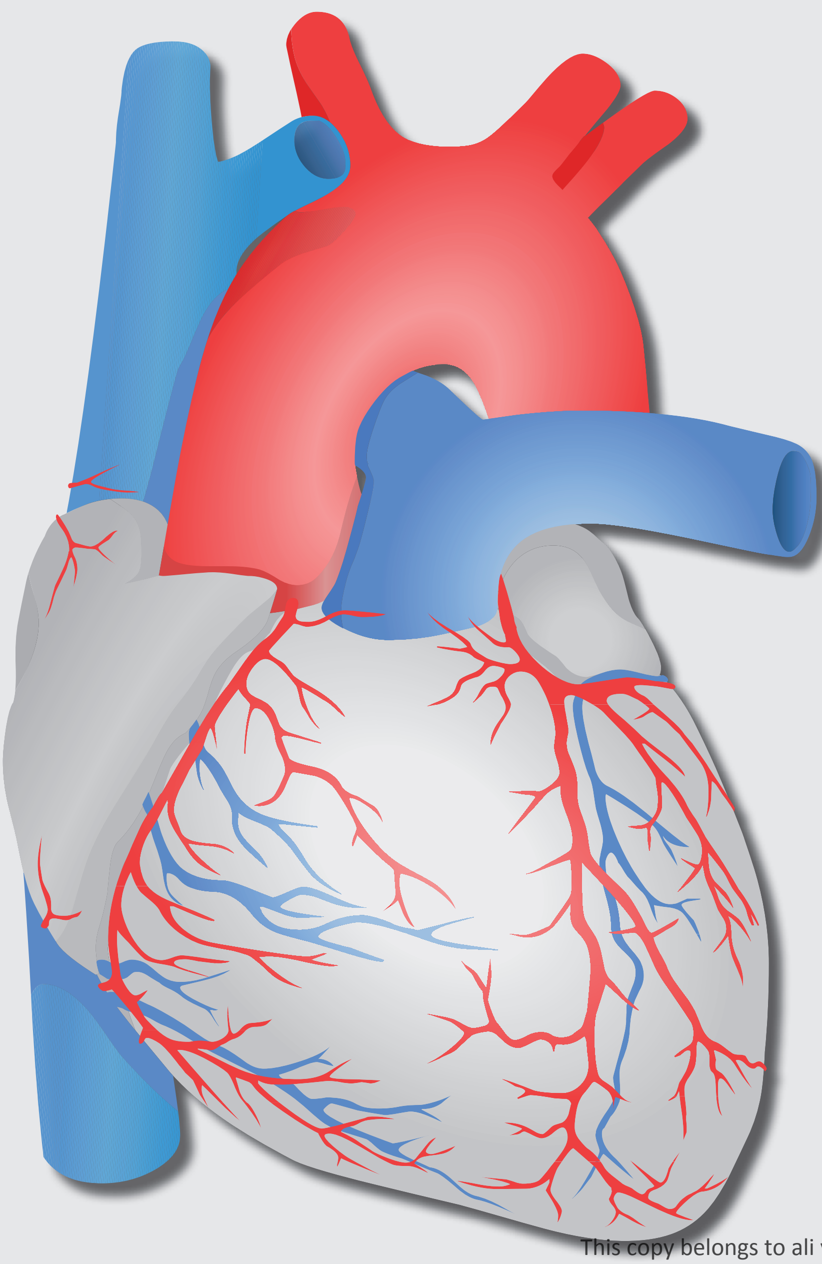


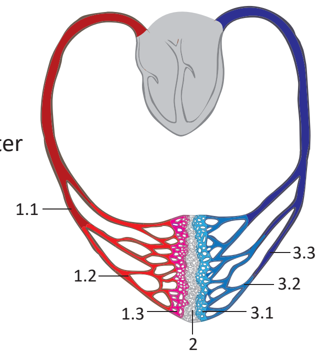
There are three types of blood vessels: **arteries, veins, and capillaries**. **Arteries** carry blood **from the heart to the organs**. **Veins** drain blood **from the organs** and direct it back **to the heart**. **Capillaries** form a **network of thin-walled vessels** in which oxygen and nutrient exchange between the blood and tissues takes place. All these types of vessels are organised in the **pulmonary and systemic circulation**. Direct connections between vessels of approximately the same calibre are called **anastomoses**.

Types of blood vessels

1 **Arteries** (*arteriae*)
- 1.1 **Large arteries** – elastic arteries
  – the pulmonary trunk and the aorta with its major branches
- 1.2 **Medium-sized and small arteries** – muscular arteries
  – regulate blood pressure by changing the size of their diameter
  – constitute the majority of arteries
- 1.3 **Arterioles** (arteriolae) – 0.1–0.2 mm in diameter

2 **Capillaries** (*vasa capillaria*)

3 **Veins** (*venae*)
- 3.1 **Venules** (*venulae*) – 0.2–1 mm in diameter
- 3.2 **Small and medium-sized veins** – 1–9 mm in diameter
  – the most common type of vein, contain valves
- 3.3 **Large veins** – the superior and inferior vena cava and their major tributaries



Histology

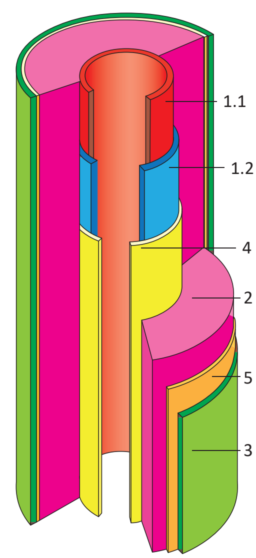
General anatomy of the vascular wall

1 **Intima** (*tunica intima*) – the innermost layer in contact with the blood
- 1.1 **Endothelium** – a single layer of flat endothelial cells
- 1.2 **Subendothelial layer** (*stratum subendotheliale*)
  – loose connective tissue

- 2 **Media** (*tunica media*) – the middle layer
  – contains reticular and elastic fibres
  and circularly arranged smooth muscle cells
- 3 **Adventitia** (*tunica adventitia/externa*) – the outermost layer
  – contains longitudinally arranged collagen and elastic fibres
  – the fibres cross-link and attach to the surrounding connective tissue

Specifics of the structure of the arterial wall

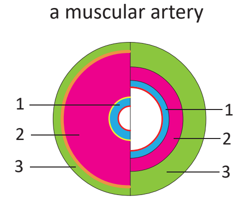
– the media is the thickest layer due to its smooth muscle component
- 4 **Internal elastic lamina** (*membrana elastica interna*)
  – a layer of elastic fibres on the inner side of the media
  of small and medium-sized arteries
- 5 **External elastic lamina** (*membrana elastica externa*)
  – a layer of elastic fibres on the outer side of the media
  of small and medium-sized arteries



Longitudinal section of a muscular artery

Specifics of the structure of the venous wall

– adventitia is the thickest layer
– venous valves are extensions of the intima
located in small and medium-sized veins



Cross-section of artery (left) and vein (right)

Anastomoses

Shunts (anastomoses) are direct connections between vessels.
They dilate to change the pattern of blood distribution in various tissues of the body.

1 **Arterio-venous (a-v) anastomoses** – directly connect arterioles and venules
– are located in the skin of the palms and fingertips and function to regulate body temperature
– are located in the kidneys, lungs and thyroid gland, where they participate in the regulation of blood pressure

2 **Arterio-arterial (a-a) anastomoses** – directly connect small and medium-sized arteries
– are located in the gastrointestinal tract
– form the palmar and plantar arches in the hands and feet
– the circle of Willis is a special kind of arterio-arterial anastomosis

3 **Veno-venous (v-v) anastomoses** – directly connect small and medium-sized veins
– constitute the porto-caval and the cavo-caval anastomoses between the portal vein and inferior vena cava, and inferior and superior vena cavae, respectively

Cor is the Latin term for heart.
Kardia is the Greek term for heart.
Vas is the Latin term for vessel (plural: vasa). The term vascularis means vascular.
Angeion is Greek for vessel.

The circulatory system consists of the heart and the blood and lymphatic vessels. The cardiovascular system consists of the heart and blood vessels.

Types of capillaries:
Continuous capillaries are lined by a smooth layer of endothelium that does not have any pores (fenestrations). This type of capillary is found in the majority of organs, including the brain and muscles.
Fenestrated capillaries with diaphragms are lined by endothelium that is decorated by small pores covered by diaphragms. They are located in organs where rapid exchange between tissue and blood takes place, such as in the intestines and endocrine glands.
Fenestrated capillaries without diaphragms are found in the glomeruli of the kidney
Sinusoids are characterised by a layer of endothelium with gaps between the endothelial cells and many fenestrations. They are found in the haematopoietic organs (liver, spleen, blood marrow, and dental pulp).

The thick muscular wall of arteries tends to maintain a circular lumen. Veins have a much thinner muscular wall and so the shape of their lumen often changes due to pressure from surrounding structures.

The vascular wall is supplied by the vasa vasorum and innervated by the nervi vasorum. The inner part of the vascular wall derives its nutrients by diffusion from the blood.

Clinical notes

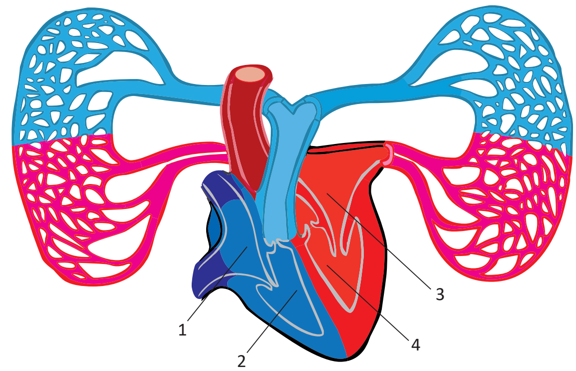
Formation of a blood clot (thrombus) in the arterial or deep venous circulation is a clinically serious state. The thrombus may break off the vessel wall and block a vessel down stream to it. Such a complication is known as a thromboembolism.

Transposition of the great arteries is a congenital heart defect where the aorta is connected to the right ventricular outflow tract and the pulmonary trunk to the left ventricular outflow tract. The pulmonary and systemic circulations are therefore completely separated unless a ventricular septal defect allows for blood mixing, which fortunately occurs in most cases.

The **anatomy of the heart wall** resembles the large elastic arteries; its **internal and middle layers consist of endothelial cells and thick muscle tissue, respectively**. During development, the heart proceeds through three stages: **formation of the heart tubes, loop formation and septation.** A series originally paired vessels form the aortic arch with its branches. **The first contractions of the heart tube can be observed as early as the 20th day of intrauterine life.**

Flow of blood through the heart and the pulmonary circulation

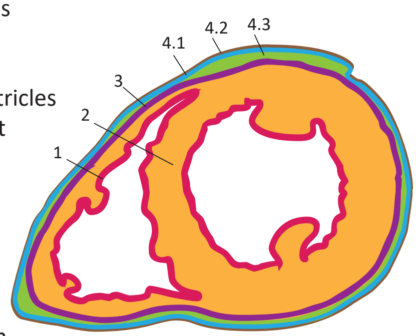
- Superior and inferior vena cava (*vena cava superior et inferior)*
- 1 **Right atrium** (*atrium dextrum*)
- 2 **Right ventricle** (*ventriculus dexter*)
- Pulmonary trunk (*truncus pulmonalis*)
  Pulmonary capillaries
- Pulmonary veins (*venae pulmonales*)
- 3 **Left atrium** (*atrium sinistrum*)
- 4 **Left ventricle** (*ventriculus sinister*)
- Aorta



Histology

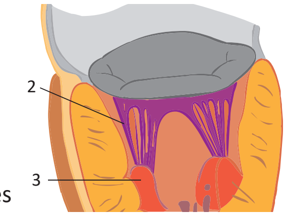
Heart wall

- 1 **Endocardium** – the innermost layer of the heart wall lining the chambers of the heart
  – is composed of endothelium and connective tissue; forms the entire mass of the heart valves
  - 1.1 **Subendocardial layer** (*stratum subendocardiale*) – a layer of connective tissue containing part of the conducting system of the heart
- 2 **Myocardium** – consists of two layers of cardiac muscle in the atria and three layers in the ventricles
  - 2.1 **Vortex of heart** (*vortex cordis*) – the spiral-shaped musculature of the apex of the heart
- 3 **Epicardium** – the visceral layer of pericardium, a serosal membrane
  – lies over the heart and the vessels that supply the myocardium with oxygen and nutrients
  - 3.1 **Subepicardial layer** (*lamina subepicardiaca*)
    – consists of adipose and connective tissues embedding the vessels of the heart
- 4 **Pericardium** – a double-layered sac covering the heart in the middle inferior mediastinum
  – forms an impression on the diaphgram called the *area pericardiaca*
  - 4.1 **Serous pericardium** (*pericardium serosum*) – the inner layer of the parietal pericardium
    – is reflected onto the heart forming the epicardium (the visceral pericardium)
  - 4.2 **Fibrous pericardium** (*pericardium fibrosum*) – the outer layer of the parietal pericardium
    – is composed of dense connective tissue
  - 4.3 **Pericardial cavity** (*cavitas pericardiaca*)
    – a potential space between the epicardium and the (visceral) serous pericardium
    – contains a small amount of fluid, which allows smooth movements of the heart



Transverse section of the heart at the level of the ventricles, inferior view

Heart valves (*valvae cordis*) – belong to the endocardium, are composed of three layers of avascular tissue:
a fibrous layer (*lamina fibrosa*) of dense connective tissue lined by endothelium on both sides
– commissures are junctions of the adjacent cusps of valves, formed by fibroelastic tissue
1 **Fibrous rings** (*anulus fibrosus*) – part of the cardiac skeleton that provide the attachment of the heart valves



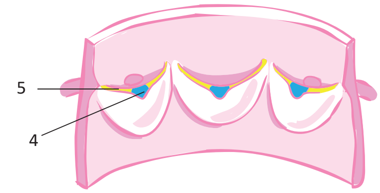
Atrioventricular valve

Characteristics of the atrioventricular (AV) valves

- 2 **Tendinous cords** *(chordae tendineae)* – anchor the AV valves to the papillary muscles
- 3 **Papillary muscles** (*musculi papillares*) – prevent the valves from prolapsing into the atria

Characteristics of the semilunar valves

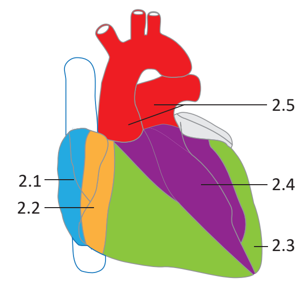
- 4 **Nodules** (*noduli*) – central thickenings of the semilunar valvules
- 5 **Lunules** (*lunulae*) – peripheral attenuations of the semilunar valvules



Semilunar valve

Development

1. **Early development** – mesodermal cells of the cardiogenic zone give rise to two heart tubes that fuse to form a single heart tube
2. **Loop formation** – the heart tube bends ventrally and then to the right, forming the first heart cavities
   – the heart tube is caudally attached to the septum transversum (the precusor of the central tendon of the diaphragm) and cranially to the aortic arches
   - 2.1 **Sinus venosus** – the future inflow tract of the right atrium (does not originate from the heart tube)
   - 2.2 **Common atrium** (*atrium commune*) – the future right atrium
   - 2.3 **Embryonic common ventricle** (*ventriculus communis*) – the future inflow tract of the ventricles
   - 2.4 **Conus arteriosus** (*conus arteriosus*) – the future outflow tract of the ventricles
   - 2.5 **Truncus arteriosus** (*truncus arteriosus*) – the future aorta and pulmonary trunk
3. **Septation** – formation of valves and septa
   – divides the heart into four chambers and separates the aorta from the pulmonary trunk

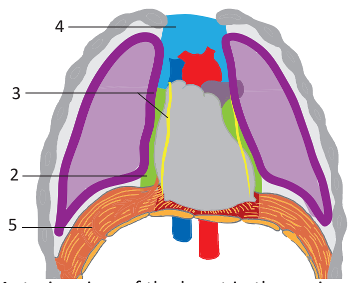


Heart divided into its original developmental parts

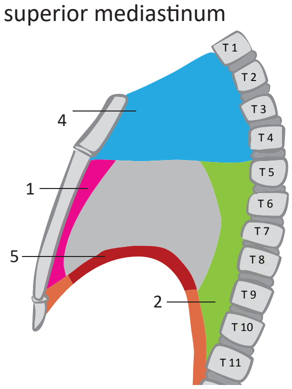
The heart is the main organ of the cardiovascular system. It is located **behind the sternum in the middle inferior mediastinum**. One-third of the heart is positioned to the right of the median plane whilst the other two-thirds are on the left side. **The cardiac axis** connects the opening of the superior vena cava with the apex of the heart; it **directs diagonally, ventrally, and caudally**. The heart consists of four chambers separated by septa and valves. **From the clinical and functional viewpoint, the heart is divided into the right heart and the left heart. The pericardium** is a firm sac **composed of two sheets** that are separated by **15–50 ml of serous fluid, enabling smooth heart movements**. Anteriorly, the heart is positioned in the pericardium freely while posteriorly it is attached by the main vessels entering and leaving the heart.

Topography of the heart in the pericardium

- 1 **Ventrally:** anterior inferior mediastinum and sternum
- 2 **Dorsally:** posterior inferior mediastinum and vertebral column
- 3 **Laterally:** descending phrenic nerves with the pericardiacophrenic vessels and pericardium lining the mediastinal pleura
- 4 **Cranially:** the aortic arch, arising from the heart and heading to the superior mediastinum
- 5 **Caudally:** central tendon of the diaphragm



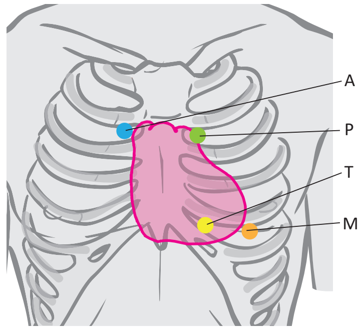
Anterior view of the heart in the pericardium



Sagittal section of the mediastinum

Projection of the heart onto the anterior thoracic wall

- **The projection of the heart on the anterior throacic wall** is determined by four points:
  - **2nd intercostal space on the right**, 1 cm lateral to the sternum
  - **5th intercostal space on the right** near the linea sternalis
  - **5th intercostal space on the left** medial to the midclavicular line corresponds to the projection of the apex of the heart
  - **2nd intercostal space on the left side** 2 cm lateral to the sternum



Projection of the heart and auscultation sites of the heart valves

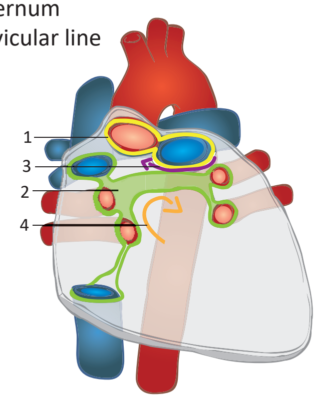
Auscultation sites of the heart valves

- A – **Aortic valve:** 2nd right intercostal space close to the sternum
- P – **Pulmonary valve:** 2nd left intercostal space close to the sternum
- T – **Tricuspid valve:** 5th right or left intercostal space close to the sternum
- M – **Mitral valve:** 5th left intercostal space just medial to the midclavicular line

Pericardium

The pericardium forms several structures where its parietal layer and visceral layers fuse:

- 1 **Porta arteriarum** – a cranially positioned opening for the aorta and pulmonary trunk arising from the heart
- 2 **Porta venarum** – a caudally positioned opening for the two caval veins and the four pulmonary veins entering the heart
- 3 **Transverse pericardial sinus** (*sinus transversus pericardii*) – a space on the posterior surface of the pericardium between porta arteriarum and porta venarum
- 4 **Oblique pericardial sinus** (*sinus obliquus pericardii*) – a space beneath the area of the porta venarum



Anterior view of the posterior part of the pericadrium with the heart removed

The average weight of a human heart is 290–350 g.

The location of the heart depends on age, position of the diaphragm, the respiratory movements, the shape of the thorax and body position. In a short and wide thorax with a high position of the diaphragm the heart is positioned more horizontally. In a narrow and long thorax the heart is positioned more longitudinally.

Pendulum heart *(cor pendulum)* is a term for a heart located in a narrow and long thorax; the heart looks like it's hanging down from the great vessels.

The auscultation points are also termed Testut's points. Connecting these points with a line traces the circumference of the projection of the heart onto the thoracic wall.

Viscerosensory innervation of the heart is provided by branches of the phrenic nerves that innervate the pericardium.

The pericardiacophrenic arteries accompanying the phrenic nerves provide the arterial supply of the pericardium.

Clinical notes

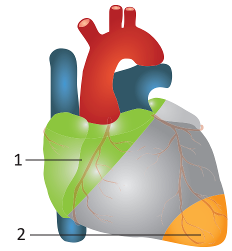
Cardiac tamponade is a pathological disorder in which fluid accumulates in the pericardial cavity causing the heart to be unable to adequately distend during diastole. The pericardial cavity can expand to contain 1000 ml of fluid. If the increase of fluid occurs slowly over a long period time there may be no clinical manifestations. An acute tamponade is caused by a sudden increase of fluid, by approximately tens of millilitres (max. 200 ml) and is a medical emergency. This situation may occur due to rupture of free myocardial wall over a postischemic scar or as a result of rupture or dissection of the aorta.

Pericardiocentesis is a surgical procedure in which fluid is removed from the pericardial cavity. It is performed with a special needle that is inserted between the xiphoid process and the left costal margin within the infrasternal angle under ultrasound control. It is also performed, but less commonly, in the inferior interpleural area in the 5th left intercostal space near the left sternal border.

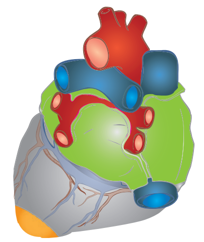
Structures

General parts

- 1 **Base of heart** (*basis cordis*) – faces superiorly, posteriorly, and medially
  – the location of the great vessels entering and leaving the heart
  – its superior margin reaches the border of the superior mediastinum
- 2 **Apex of heart** (*apex cordis*) – faces inferiorly, anteriorly, and laterally
  – touches the thoracic wall at the 5th intercostal space in the midclavicular line

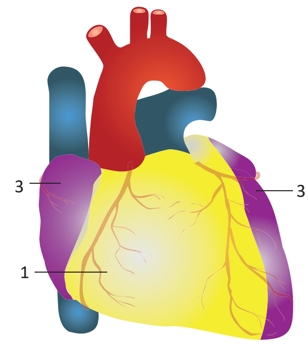


Anterior view

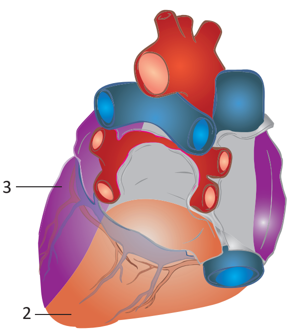

Posterior view

Surfaces

- 1 **Anterior surface** (*facies anterior/sternocostalis*)
  – is formed mainly by the right ventricle and partially by the left ventricle
- 2 **Inferior surface** (*facies inferior/diaphragmatica*) – faces the diaphragm
  – is formed by the posterior surfaces of both ventricles
- 3 **Right and left pulmonary surfaces** (*facies pulmonalis dextra et sinistra*)
  – the right pulmonary surface is formed by the right atrium
  – the left pulmonary surface is formed by the left atrium and left ventricle
  3.1 **Cardiac impression** (*impressio cardiaca pulmonis*)
  – the impression of the pulmonary surfaces of the heart on the lungs



Surfaces, anterior view



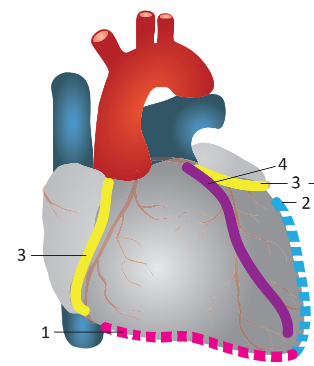
Surfaces, posterior view

Borders

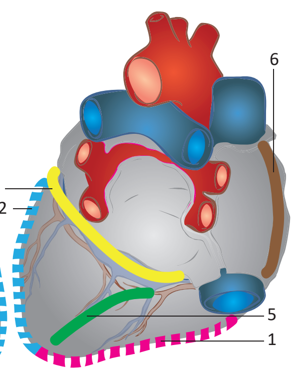
- 1 **Right border** (*margo dexter/acutus*) – a sharp margin on the right ventricle
- 2 **Left border** (*margo sinister/obtusus*) – a rounded margin on the left ventricle

Sulci

- 3 **Coronary sulcus** (*sulcus coronarius*) – a groove on the surface of the heart separating the atria from the ventricles
  – contains the right coronary artery, the circumflex branch of the left coronary artery, the great cardiac vein and the coronary sinus
- 4 **Anterior interventricular sulcus** (*sulcus interventricularis anterior*)
  – a groove on the anterior surface of the heart between the right and left ventricles
  – is formed by the interventricular septum
  – contains the anterior interventricular branch and the great cardiac vein
- 5 **Posterior interventricular sulcus** (*sulcus interventricularis posterior*)
  – a groove on the posterior surface of heart between the right and left ventricles
  – is formed by the interventricular septum
  – contains the posterior interventricular branch and middle cardiac vein
- 6 **Sulcus terminalis cordis** – is formed by the crista terminalis
  – a groove close to the opening of the superior and inferior vena cava
  – the border between the proper right atrium and the sinus venosus



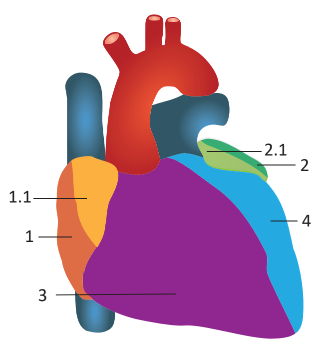
Borders and sulci, anterior view



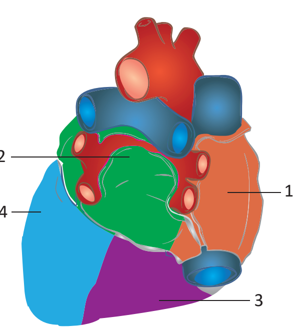
Borders and sulci, posterior view

Heart chambers

- 1 **Right atrium** (*atrium dextrum*)
  – receives oxygen-poor blood from the systemic circulation
  - 1.1 **Right auricle** (*auricula dextra*)
- 2 **Left atrium** (*atrium sinistrum*)
  – receives oxygen-rich blood from the pulmonary circulation
  - 2.1 **Left auricle** (*auricula sinistra*)
- 3 **Right ventricle** (*ventriculus dexter*) – receives blood from the right atrium and pumps it to the pulmonary circulation via the pulmonary trunk
- 4 **Left ventricle** (*ventriculus sinister*) – receives blood from the left atrium and pumps it to the systemic circulation via the aorta



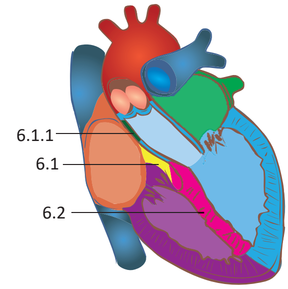
Heart chambers, anterior view



Heart chambers, posterior view

Cardiac septa

5 **Interatrial septum** (*septum interatriale*) – a thin septum separating the right and left atria
6 **Interventricular septum** (*septum interventriculare*)
– a thick septum separating the right and left ventricles
- 6.1 **Membranous part** (*pars membranacea*) – the superior thin fibrous part between the inflow tract of the right ventricle and the outflow tract of the left ventricle
  - 6.1.1 **Atrioventricular septum** (*septum atrioventriculare*)
    – part of the membranous part of interventricular septum, between the right atrium and the outflow tract of the left ventricle
- 6.2 **Muscular part** (*pars muscularis*) – the inferior thick muscular part



Frontal section of the heart showing the heart chambers and septa

Blood from the systemic circulation flows **through the superior and inferior vena cava into the right atrium**. The **tricuspid valve** directs blood from the right atrium into the right ventricle. The initial parts of the conducting system of the heart, **the sinoatrial and atrioventricular nodes**, are located in the wall of the right atrium.

Basic structures

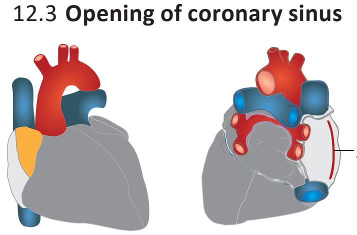
- 1 **Right auricle** (*auricula dextra*) – an outpocketing appendage of the right atrium
  - 1.1 **Pectinate muscles** (*musculi pectinati*) – comb-like muscle fascicles inside the right auricle
- 2 **Interatrial septum** (*septum interatriale*) – a thin barrier separating the right and left atria
  - 2.1 **Oval fossa** (*fossa ovalis*) – an oval-shaped depression in the interatrial septum – a remnant of the septum primum and septum secundum that merge shortly after birth, closing the foetal interatrial shunt (*foramen ovale*)
    - 2.1.1 **Border of oval fossa** (*limbus fossae ovalis*) – an elevated ridge around the margin of the oval fossa
- 3 **Atrioventricular septum** (*septum atrioventriculare*) – a fibrous barrier between the right atrium and the outflow tract of the left ventricle
- 4 **Crista terminalis** – a muscular crest inside the right atrium
  – separates the sinus of the venae cavae from the proper right atrium
- 5 **Sulcus terminalis cordis** – a groove on the external surface of the right atrium corresponding to the crista terminalis
- 6 **Intervenous tubercle** (*tuberculum intervenosum*)
  – an elevated muscular ridge between the superior and inferior vena cava
  – directs blood from the superior vena cava to the right ventricle

Openings in the right atrium

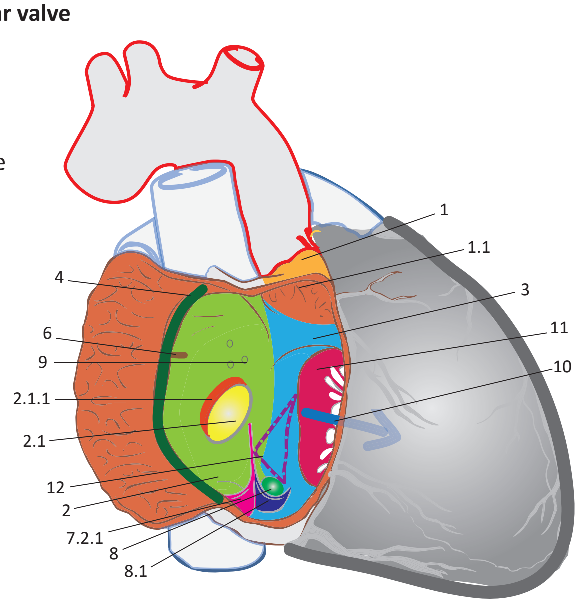
- 7 **Sinus of venae cavae** (*sinus venarum cavarum*)
  – the posterior part of the right atrium where the superior and inferior vena cava enter
  - 7.1 **Opening of superior vena cava** (*ostium venae cavae superioris*)
  - 7.2 **Opening of inferior vena cava** (*ostium venae cavae inferioris*)
    - 7.2.1 **Valve of inferior vena cava / Eustachian valve** (*valvula venae cavae inferioris*)
      –a semilunar fold at the opening of the inferior vena cava
      –directs blood through foramen ovale to the left atrium during prenatal life
- 8 **Opening of coronary sinus** (*ostium sinus coronarii*)
  – the opening of the main vein draining blood from the heart
  - 8.1 **Valve of coronary sinus / Thebesian valve** (*valvula sinus coronarii*)
    – a fold covering the opening of the coronary sinus
- 9 **Openings of smallest cardiac veins** (*foramina venarum minimarum*)
- 10 **Right atrioventricular orifice** (*ostium atrioventriculare dextrum*) – an opening between the right atrium and the right ventricle covered by the tricuspid valve

Other structures

- 11 **Tricuspid valve / right atrioventricular valve** (*valva atrioventricularis dextra / tricuspidalis*)
- 12 **Triangle of Koch** (*trigonum nodi atrioventricularis*)
  – location of the atrioventricular node
  – its borders are:
  - 12.1 **Septal cusp of tricuspid valve**
  - 12.2 **Tendon of valve of inferior vena cava / Tendon of Todaro** (*tendo valvulae venae cavae inferioris*)
  - 12.3 **Opening of coronary sinus**



Anterior and posterior views of the heart, right ventricle in bright color



Open right ventricle after cutting off the right auricle

The anterior veins of the right ventricle are small veins on the anterior surface of the heart that open directly into the right atrium.

The right ventricular outflow tract is also called the conus arteriosus (*infundibulum / pars glabra*) because of its funnel-like shape narrowing cranially towards the opening of the pulmonary trunk.

The stroke volume is the volume of blood ejected in each heartbeat.

The thickness of the wall of the right ventricle is 3–4 mm. The left ventricular wall can be up to three times thicker (7–11 mm).

The septal papillary muscle is made up of several smaller muscles.

The crista terminalis separates the sinus venosus from the common atrium during prenatal development.

Clinical notes

Right heart failure results in blood congestion in the systemic circulation and an elevation in central venous pressure. This condition manifests with oedema, which accumulates around the ankles (perimalleolar oedema) due to gravity and elevated jugular venous pressure detected by increased filling of the external jugular veins. Blood also accumulates in the portal circulation, which leads to enlargement of the liver and spleen (hepatosplenomegaly) and an accumulation of fluid in the peritoneal cavity (ascites).

Tricuspid valve stenosis is a relatively infrequent condition in which the orifice of the tricuspid valve is narrowed, usually due to endocarditis in intravenous drug users.
Tricuspid insufficiency is a condition in which blood regurgitates through the tricuspid valve to the right atrium. In most cases, it is caused by increased pressure in the pulmonary circulation that results in dilation of the right fibrous ring and tricuspid insufficiency.

Pulmonary stenosis is a narrowing of the right ventricular outflow tract in the area of the pulmonary valve and usually occurs as a part of complex congenital heart defects.
Pulmonary valve insufficiency is a relatively rare condition in which blood regurgitates from the pulmonary trunk to the right ventricle. It is caused by a relative dilation of the right ventricle (result of volume overload and pulmonary hypertension) in adult patients treated for complex congenital heart defect in childhood.

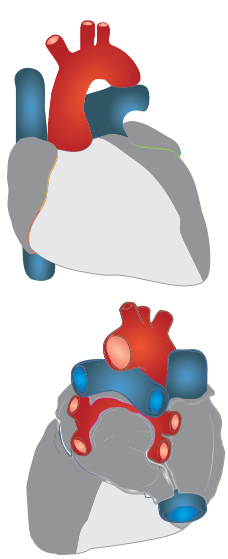
Blood enters the right ventricle through the tricuspid valve. Blood is directed first through **the trabecular inflow tract, composed of muscular bundles (trabeculae carnae)**, and then through **the smooth outflow tract (conus arteriosus)**. Blood flows through the pulmonary valve to the pulmonary trunk and consequently throughout the entire pulmonary circulation where gas exchange takes place. The right ventricle pumps against the relatively low pulmonary blood pressure of around 20 mmHg.

Inflow tract

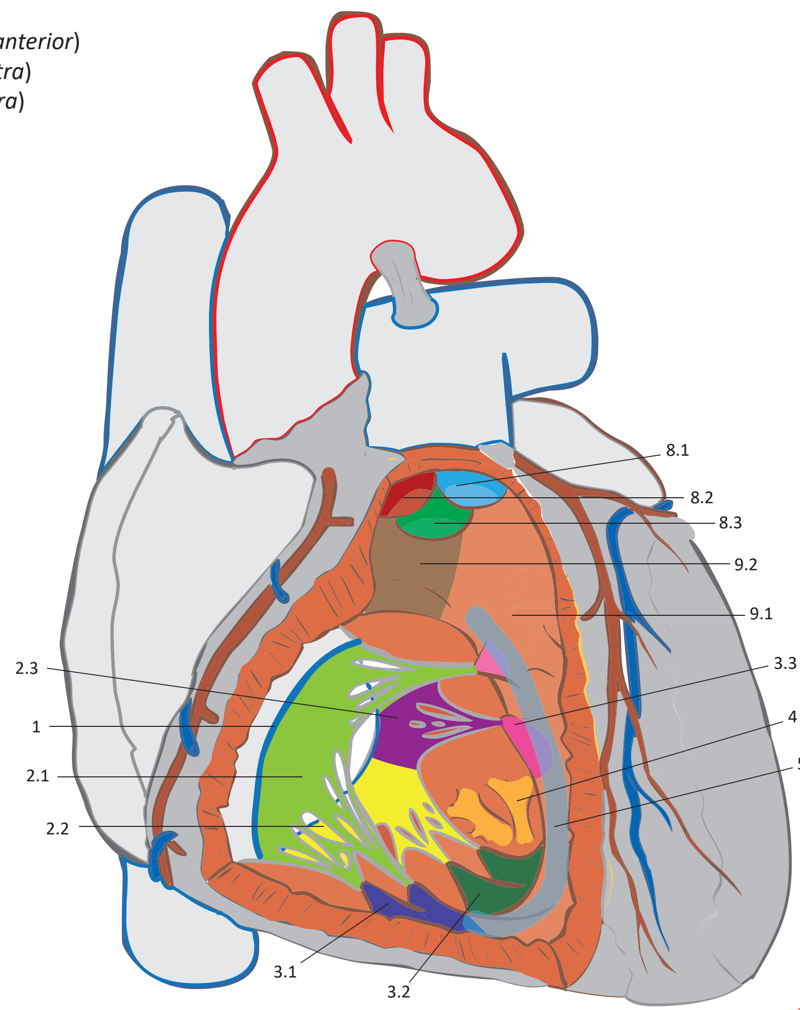
- 1 **Right atrioventricular orifice** (*ostium atrioventriculare dextrum*) – the opening between the right atrium and the right ventricle
- 2 **Tricuspid valve / right atrioventricular valve** (*valva atrioventricularis dextra/tricuspidalis*) – prevents retrograde flow from the right ventricle to the right atrium
  - 2.1 **Anterior cusp** (*cuspis anterior*) – is anchored to the anterior and septal papillary muscles
  - 2.2 **Posterior cusp** (*cuspis posterior*) – is anchored to the posterior and septal papillary muscles
  - 2.3 **Septal cusp** (*cuspis septalis*) – is anchored to the posterior and septal papillary muscles
- 3 **Papillary muscles** (*musculi papillares*) – muscles originating from myocardium, covered with endocardium and attached to the cusps of the tricupid valve by tendinous cords (*chordae tendineae*)
  - 3.1 **Anterior papillary muscle** (*musculus papillaris anterior*) – is attached to the anterior cusp
  - 3.2 **Posterior papillary muscle** (*musculus papillaris posterior*) – is attached to the posterior and septal cusps
  - 3.3 **Septal papillary muscle** (*musculus papillaris septalis*) – is attached to the anterior, posterior, and septal cusps
- 4 **Trabeculae carnae** – thick muscular bundles in the right ventricular inflow tract

Outflow tract – conus arteriosus

- 5 **Septomarginal trabecula / moderator band** (*trabecula septomarginalis*)
  – a muscular bundle connecting the interventricular septum with the anterior papillary muscle
  – the right bundle branch of the conducting system of the heart runs within the septomarginal trabecula
- 6 **Supraventricular crest** (*crista supraventricularis*) – a crest dividing the right ventricle into inflow and outflow tracts
- 7 **Opening of pulmonary trunk** (*ostium trunci pulmonalis*) – leads to the pulmonary trunk and contains the pulmonary valve
- 8 **Pulmonary valve** (*valva trunci pulmonalis*) – is composed of the three semilunar cusps
  – prevents retrograde flow of blood from the pulmonary trunk into the right ventricle
  - 8.1 **Anterior semilunar cusp** (*valvula semilunaris anterior*)
  - 8.2 **Right semilunar cusp** (*valvula semilunaris dextra*)
  - 8.3 **Left semilunar cusp** (*valvula semilunaris sinistra*)
- 9 **Interventricular septum** (*septum interventriculare*)
  - 9.1 **Muscular part** (*pars muscularis*)
  - 9.2 **Membranous part** (*pars membranacea*)
    – part of the cardiac skeleton



Anterior and posterior views of the heart, right ventricle in bright color



Oxygenated blood from the pulmonary circulation flows through **the four pulmonary veins to the left atrium** and then through **the mitral valve to the left ventricle.**

Basic structures

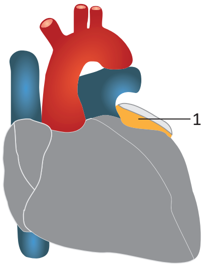
- 1 **Left auricle** (*auricula sinistra*) – an outpocketing appendage of the left atrium
  - 1.1 **Pectinate muscles** (*musculi pectinati*)
    – little muscular fascicles organised in ridges inside the left auricle
- 2 **Interatrial septum** (*septum interatriale*) – a septum between the right and left atria
- 3 **Valve of foramen ovale** (*valvula foraminis ovalis*)
  – a circular thinned area in the interatrial septum
  – the original valve covering the foramen ovale
  – postnatally merges with the primary interatrial septum and thus can be found on the opposite side of the oval fossa

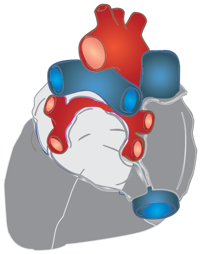
Openings in the left atrium

- 4 **Openings of pulmonary veins** (*ostia venarum pulmonalium*)
  – openings of the four pulmonary veins, two right and two left veins
- 5 **Openings of smallest cardiac veins** (*ostia venarum minimarum*)
- 6 **Left atrioventricular orifice** (*ostium atrioventriculare sinistrum*)
  – an opening between the left atrium and the left vetricle containing the mitral valve

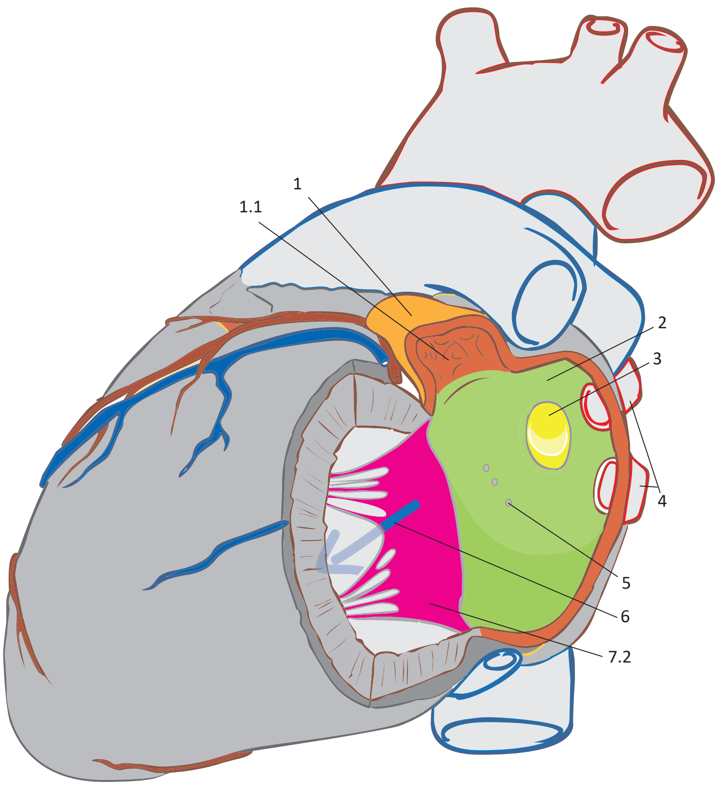
Other structures

- 7 **Mitral valve / left atrioventricular valve** (*valva atrioventricularis sinistra / valva mitralis*)
  7.1 **Anterior cusp** (*cuspis anterior*)
  7.2 **Posterior cusp** (*cuspis posterior*)





Anterior and posterior views of the heart; left atrium in bright color



The posterior aortic sinus of Valsalva (*sinus aortae posterior Valsalvae*) is a noncoronary sinus, from which no coronary artery arises.

Normal blood pressure values are around 120/80 mmHg. Elevated blood pressure (hypertension) is defined as blood pressure over 140/90 mmHg. Blood pressure below 90/60 mmHg is considered abnormally low (hypotension).

Bicuspid valve is an obsolete term for the mitral valve.

Clinical notes

Left-sided heart failure leads to congestion of blood and elevated blood pressure in the pulmonary circulation. This situation results in lung oedema presenting as dyspnoea that is most prominent when lying down. Dyspnoea is relieved when the patient sits or stands up (orthopnoea).

Atrial fibrillation is a heart rhythm disorder in which the atria contract irregularly, chaotically, and thus ineffectively. This causes a change of blood flow in the atrial auricle from laminar to turbulent with an increased risk of blood clot formation. A blood clot may be released to the systemic circulation and obstruct an artery supplying the brain and/or other organs (thromboembolism), leading to a complete or partial loss of blood supply (ischaemia).

Mitral stenosis is described as a narrowing of the orifice of the mitral valve. It can lead to the accumulation of blood in the pulmonary circulation and lung oedema, which presents as dyspnoea. It is most commonly occurs as a sequela of rheumatic heart disease.

Mitral regurgitation (insufficiency) leads to reverse blood flow from the left ventricle to the left atrium during systole. It is one of the most common acquired valve defects, usually caused by a degenerative process of the valve. Mitral regurgitation often results in left-sided heart failure.

Aortic valve stenosis is a narrowing of the orifice of the aortic valve, usually caused by degenerative calcification (a process similar to atherosclerosis), thus presenting mostly in older age. It is a common acquired valve defects.

Aortic insufficiency leads to reverse flow from the aorta to the left ventricle. Most commonly, this is caused by degeneration or dilatation of the aortic ring and/or by other pathologic states.

Blood flows **from the left atrium through the mitral valve to the left ventricle**. It flows through the left ventricular **inflow tract composed of muscular bundles** (trabeculae carnae) and then through the **smooth left ventricular outflow tract** (the aortic vestibule). Blood is ejected through the aortic valve in**to the ascending aorta and then flows throughout** the entire systemic circulation. The left ventricle **pumps blood against the relatively high systemic blood pressure**.

Inflow tract

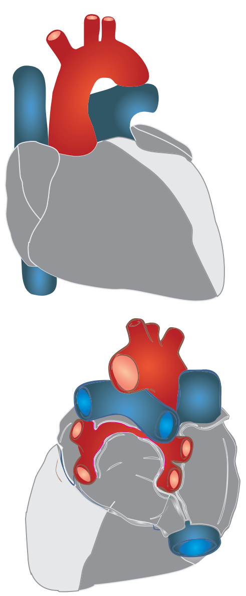
1 **Left atrioventricular orifice** (*ostium atrioventriculare sinistrum*)
– an opening between the left atrium and the left ventricle covered by the mitral valve
2 **Mitral valve / left atrioventricular valve** (*valva atrioventricularis sinistra / valva mitralis*)
– a valve preventing retrograde flow from the left ventricle to the left atrium
- 2.1 **Anterior cusp** (*cuspis anterior*) – is anchored to the anterior and posterior papillary muscles
- 2.2 **Posterior cusp** (*cuspis posterior*) – is anchored to the posterior and anterior papillary muscles

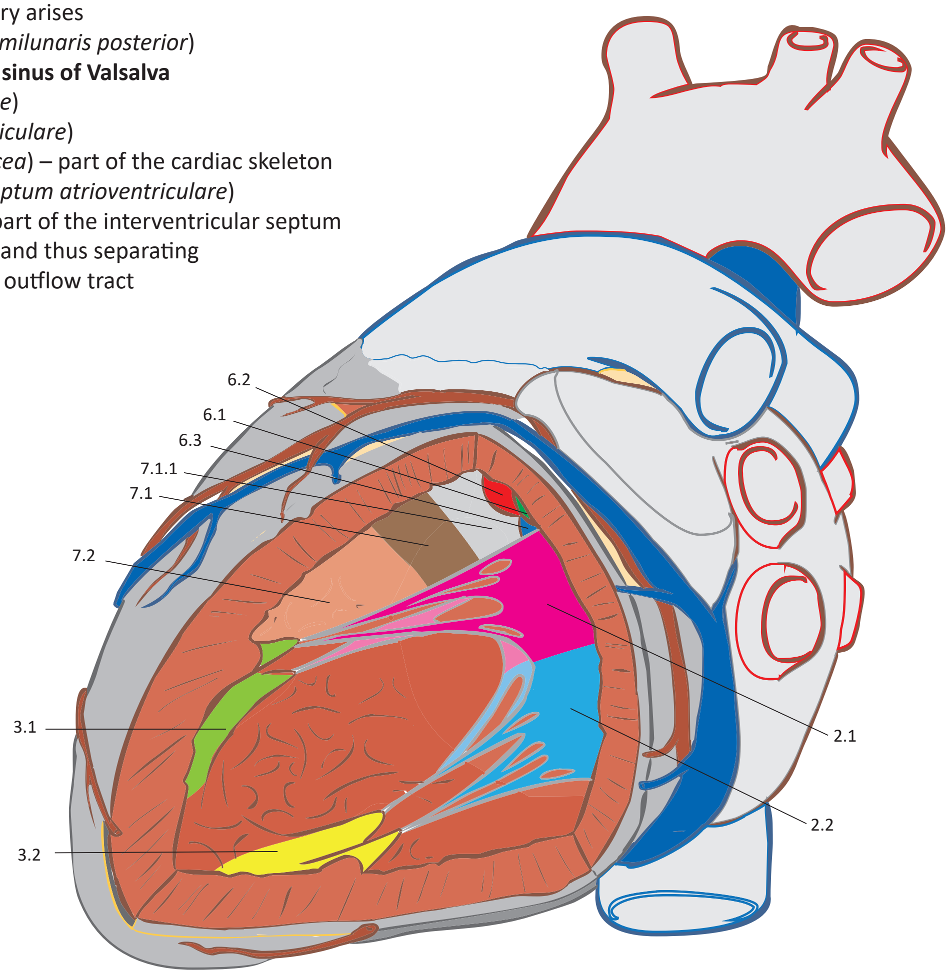
3 **Papillary muscles** (*musculi papillares*) – valve muscles originating from myocardium and covered with endocardium
– attached to the mitral valve cusps by tendinous cords (*chordae tendineae*)
- 3.1 **Anterior papillary muscle** (*musculus papillaris anterior*) – is attached to the anterior and posterior cusps
- 3.2 **Posterior papillary muscle** (*musculus papillaris posterior*) – is attached to the posterior and anterior cusps

Outflow tract

4 **Aortic vestibule** (*vestibulum aortae*) – the smooth part of the left ventricle below the aortic valve
5 **Aortic orifice** (*ostium aortae*) – the left ventricular opening to the ascending aorta containing the aortic valve
6 **Aortic valve** (*valva aortae*) – is composed of three semilunar cusps
– prevents reverse flow from the aorta to the left ventricle
- 6.1 **Left semilunar cusp** (*valvula semilunaris sinistra*)
  - 6.1.1 **Left (coronary) aortic sinus of Valsalva** (*sinus aortae sinister Valsalvae*)
    – a dilation of the aortic wall behind the left semilunar cusp where the left coronary artery arises
- 6.2 **Right semilunar cusp** (*valvula semilunaris dextra*)
  - 6.2.1 **Right (coronary) aortic sinus of Valsalva** (*sinus aortae dexter Valsalvae*)
    – a dilation of the aortic wall behind the right semilunar cusp where the right coronary artery arises
- 6.3 **Posterior semilunar cusp** (*valvula semilunaris posterior*)
  - 6.3.1 **Posterior (noncoronary) aortic sinus of Valsalva** (*sinus aortae posterior Valsalvae*)

7 **Interventricular septum** (*septum interventriculare*)
- 7.1 **Membranous part** (*pars membranacea*) – part of the cardiac skeleton
  - 7.1.1 **Atrioventricular septum** (*septum atrioventriculare*)
    – part of the membranous part of the interventricular septum overlying the right atrium and thus separating it from the left ventricular outflow tract
- 7.2 **Muscular part** (*pars muscularis*)


Anterior and posterior views of the heart; left ventricle in bright color



The valves are fibrous plates separating the atria from the ventricles and the ventricles from the great vessels leaving the heart. **They ensure unidirectional flow of blood**. Characterised by their shape, location, and dynamics, they are **divided into cuspidate and semilunar valves**. **The cuspidate valves** are opened by blood flowing **from the atrium into the ventricle** and are closed during ventricular systole. **The semicircular valves are pushed open as blood flows into the aorta and pulmonary trunk during ventricular systole**. During ventricular diastole blood flows into the heart and the semilunar valves close.

Cuspidate valves

1 **Tricuspidal valve / right atrioventricular valve** (*valva atrioventricularis dextra* / *valva tricuspidalis*)
– is composed of three cusps, which are attached to papillary muscles by tendinous cords
– separates the right atrium from the right ventricle
- 1.1 **Anterior cusp** (*cuspis anterior*) – is anchored to the anterior and septal papillary muscles by tendinous cords
- 1.2 **Posterior cusp** (*cuspis posterior*) – is anchored to the posterior and septal papillary muscles by tendinous cords
- 1.3 **Septal cusp** (*cuspis septalis*) – is anchored to the anterior, posterior and septal papillary muscles by tendinous cords

– the individual cusps are connected at the commissures
– there are three commisures: anteroseptal, posteroseptal, and anteroposterior

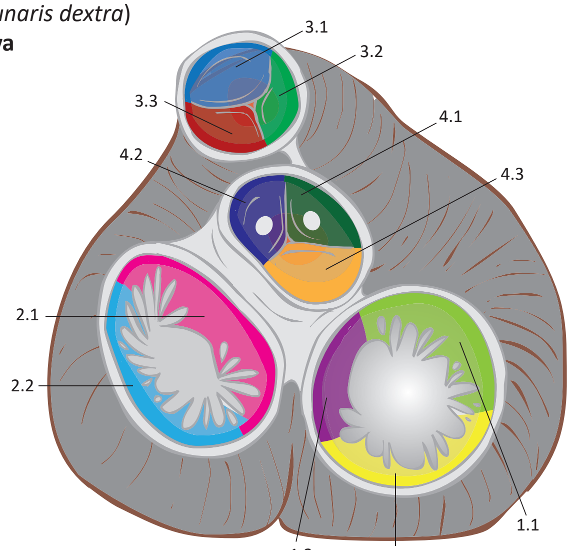
2 **Mitral valve / left atrioventricular valve**
(*valva atrioventricularis sinistra / valva bicuspidalis/mitralis*)
– contains two cusps that are attached to the papillary muscles by tendinous cords
– separates the left atrium and left ventricle
- 2.1 **Anterior cusp** (*cuspis anterior*) – anchored to the anterior and posterior papillary muscles by tendinous cords
- 2.2 **Posterior cusp** (*cuspis posterior*) –anchorded to the posterior and anterior papillary muscles by tendinous cords

– the cusps are connected at the anterolateral and posteromedial commissures

Semilunar valves

3 **Pulmonary valve** (*valva trunci pulmonalis*) – consists of three semilunar-shaped cusps
– is located between the right ventricle and the pulmonary trunk
- 3.1 **Anterior semilunar cusp** (*valvula semilunaris anterior*)
- 3.2 **Right semilunar cusp** (*valvula semilunaris dextra*)
- 3.3 **Left semilunar cusp** (*valvula semilunaris sinistra*)

4 **Aortic valve** (*valva aortae*) – is formed by three semilunar cusps
– is located between the left ventricle and the ascending aorta
- 4.1 **Right semilunar cusp** (*valvula semilunaris dextra*)
  - 4.1.1 **Right aortic sinus of Valsalva** (*sinus aortae dexter*)
    – a widening of the aorta containing the ostium of the right coronary artery
- 4.2 **Left semilunar cusp** (*valvula semilunaris sinistra*)
  - 4.2.1 **Left aortic sinus of Valsalva** (*sinus aortae sinister*)
    – a widening of the aorta containing the ostium of the left coronary artery
- 4.3 **Posterior semilunar cusp** (*valvula semilunaris posterior*)
  - 4.3.1 **Posterior aortic sinus of Valsalva** (*sinus aortae posterior*)



The right and left fibrous rings are oval and flat in comparison to the aortic and pulmonary rings, which are circle and garland-shaped (with three projections).

The tendon of Todaro (*tendo valvulae venae cavae inferioris*) originates in the right fibrous triangle, runs through the myocardium of the posterior wall of the right ventricle and circumscribes the atrioventricular node.

The tendon of infundibulum is tendon connecting the pulmonary ring (ventrally) and aortic ring (dorsally).

The membranous part of the interventricular septum (*pars membranacea septi interventricularis*) is the only part of the cardiac skeleton, which is vertically oriented.

The existence of specialised pathways spreading impulses from the right to the left atrium is disputable. The only exception is the interatrial bundle of Bachmann, which is, most likely, formed by non-pacemaking cardiomyocytes.

The sympathetic system increases the strength and frequency of myocardial contractions and the conductivity and excitability of the myocardium (positive inotropic, chronotropic, dromotropic, and bathmotropic effects, respectively).

The parasympathetic system decreases the strength and frequency of myocardial contractions and the conductivity and excitability of the myocardium (negative inotropic, chronotropic, dromotropic, and bathmotropic effects, respectively).

Clinical notes

A heart block is a block in the conducting system of the heart, which prevents the propagation of action potential through certain parts of the heart. Heart blocks are differentiated by their ECG profiles and clinical manifestations.

Sick sinus syndrome is a disorder of the sinu-atrial node (the main pacemarker of the heart). Many different patterns of the heart rhythm are typical for this syndrome.

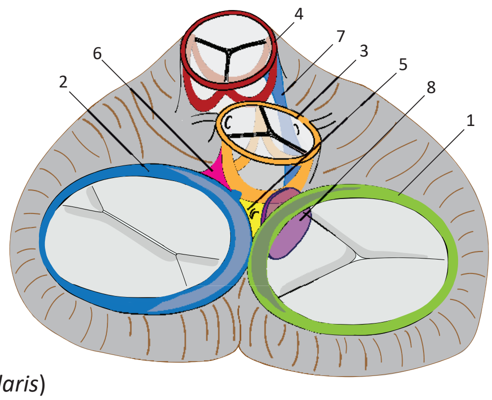
Atrioventricular blockages represent affections of the atrioventricular communication and action potential transmission. Three stages of AV block are distinguished: 1st, 2nd and 3rd degree block

Types of bundle branch block: right block, left anterior hemiblock and left posterior hemiblock.

The **cardiac skeleton** is a fibrous structure that **surrounds the heart valves** and serves as an **attachment for the myocardial muscle**. It is a poor conductor of electricity and thus ensures that electrical impulses pass from the atria to the ventricles only through the atrioventricular bundle. The components of the cardiac skeleton are the **fibrous rings of the valves, the fibrous triangles, the membranous part of the interventricular septum**, which is located perpendicularly to the plane of the valves and the **tendon of the infundibulum**.

Fibrous rings

- 1 **Right fibrous ring** (*anulus fibrosus dexter*) – the fibrous ring of the tricuspid valve
- 2 **Left fibrous ring** (*anulus fibrosus sinister*) – the fibrous ring of the mitral valve
- 3 **Aortic ring** (*anulus aorticus*) – the fibrous ring of the aorta
- 4 **Pulmonary ring** (*anulus trunci pulmonalis*) – the fibrous ring of the pulmonary valve



Other components

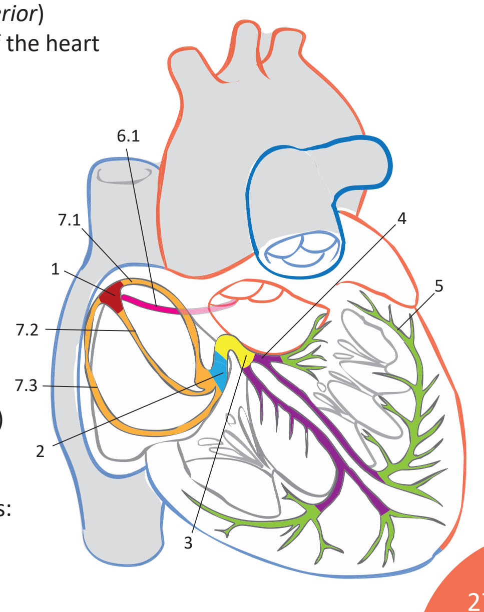
- 5 **Right fibrous triangle** (*trigonum fibrosum dextrum*) – situated between the right, left, and aortic fibrous rings, the atrioventricular bundle passes through this triangle
- 6 **Left fibrous triangle** (*trigonum fibrosum sinistrum*) – situated between the left and aortic fibrous rings
- 7 **Tendon of infundibulum** (*tendo infundibuli*)
- 8 **Membranous part of interventricular septum** (*pars membranacea septi interventricularis*)

3.7 **Conducting system of the heart** – Complexus stimulans cordis / systema conducens cordis

The heart has the capacity to **generate and conduct electric impulses**. These impulses are **produced in the sinu-atrial node** in the right atrium. They pass through the atria to the **atrioventricular node** and then pass through the **bundle of His** to reach the ventricles. After passing through the atrioventricular septum, the bundle of His divides into the **right and left bundles of Tawara,** which carry the impulses down the interventricular septum and into the left and right ventricles. The final branches of the cardiac conducting system are the **subendocardial branches of Purkinje**, which spread the impulses throughout the ventricular muscle.

Parts

- 1 **Sinu-atrial node of Keith-Flack / SA node** *(nodus sinuatrialis)*
  – is located in the right atrium between the superior vena cava and the right auricle
  – the main pacemaker of the heart, action potentials spread through atrial myocardium and interatrial connections to the AV node
- 2 **Atrioventricular node of Aschoff-Tawara / AV node** *(nodus atrioventricularis)* – is located in the right atrium within the triangle of Koch
  – is responsible for slowing SA impulses and passing transmission onto the atrioventricular fascicle
- 3 **Atrioventricular fascicle of Kent-Gaskell-His / Bundle of His** *(fasciculus atroventricularis)* – the continuation of the AV node, runs through the right fibrous triangle into the posterior part of the membranous part of the interventricular septum, then continues caudally and enters the muscular part of the interventricular septum, where it divides into the bundles of Tawara
- 4 **Bundles of Tawara** *(crus dextrum et sinistrum fasciculi atroventricularis)* – run through the interventricular septum to the heart apex where they divides into the subendocardial branches, the right bundle runs through the septomarginal trabecula of the right ventricle
  – the left bundle divides into an anterior and a posterior branch (*limbus anterior et posterior*)
- 5 **Subendocardial branches of Purkinje** *(rami subendocardiales)* – spread from the apex of the heart in the subendocardial layer to all the cardiac muscle cells of the ventricles
- 6 **Interconnections between the SA node and left atrium**
  - 6.1 **Interatrial fascicle of Bachmann** *(fasciculus interatrialis)*
- 7 **Accessory interconnections between the SA node and AV node**
  - 7.1 **Anterior internodal bundle of James**
  - 7.2 **Middle internodal bundle of Wenckebach**
  - 7.3 **Posterior internodal bundle of Thorel**



Autonomic innervation of the heart

Sympathetic system – cardiac nerves (*nervi cardiaci*)
– arise from all three cervical and a few thoracic sympathetic ganglia
Parasympathetic system – cardiac branches (*rami cardiaci*) – arise from the vagus nerve (n. X)
– fibres from the right vagus nerve primarily enter the SA node
– fibres from the left vagus nerve primarily enter the AV node
The cardiac plexus is a **mixed sympathetic and parasympathetic plexus** divided into two parts:
– the superficial cardiac plexus is located between the aorta and the pulmonary trunk
– the deep cardiac plexus is located between the aorta and the tracheal bifurcation

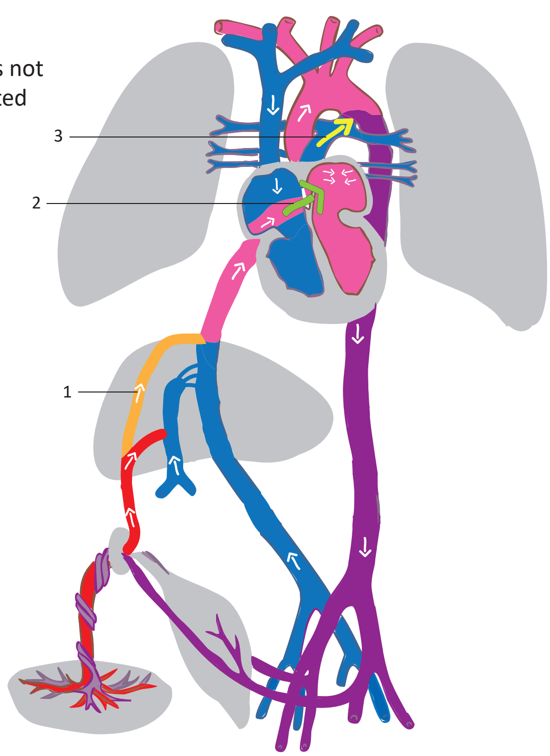
The **foetal circulation** is derived from the maternal circulation, hence the term **"foetoplacental" circulation**. In the placenta, maternal blood does not directly mix with foetal blood; it is separated from it by the **placental barrier**. Oxygen and nutrients are exchanged across this barrier. **Oxygenated blood from the placenta flows through the single umbilical vein to the foetus**. Less oxygenated blood from the aorta flows through the common iliac arteries, internal iliac arteries and the **two umbilical arteries** to the placenta for oxygenation. **Foetal shunts** maintain the highest possible oxygen concentration for the brain and other organs.

Basic pathway of the foetal circulation

- **Deoxygented blood**
- **Oxygenated blood:** placenta ➞ umbilical vein ➞ ductus venosus ➞ inferior vena cava ➞
- **Mixed blood (high oxygen concentration):** inferior vena cava ➞ right atrium ➞ foramen ovale ➞ left atrium ➞ left ventricle ➞
- **Mixed blood (low oxygen concentration):** descending aorta ➞ common iliac arteries ➞ internal iliac arteries ➞ umbilical arteries ➞ placenta

Foetal shunts

- 1 **Ductus venosus / Arantius' duct**
  - **a shunt between the umbilical vein and the inferior vena cava**, bypassing the liver
  - despite the fact that the liver is an important organ during development, it is necessary to keep blood oxygenated to the greatest possible extent for the brain, which is why the liver circulation is bypassed
  - postnatally, the remnant of the ductus venosus is the ligamentum venosum, located on the visceral surface of the liver in the fissure for the ligamentum venosum
- 2 **Foramen ovale**
  - **a shunt between the right and left atria**, bypassing the pulmonary circulation
  - the pulmonary circulation is minimal during prenatal development, so most of the oxygenated blood is directed to the brain
  - postnatally, it is closed and the only remnant left is the oval fossa in the right atrium and the valve of the foramen ovale in the left atrium
- 3 **Ductus arteriosus / ductus Botalli**
  - **a shunt between the pulmonary trunk and the initial segment of the descending aorta,** bypassing the pulmonary circulation and the aortic arch
  - deoxygenated blood from the head, neck and upper limbs returning via the superior vena cava to the heart does not flow to the lungs as they are not ventilated
  - instead this blood flows through the ductus arteriosus to the initial segment of the descending aorta and its branches to the placenta to be oxygenated
  - due to this shunt, blood from the right ventricle does not mix with highly oxygenated blood for the head and brain
  - the ductus ateriosus closes postnatally forming the ligamentum arteriosum



The foramen ovale is a unidirectional blood shunt covered by a valve composed of the septum primum and septum secundum. After the first few postnatal breaths, the left atrial pressure abruptly increases, which pushes the septum primum onto the secundum closing the foramen ovale. In 20–25 % of people, the closure is incomplete and blood keeps shunting through an opening in the interatrial septum (patent foramen ovale, FOA, *foramen ovale patens/pervium*). Under physiological conditions, this shunt causes no clinical symptoms unless haemodynamically significant. In such case, the clinical situation is described as a persistent foramen ovale (*foramen ovale apertum/persitents*).

Right and left coronary artery dominance depends on whether the posterior interventricular branch originates from the right, or left coronary artery, respectively. In one third of cases, both arteries equally supply the posterior interventricular branch. In 50 % of cases, the right coronary artery is dominant with a thick right posterolateral branch.

The coronary arteries, in the subepicardial layer on the heart's surface, have a tortuous course. This allows them to elongate and shorten as the heart contracts and dilates.

Lymph drainage from the heart goes from the left atrium and ventricle via the left lymphatic trunk to the inferior tracheobronchial nodes (and then to the right lymphatic duct). From the right atrium and ventricle, lymphatic drainage is directed through the right lymphatic trunk to the anterior mediastinal nodes (and then to the thoracic duct).

The oblique vein of the left atrium of Marshal (*v. obliqua atrii sinistri*) is a remnant of the left superior vena cava.

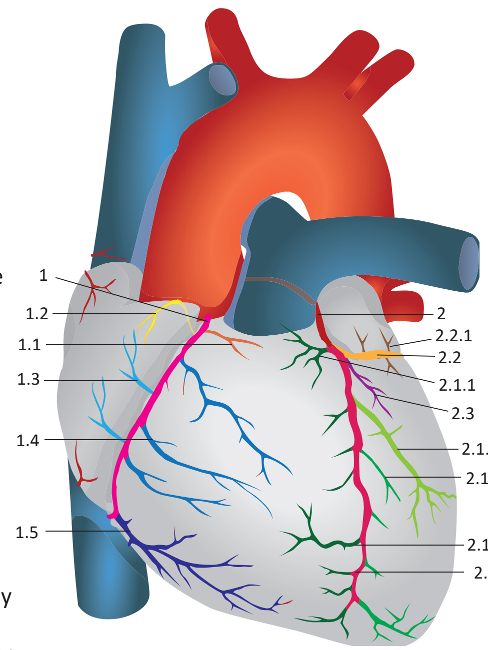
Clinical notes

Ischemic heart disease is the result of coronary artery disease on the heart. Coronary artery disease results from atherosclerosis obstructing the lumen of the coronary arteries. The extent of the obstruction leads to variable range of clinical entities: stable/unstable angina, myocardial infarction, sudden cardiac death and heart failure.

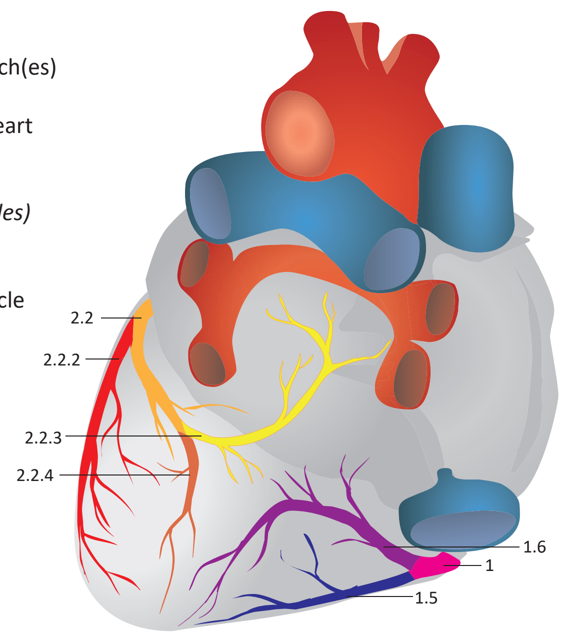
Angina pectoris is a form of ischaemic heart disease that classically presents with chest pain or discomfort.

The coronary arteries supply myocardium with oxygenated blood during diastole of the ventricles. They originate in the **right and left aortic sinuses of Valsalva** in the aortic bulb. Both coronary arteries pass in the coronary sulcus. **The main stem of the left coronary artery is short and divides early** in its course **into the anterior interventricular branch,** running on the anterior surface of the heart, **and the circumflex branch,** running posteriorly and to the left side of the heart. **The right coronary artery** passes across the right heart border and **terminates as the posterior interventricular branch.**

- 1 **Right coronary artery** (RCA) (*arteria coronaria dextra, ACD*)
  – arises from the right aortic sinus
  – runs in the coronary sulcus between the right auricle and the right ventricle
  - 1.1 **Conus branch** (*ramus coni arteriosi*) – the first branch
    – supplies the right ventricular outflow tract
  - 1.2 **Sinu-atrial nodal branch** (*ramus nodi sinuatrialis*) – supplies the SA node
  - 1.3 **Atrial branches** (*rami atriales*)
    – supply the right atrium and part of the left atrium
  - 1.4 **Ventricular branches** (*rami ventriculares*)
    – small branches supplying the anterior and posterior walls of the right ventricle
  - 1.5 **Right marginal branch** (*ramus marginalis dexter, RMD*)
    – a large branch for the right ventricle, passes on the right heart border
  - 1.6 **Posterior interventricular branch**
    (*ramus interventricularis posterior, RIP, RIVP*)
    – the first terminal branch, runs in the posterior interventricular sulcus
  - 1.7 **Right posterolateral branch** (*ramus posterolateralis dexter, RPLD*)
    – the second terminal branch, supplies the right part of the posterior wall of the left ventricle
- 2 **Left coronary artery** (LCA) (*arteria coronaria sinistra, ACS*) – a 1–2cm long stem that arises from the left aortic sinus and divides into two main branches: circumflex branch and anterior interventricular branch, intermediate branch additively
  - 2.1 **Anterior interventricular branch / left anterior descending artery** (LAD) (*ramus interventricularis anterior, RIA*) – descends in the anterior interventricular sulcus to the apex of the heart and supplies the left ventricle
    - 2.1.1 **Conus branch** (*ramus coni arteriosi*) – the first branch
      – supplies the left ventricular outflow tract
    - 2.1.2 **Diagonal branch(es)** (*ramus diagonalis, RD*) – the thickest branch(es)
      – supply the anterior wall of the left ventricle
      – run diagonally across the ventricle towards the apex of the heart
    - 2.1.3 **Anterior ventricular branches** (*rami ventriculares anteriores*)
      – supply the left ventricle
    - 2.1.4 **Interventricular septal branches** (*rami interventriculares septales)*
  - 2.2 **Circumflex branch** (*ramus circumflexus*, *RCX*)
    – the second branch of the left coronary artery
    – runs in the coronary sulcus between the left auricle and the left ventricle
    – supplies the left atrium and ventricle
    - 2.2.1 **Atrioventricular branches** (*rami atrioventriculares*)
      – supplies the left atrium and ventricle
    - 2.2.2 **Left marginal branch** (*ramus marginalis sinister*, *RMS*)
      – supplies the lateral wall of the left ventricle
    - 2.2.3 **Atrial branches** (*rami atriales*) – supplies the left atrium
    - 2.3.4 **Left posterolateral branch**
      (*ramus posterolateralis sinister*, *RPLS*)
      – the terminal branch on the left part of the posterior wall of the left ventricle
  - 2.3 **Intermediate branch** (*ramus intermedius*, *RIM*)
    – a variable branch (25 %) originating from the LCA



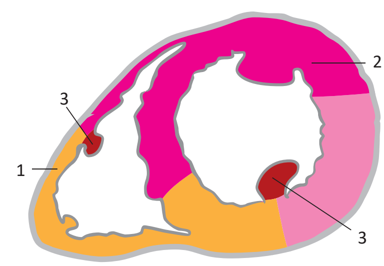
Anterior view of the heart



Posterior view of the heart

Distribution of the blood supply to the heart

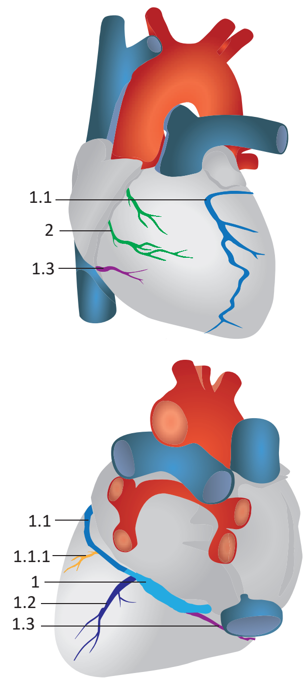
- 1 **Right coronary artery** – supplies the right atrium and ventricle, posterior 1/3 of the interventricular septum, and the conducting system of the heart to the level of the proximal part of the bundle branches of Tawara
- 2 **Left coronary artery** – supplies the left atrium and ventricle and the anterior 2/3 of the interventricular septum
- 3 **Common supply from both arteries** – right anterior papillary muscle and left posterior papillary muscle



Transverse section of the heart at the level of the ventricles; inferior view

Venous blood is drained from the heart wall in three paths. The main venous drainage is via the coronary sinus, which opens into the right atrium. The anterior veins of the right ventricle drain blood from the right ventricular wall directly into the right atrium. **The Thebesian veins** direct blood individually into all four chambers of the heart.

- 1 **Coronary sinus** (*sinus coronarius*)
  – is located in the coronary sulcus on the inferior wall of the heart
  – collects blood from its tributaries and directs it to the right atrium
  - 1.1 **Great cardiac vein** (*vena cardiaca/cordis magna*)
    – begins in the anterior interventricular sulcus and collects blood from the area supplied by the left coronary artery
    - 1.1.1 **Posterior vein of left ventricle** (*vena posterior ventriculi sinistri*)
  - 1.2 **Middle cardiac vein** (*vena cardiaca/cordis media*)
    – courses through the posterior interventricular sulcus and drains blood from the area supplied by the posterior interventricular branch
  - 1.3 **Small cardiac vein** (*vena cardiaca/cordis parva*)
    – usually a tiny vein draining the area of the right heart border
- 2 **Anterior veins of right ventricle** (*venae ventriculi dextri anteriores*)
- 3 **Small cardiac veins / Thebesian veins** (*venae cardiacae/cordis minimae*)



Anterior and posterior view of the heart

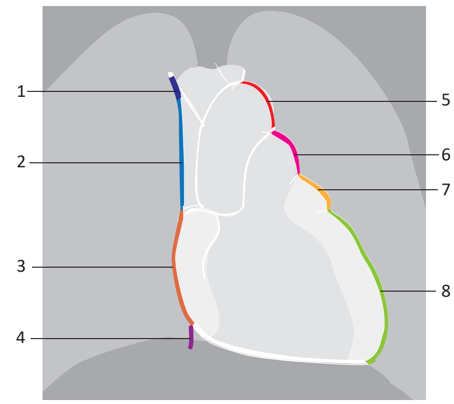
3.11 Clinical anatomy of the heart

X-Ray view

Anteroposterior (AP) projection

Right margin
- 1 Right brachiocephalic vein
- 2 Superior vena cava
- 3 Right atrial margin
- 4 Inferior vena cava (in inspiration)

Left margin
- 5 Ascending aorta (posterior wall)
- 6 Left pulmonary artery
- 7 Left auricle
- 8 Left border of left ventricle

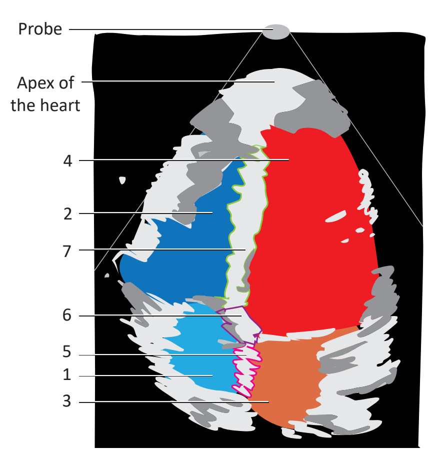


Anteroposterior projection, X-Ray

Echocardiography (ECHO)

– a non-invasive investigation of the heart using ultrasound
– the transthoracic projection through the anterior thoracic wall is the most common projection

Four-chamber projection
– a probe facing the apex of the heart can view the following structures simultaneously:
- 1 **Right atrium (with tricuspid valve)**
- 2 **Right ventricle**
- 3 **Left atrium (with mitral valve)**
- 4 **Left ventricle**
- 5 **Interatrial septum**
- 6 **Atrioventricular septum**
- 7 **Interventricular septum**



Apical four-chamber projection, ECHO

It is estimated that 40 % of venous blood from the heart is drained via the small cardiac veins and the anterior veins of the right ventricle (*vv. ventriculi dextri anteriores*).

The right and left oblique projections can be used to compliment the usual AP projection as part of the X-ray study of the heart.

Aortic arch variations:
– vertebral artery (*a. vertebralis*) (3 %)
– thyroid ima artery (*a. thyroidea ima*) (2 %)
– arteria lusoria (*a. lusoria*) (1.5 %)
– right aortic arch (*arcus aortae dexter*) (0.1 %)

The arteria lusoria is a variation of the right subclavian artery originating as the last branch of the aortic arch (after the origin of the left subclavian artery). Due to its course (posterior to the oesophagus), it may compress the oesophagus, which leads to problems with swallowing (dysphagia lusoria).

Clinical notes

Myocardial infarction is caused by an obstruction of blood flow in the coronary artery (usually by a thrombus) leading to ischemia and necrosis of the given segment of myocardium. Acute chest pain is the typical presenting symptom. However, it may be absent in some cases.

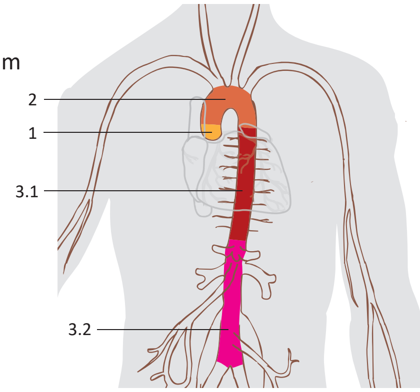
In myocardial infarction, pain radiates to the Head's zones of the T1–T4 spinal cord segments (the thoracic wall and the left side of the upper limb, including the little finger).

Acute treatment of myocardial infarction comprises:
PCI (percutaneous coronary intervention) – a dilation of the narrowed arterial segment by a catheter with an inflatable balloon at its tip, inserted via the femoral or radial artery, followed by placement of a bare metal (or drug-eluting) stent (a circular, net-like prosthesis used to mechanically dilate the artery).
CABG (coronary artery bypass grafting) – vascular bypass of a narrowed arterial segment using an autologous venous (great saphenous vein) or arterial (internal thoracic or radial artery) graft.
Thrombolysis – is indicated when PCI is not available.

Transoesophageal echocardiography: as the left atrium lies right in front of the oesophagus, an ECHO probe can be inserted into the oesophagus to give a better view of the left atrium than a transthoracic probe.

The aorta is the **longest** and **largest artery** of the human body. **The is an elastic artery**, which helps it maintain a **continuous flow of blood**. It originates in the left ventricle. Near its origin**, it gives off the coronary arteries** that supply the heart. The aorta then **continues into the cranially convex aortic arch**, which gives off branches supplying the head, neck, and upper extremities. The aorta then **descends in the thoracic cavity** and gives off branches for the thoracic organs, thoracic wall and back. It then passes through the diaphragm in front of the vertebral column. **The last part of the aorta** runs in the abdominal cavity; its branches supply the abdominal organs, abdominal wall, and back. Finally, **the aorta bifurcates into the right and left common iliac arteries,** which supply the pelvis and lower extremities.

- 1 **Ascending aorta** (*aorta ascendens*) – T4–T3
  – arises from the left ventricle and continues as the aortic arch in the superior mediastinum
- 2 **Aortic arch** (*arcus aortae*) – to the top of T2
  – passes dorsally and to the left in the superior mediastinum
- 3 **Descending aorta** (*aorta descendens*) – T3–L4
  - 3.1 **Thoracic aorta** (*aorta thoracica*) – T3–T12
    – passes on the left side of the vertebral column and then in front of vertebrae in the inferior mediastinum
    – passes through the aortic hiatus of the diaphragm at the level of T12 and changes here into the abdominal aorta
  - 3.2 **Abdominal aorta** (*aorta abdominalis*) – T12–L4
    – passes in front of the vertebral column inside the retroperitoneal space
    – bifurcates into the right and left common iliac arteries at the level of L4



4.2 **Ascending aorta and aortic arch** – Aorta ascendens et arcus aortae

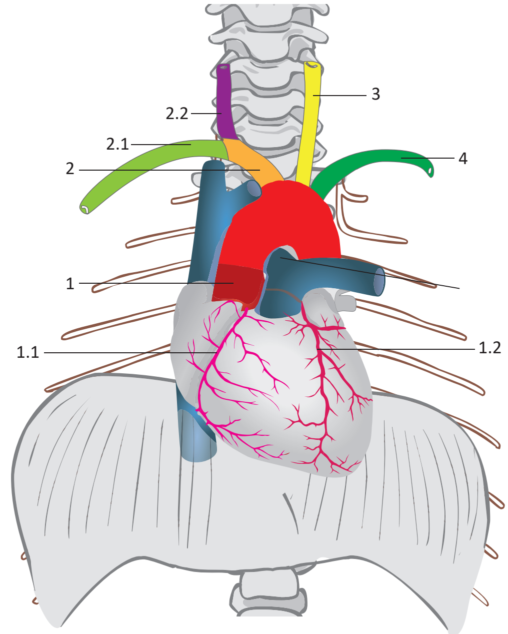
Blood flows **from the left ventricle** through the aortic valve **into the ascending aorta**. **The ascending aorta gives off the right and left coronary arteries in the aortic bulb**. It then continues as **the aortic arch,** which gives off branches supplying the head, neck, and upper extremities. The aortic arch turns dorsally and to the left, approaches the vertebral column and continues as **the descending aorta**.

Ascending aorta (*aorta ascendens*)

- 1 **Aortic bulb** (*bulbus aortae*) – the enlarged initial segment of the aorta
  – contains the aortic sinuses of Valsalva (*sinus aortae*) from which the coronary arteries arise
  - 1.1 **Right coronary artery** (*arteria coronaria dextra*) – arises from the right aortic sinus and supplies the right atrium and ventricle, posterior third of the interventricular septum, anterior papillary muscle of the right ventricle and posterior papillary muscle of the left ventricle
  - 1.2 **Left coronary artery** (*arteria coronaria sinistra*) – arises from the left aortic sinus and supplies the left atrium and ventricle, anterior two thirds of the interventricular septum, anterior papillary muscle of the right ventricle and posterior papillary muscle of the left ventricle

Aortic arch (*arcus aortae*)

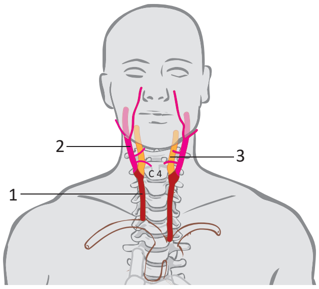
- 2 **Brachiocephalic trunk** (*truncus brachiocephalicus*) – the first branch
  – an unpaired short artery on the right
  – originates behind the manubrium of the sternum and passes behind the right sternoclavicular joint where it divides into the right common carotid artery and the right subclavian artery
  - 2.1 **Right subclavian artery** (*arteria subclavia dextra*)
    – arises near the 1st rib on the right and passes into the scalene fissure
    – its branches supply the right upper extremity, part of the neck, head, brain, and the superior part of the thorax with part of the back
  - 2.2 **Right common carotid artery** (*arteria carotis communis dextra*)
    – arises near the head of the 1st rib on the right and passes through the visceral space of the neck
    – its branches supply part of the neck, head, and brain
- 3 **Left common carotid artery** (*arteria carotis communis sinistra*)
  – originates near the head of the 1st rib and passes through the visceral space of the neck
  – its branches supply part of the neck, head, and brain
- 4 **Left subclavian artery** (*arteria subclavia sinistra)*
  – arises near the 1st rib on the left and passes into the scalene fissure
  – its branches supply the left upper extremity, head, brain, part of the neck and the superior part of the thorax and back



4.3 **Common carotid artery** – Arteria carotis communis

The right common carotid artery arises with the right subclavian artery from the brachiocephalic trunk. **The left common carotid artery arises directly from the aortic arch**. Each common carotid artery passes cranially along the trachea and bifurcates into the internal and external carotid arteries **within the carotid triangle** at the level of the superior margin of the thyroid cartilage (at the level of C4). **The common carotid artery** gives off no other branches as it **travels with the internal jugular vein, the vagus nerve** and the superior branch of the deep cervical ansa in the fibrous carotid sheath (*vagina carotica*).

- 1 **Common carotid artery** (*arteria carotis communis*)
  – ascends along the trachea
  – bifurcates within the carotid triangle
- 2 **External carotid artery** (*arteria carotis externa*)
  – passes ventromedially
  – has eight main branches within the neck and head
- 3 **Internal carotid artery** (*arteria carotis interna*)
  – passes dorsolaterally giving off no branches within the neck
  – enters the cranium to give off branches supplying the brain and eyeball



4.4 **External carotid artery** – Arteria carotis externa

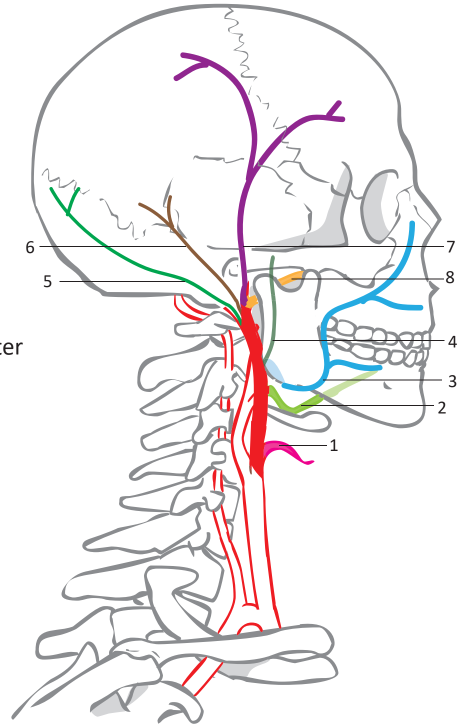
The external carotid artery supplies the head, the majority of the organs and muscles of the **anterior part of the neck** and some of the **nuchal muscles**. It ascends through the carotid triangle, courses behind the **stylohyoid** and **posterior belly of the digastric** and pierces the **styloid septum** to enter the **prestyloid space**. It then enters the **retromandibular fossa** close to the angle of the mandible. It bifurcates into its terminal branches close to the neck of the mandible.

Ventral branches
- 1 **Superior thyroid artery** (*arteria thyroidea superior*)
  – the first branch of the external carotid artery
  – descends to the thyroid gland
- 2 **Lingual artery** (*arteria lingualis*)
  – the second branch
  – ascends to the tongue
- 3 **Facial artery** (*arteria facialis*) – the third branch
  – ascends across the face towards the orbit

Medial branches
- 4 **Ascending pharyngeal artery** (*arteria pharyngea ascendens*)
  – the only medial branch,
  – supplies the pharynx, middle ear, and dura mater

Dorsal branches
- 5 **Occipital artery** (*arteria occipitalis*)
  – passes to and supplies the occiput
- 6 **Posterior auricular artery** (*arteria auricularis posterior*)
  – passes right behind the auricle supplies the mastoid air cells, parotid gland and middle ear,

Terminal branches
- 7 **Superficial temporal artery** (*arteria temporalis superficialis*)
  – passes to and supplies the temple, vertex, and forehead
- 8 **Maxillary artery** (*arteria maxillaris*)
  – passes through the infratemporal fossa and the pterygopalatine fossa
  – supplies the masticatory muscles, dura mater, mandible and lower teeth, maxilla and upper teeth, middle ear, nasopharynx and palate and deep facial layers



The carotid sinus (*sinus caroticus*) is an enlargement located at either the bifurcation of the common carotid artery or at the beginning of the internal carotid artery. Free nerve endings are located in the wall of the carotid sinus, which monitor blood pressure. Compression of the wall can cause a reflex decrease in blood pressure.

The carotid body (*glomus caroticum*) is located in the carotid bifurcation. It is a chemoreceptor, which detects changes in the internal environment (mainly partial pressure of oxygen, concentration of carbon dioxide and changes in pH).

The facial and lingual artery can arise from a common origin called the linguofacial trunk (*truncus linguofacialis*). This is important to known when performing catheterisation.

The sternocleidomastoid artery can arise directly from the external carotid artery as its lateral branch in about 25 % of cases.

Internal carotid artery page 472.

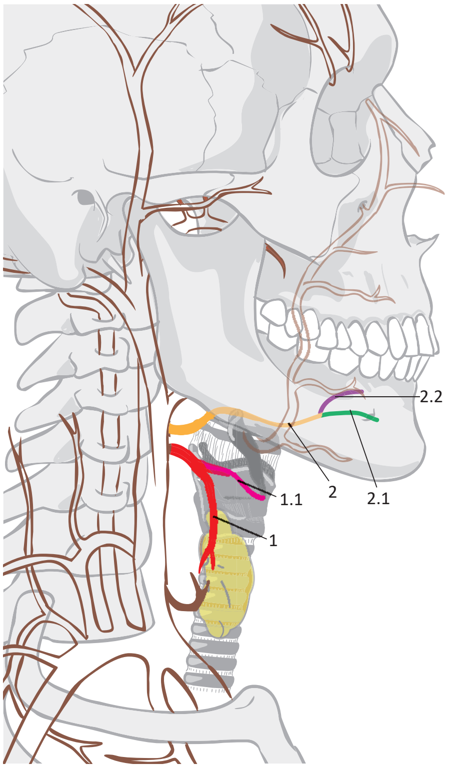
Clinical notes

The common carotid artery can be examined with a stethoscope. Such an examination is called auscultation and allows the detection of an abnormal sound called a bruit, which indicates the presence of stenosis. The bruit manifests as a regular rustling sound during each heart contraction and can indicate stenosis of the artery even when the pulse is normally palpable.

Carotid ultrasonography is an ultrasonographic procedure that shows the quality of blood flow within the carotid arteries. It measures the luminal diameter (6–7 mm), blood flow and the intima-media thickness (IMT), which is a marker of atherosclerosis.

Carotid stenosis is, in the majority of cases, caused by an atherosclerotic plaque. A piece of the plaque can break off and travel through the circulation to the blood vessels in the brain. The pieces can lodge in the vessel wall and restrict blood flow, causing a thrombembolic stroke.

The carotid tubercle / tubercle of Chassaignac is a synonym for the anterior tubercle of the transverse process of the sixth cervical vertebra. The common carotid artery can be compressed against it.



Superior thyroid artery (*arteria thyroidea superior*)

Course

– the first branch of the external carotid artery
– passes ventrally then descends to the anterior surface of the thyroid gland

- 1 **Superior thyroid artery** (*arteria thyroidea superior*)
  - 1.1 **Superior laryngeal artery** (*arteria laryngea superior*)
    – supplies the larynx and accompanies the superior laryngeal nerve as it passes through the thyrohyoid membrane
  - 1.2 **Anterior, posterior, and lateral branches** (*ramus anterior, posterior et lateralis*) – supply the thyroid gland
  - 1.3 **Infrahyoid, sternocleidomastoid, and cricothyroid branches** (*ramus infrahyoideus, sternocleidomastoideus et cricothyroideus*)
    – supply the sternocleidomastoid and infrahyoid muscles

Lingual artery (*arteria lingualis*)

Course

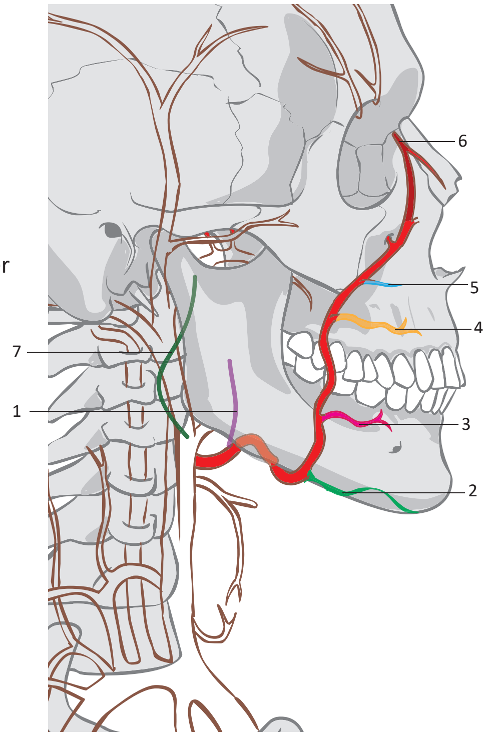
– arises from external carotid artery at the level of the lesser horn of the hyoid bone
– passes through the triangle of Pirogov and the angle of Béclard
– then it runs inferiorly to the hyoglossus passing through the paralingual canal to the tongue

- 2 **Lingual artery**
  - 2.1 **Sublingual artery** (*arteria sublingualis*)
    – supplies the sublingual gland and the floor of the oral cavity
    – arises from the lingual artery before entering the tongue
    – passes along the floor of the oral cavity below the sublingual gland
  - 2.2 **Deep lingual artery** (*arteria profunda linguae*) – passes inside the tongue between the genioglossus and the inferior longitudinal muscle to the apex of the tongue
  - 2.3 **Suprahyoid branch and dorsal lingual branches** (*ramus suprahyoideus et rami dorsales linguae*) – supply the oral part of the pharynx

Facial artery (*arteria facialis*)

Course

– the third branch from the external carotid artery, passes through the submandibular triangle to the ramus of the mandible
– then it passes behind (medially to) the stylohyoid and posterior belly of the digastric and either through or under the submandibular gland (the facial vein passes superficially to both the gland and muscles)
– reaches the face along the anterior margin of the masseter and has a tortuous course along the oral and nasal angles
– its terminal branch, the angular artery, ascends to the medial margin of the orbit where it anastomoses with the branches of the ophthalmic and infraorbital arteries



Branches and areas supplied

- 1 **Ascending palatine artery** (*arteria palatina ascendens*)
  – supplies the isthmus of the fauces, pharynx, and palate
  – passes cranially along the pharyngeal wall within the prestyloid space
- 2 **Submental artery** (*arteria submentalis*) – supplies the suprahyoid muscles
  – passes along the inferior surface of the mylohyoid
- 3 **Inferior labial branch** (*arteria labialis inferior*) – supplies the lower lip, passes under the orbicularis oris and anastomoses with the contralateral inferior labial branch
- 4 **Superior labial branch** (*arteria labialis superior*) – supplies the upper lip, passes under the orbicularis oris and anastomoses with the contralateral superior labial branch
- 5 **Lateral nasal branch** (*ramus lateralis nasi*) – supplies the nasal wing
- 6 **Angular artery** (*arteria angularis*) – supplies the medial angle of the eye
  – passes along the nasal wing to the medial angle of the eye

Ascending pharyngeal artery (*arteria pharyngea ascendens*)

Course

– passes medially, between the external and internal carotid arteries
– its branches feed the pharyngeal wall, cranial base, tympanic cavity, and dura mater

- 7 **Ascending pharyngeal artery**
  - 7.1 **Pharyngeal branches** (*rami pharyngeales*) – supply the pharyngeal wall
  - 7.2 **Inferior tympanic artery** (*arteria tympanica inferior*)
    – passes through the tympanic canaliculus to supply the tympanic cavity
  - 7.3 **Posterior meningeal artery** (*arteria meningea posterior*)
    – passes through the jugular foramen into the posterior cranial fossa

The occipital artery is the first dorsal branch of the external carotid artery. It supplies the external soft tissue of the occipital region and the nuchal muscles. **The posterior auricular artery is the second dorsal branch**. It supplies the auricle, mastoid cells, and part of the parotid gland.

Occipital artery (*arteria occipitalis*)

Course

– passes along the posterior margin of the posterior belly of the digastric to the base of the skull behind the mastoid process where it runs in the shallow occipital groove and enters the external soft tissue of the skull

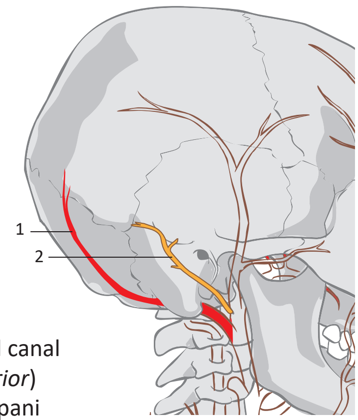
- 1 **Occipital artery**
  - 1.1 **Sternocleidomastoid branches** (*rami stermocleidomastoidei*) – supply the sternocleidomastoid
  - 1.2 **Glandular branches** (*rami glandulares*) – supply the parotid gland
  - 1.3 **Auricular branch** (*ramus auricularis*) – supplies the posterior and medial surface of the auricle

Posterior auricular artery (*arteria auricularis posterior*)

Course

– passes along the posterior belly of the digastric behind the auricle and in front of the mastoid process

- 2 **Posterior auricular artery**
  - 2.1 **Parotid branch** (*ramus parotideus*) – supplies the parotid gland
  - 2.2 **Auricular branch** (*ramus auricularis* – supplies the auricle
  - 2.3 **Stylomastoid artery** (*arteria stylomastoidea*) – passes through the stylomastoid foramen into the facial canal
    - 2.3.1 **Posterior tympanic artery** (*arteria tympanica posterior*) – passes through the canaliculus for the chorda tympani and supplies the tympanic cavity



The **stapedial branch** of the stylomastoid artery is a small artery, which is an embryological remnant of the main artery of the second pharyngeal arch.

Other branches of the maxillary artery:
1. **Pterygomeningeal artery** (*a. pterygomeningea*) is an accessory artery supplying the pterygoids and the middle cranial fossa, branching from the pterygoid part of the maxillary artery.
2. **Middle superior alveolar artery** (*a. alveolaris superior media*), vein and nerve are variable structures which leave the infra-orbital groove and descend in the lateral wall of the maxillary sinus.
3. **Pharyngeal branch** (*r. pharyngeus*) is a smaller branch from either the pterygoid part of the maxillary artery or the artery of the pterygoid canal, which runs dorsally through the palatovaginal canal to the nasopharynx.

The right and left superior labial branches of the facial artery anastomose and form an arterial circle within the area of the upper lip. Similarly, the inferior labial branches form an arterial circle within the area of the lower lip.

4.4.3 **Superficial temporal artery** – Arteria temporalis superficialis

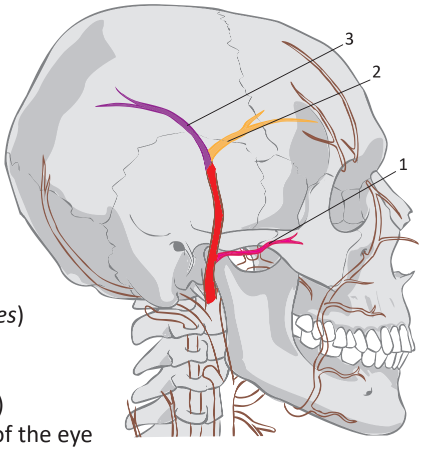
The superficial temporal artery is one of the two terminal branches of the external carotid artery. It supplies **supplies the parotid gland, auricle, temporal muscle and external soft tissue of the skull within the temporal and zygomatico-orbital regions**.

Course

– ascends behind the temporomandibular joint through the parotid gland, then it crosses the zygomatic arch, and in front of the auricle it gives off branches that travel into the frontal, temporal, and parietal regions

Branches and areas supplied

- 1 **Transverse facial artery** (*arteria transversa faciei*) – supplies the face, running along the parotid duct
- 2 **Frontal branch** (*ramus frontalis*) – supplies the external soft tissue of the skull in the frontal region
- 3 **Parietal branch** (*ramus parietalis*) – supplies the external soft tissue of the skull in the parietal region
- 4 **Parotid branches** (*rami parotidei*) – supply the parotid gland
- 5 **Anterior auricular branches** (*rami auriculares anteriores*) – supply the temporomandibular joint and the lateral surface of the external acoustic meatus
- 6 **Zygomatico-orbital artery** (*arteria zygomaticoorbitalis*) – passes along the zygomatic arch to the lateral angle of the eye
- 7 **Middle temporal artery** (*arteria temporalis media*) – perforates the temporal fascia, supplies the temporalis and ascends along the squamous part of the temporal bone in a shallow groove for the middle temporal artery



Clinical notes

The facial artery is palpable where it crosses the margin of the mandible in front of the insertion of the masseter. This is a pressure point used to stop arterial bleeding.

The superficial temporal artery is palpable in front of the auricle and above the temporomandibular joint. The most common disease affecting this artery is a type of arteritis called Horton's disease or temporal arteritis. This disease manifests as a red painful line over the course of the superficial temporal artery.

The sphenopalatine artery is often responsible for nasal bleeding (epistaxis). It can be occluded with electrocautery or embolisation.

Electrocautery refers to a process that uses electric current to destroy vessels and stop bleeding. **Embolisation** is a procedure in which a chemical substance injected by an intra-arterial catheter causes selective occlusion of a blood vessel.

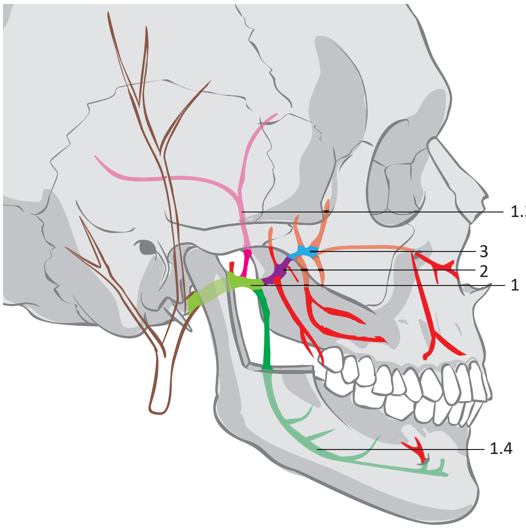
The maxillary artery is one of the two terminal branches of the external carotid artery. It is **divided into three parts within its course. It passes through the infratemporal and pterygopalatine fossae** and **supplies part of the acoustic meatus, tympanic cavity, maxilla, mandible, teeth, masticatory muscles, buccinator, part of the nasopharynx and auditory tube and the hard and soft palates.**

Course

- 1 **Mandibular part** (*pars mandibularis*)
  – runs between the neck of the mandible and the sphenomandibular ligament and ascends to the infratemporal fossa
- 2 **Pterygoid part** (*pars pterygoidea*) – usually lies between the lateral and medial pterygoid muscles
- 3 **Pterygopalatine part** (*pars pterygopalatina*) – passes through the pterygomaxillary fissure to enter the pterygopalatine fossa

Branches of the mandibular part

1.1 **Deep auricular artery** (*arteria auricularis profunda*) – supplies the wall of the external acoustic meatus
1.2 **Anterior tympanic artery** (*arteria tympanica anterior*) – supplies the tympanic cavity
– passes through the petrotympanic fissure into the tympanic cavity
1.3 **Middle meningeal artery** (*arteria meningea media*)
– passes through the foramen spinosum to enter the middle cerebral fossa
1.3.1 **Frontal, parietal, and orbital branches**
(*ramus frontalis, parietalis et orbitalis*)
– supply the dura mater and part of the bony cranial vault
1.3.2 **Petrosal branch** (*ramus petrosus*) – passes into the facial canal
1.3.3 **Superior tympanic artery** (*arteria tympanica superior*)
– supplies the tympanic cavity
– passes through the canal for the lesser petrosal nerve
1.4 **Inferior alveolar artery** (*arteria alveolaris inferior*)
– enters the mandibular canal through the mandibular foramen
1.4.1 **Mental branch** (*ramus mentalis*) – the terminal branch
– passes through the mental foramen to reach the mental region
1.4.2 **Mylohyoid branch, dental, and gingival branches**
(*ramus mylohyoideus, rami dentales et gingivales*)
– supply the mylohyoid, lower teeth and gingiva

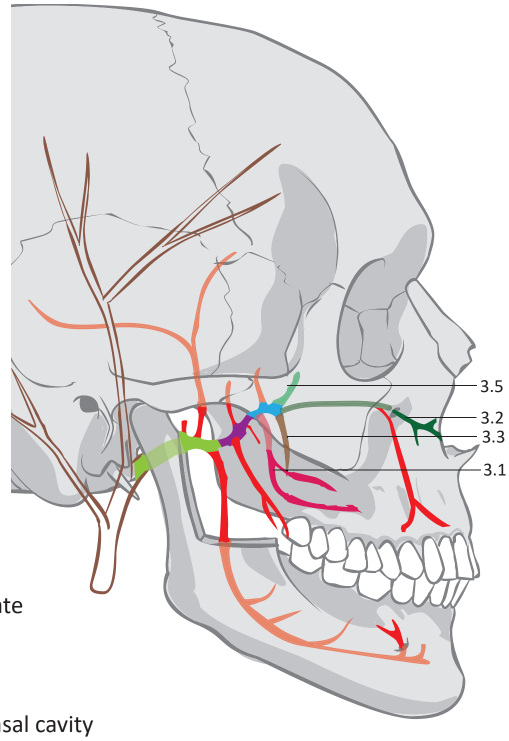


Branches of the pterygoid part

2.1 **Masseteric artery** (*arteria masseterica*) – passes through the mandibular notch and enters the medial surface of the masseter
2.2 **Deep temporal arteries** (*arteriae temporalis profundae*) – anterior and posterior
– two arteries running cranially and entering the temporalis from the medial side
2.3 **Pterygoid branches** (*aami pterygoidei*) – supply the pterygoids
2.4 **Buccal artery** (*arteria buccalis*) – passes on the lateral surface of the buccinator

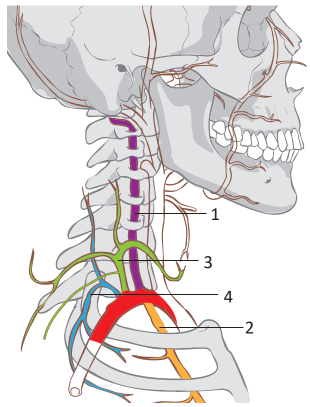
Branches of the pterygopalatine part

3.1 **Posterior superior alveolar artery** (*arteria alveolaris posterior superior*)
– passes over the maxillary tuberosity and travels through the alveolar canals to reach the tooth sockets (*alveoli*) of the upper premolars and molars
3.2 **Infraorbital artery** (*arteria infraorbitalis*) – supplies the facial area under the orbit and the anterior upper teeth
– passes through the inferior orbital fissure and along the infraorbital groove
– enters the infraorbital canal and leaves it through the infraorbital foramen
3.2.1 **Anterior superior alveolar arteries**
(*arteriae alveolares superiors anteriores*)
– branch from the infraorbital artery in the infraorbital canal and descend in osseous canals to the tooth sockets of the upper incisors and canines
3.3 **Descending palatine artery** (*arteria palatina descendens*)
– descends in the greater palatine canal to the palate
3.3.1 **Greater palatine artery** (*arteria palatina major*) – passes through the greater palatine foramen to reach the mucosa of the hard palate
3.3.2 **Lesser palatine arteries** (*arteriae palatinae minores*) – pass through the lesser palatine canals and foramina to reach the mucosa of the soft palate
3.4 **Artery of pterygoid canal** (*arteria canalis pterygoidei*)
– passes through the pterygoid canal to reach the nasopharynx
3.5 **Sphenopalatine artery** (*arteria sphenopalatina*)
– passes through the sphenopalatine foramen to reach the posterior part of the nasal cavity
3.5.1 **Posterior lateral nasal arteries and posterior septal nasal branches**
(*arteriae nasales posteriores laterales et rami septales posteriores*)
– pass to the mucosa of the lateral wall of the nasal cavity and to the nasal septum



The subclavian artery arises from the **brachiocephalic trunk** on the **right side** and directly from the **aortic arch** on the **left side**. It supplies the **inferior part of the neck**, **posterior part of the brain**, **anterior and lateral thoracic walls**, **anterior abdominal wall**, and **upper extremity**. Its course is divided into three parts.

Course
1 **Intrascalenic/thoracic part** (*pars intrascalenica*)
– arches over the plural cupula in the groove for the subclavian artery
2 **Interscalenic/scalene part** (*pars interscalenica*) – enters the scalene fissure and lies in the groove for the subclavian artery of the first rib
3 **Extrascalenic/clavicular part** (*pars extrascalenica*)
– heads under the clavicle to the axillary fossa where it continues as the axillary artery

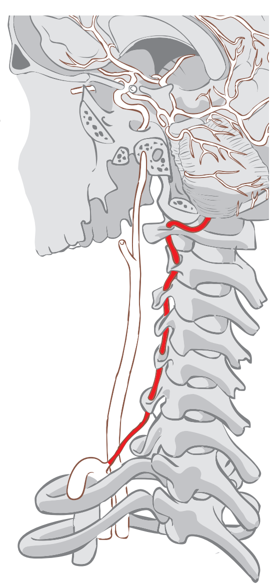


Branches and areas supplied
- 1 **Vertebral artery** (*arteria vertebralis*) – supplies part of the brain
- 2 **Internal thoracic artery** (*arteria thoracica interna*)
  – supplies the thorax and diaphragm
- 3 **Thyrocervical trunk** (*truncus thyrocervicalis*)
  – supplies the thyroid gland, larynx, lateral cervical region and its muscles and also scapular muscles
- 4 **Costocervical trunk** (*truncus costocervicalis*) – supplies the first two posterior intercostal muscles and the nuchal muscles

4.5.1 **Vertebral artery** – Arteria vertebralis

The vertebral artery is the first branch of the subclavian artery. It passes through the **transverse foramina of the cervical vertebrae** as it ascends in the neck. It supplies blood to **the circle of Willis**. The vertebral artery supplies the **posterior part of the brain, the cranial part of the spinal cord, cranial and spinal meninges and the deep cervical muscles**. Its course can be divided into four parts.

Course
1 **Prevertebral part** / V1 segment *(pars prevertebralis)*
– runs cranially between the longus coli and the scalenus anterior
2 **Transverse part** / V2 segment *(pars transversaria)* – enters the transverse foramen of the 6th cervical vertebra and ascends through the transverse foramina of the other cerivical vertebrae to the atlas
3 **Atlantic part** / V3 segment *(pars atlantica)* – after leaving the transverse foramen of the atlas, lies in the groove for the vertebral artery on the posterior arch of the atlas
– then perforates the posterior atlanto-occipital membrane and spinal dura mater and passes through the foramen magnum into the cranial cavity
4 **Intracranial part** / V4 segment *(pars intracranialis)* – is located in the posterior cerebral fossa on the clivus of the occipital bone, the two vertebral arteries fuse near the inferior margin of the pons to form the unpaired basilar artery



Extracranial branches
1 **Spinal branches** (*rami spinales*) – supply the spinal cord and its meninges
2 **Muscular branches** (*rami musculares*) – supply the deep cervical muscles

Intracranial branches
3 **Meningeal branch** (*ramus meningeus*) – runs through the foramen magnum to the cranial dura mater in the posterior cerebral fossa
4 **Posterior spinal artery** (*arteria spinalis posterior*)
– passes to the posterior surface of the medulla oblongata and descends on the posterior surface of the spinal cord usually as a paired artery
– supplies the posterior third of the spinal cord
5 **Anterior spinal artery** (*arteria spinalis anterior*) – an unpaired artery arising from two short branches from each vertebral artery
– enters the vertebral canal and descends in the anterior median fissure to supply the anterior two thirds of the spinal cord
6 **Posterior inferior cerebellar artery** „PICA" (*a. cerebelli posterior inferior*)
– a paired branch supplying the inferior surface of the cerebellum and a part of the medulla oblongata

Internal mammary artery is an obsolete term for the internal thoracic artery.

The internal thoracic artery is the only vessel in the human body with a diameter of 1,8–2,6 mm, which has an elastic wall. Smaller arteries usually have muscular walls. The internal thoracic artery is an ideal graft for cardiac bypass because it is only rarely affected by atherosclerosis.

The branching pattern of the transverse cervical artery is highly variable. A deep branch may be present as an individual branch of the extrascalenic part of the subclavian artery in 33 % of cases.

Clinical notes

The internal thoracic artery can be used as a graft for bypassing a stenotic coronary vessel in cardiosurgery.

LIMA-RIA-bypass means a bypass from the left internal thoracic artery to the anterior interventricular branch of the left coronary artery.

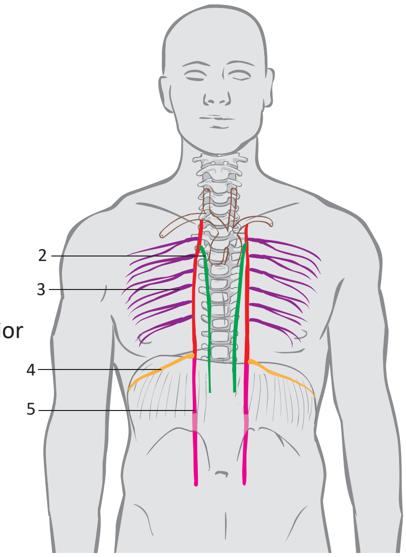
The inferior thyroid artery crosses the recurrent laryngeal nerve at a right angle. Its arterial and nervous branches are often interwoven. Increased attention should be paid to the recurrent laryngeal nerve during thyroidectomy as its ligation carries a high risk of nervous damage with subsequent loss of voice and hoarseness.

Vertebrobasilar insufficiency (beauty parlour syndrome or syndrome of viewing cathedrals) is a pathological condition caused by decreased blood flow in the posterior part of the brain. It can be caused by congenital arterial stenosis and atherosclerosis and manifests as vertigo, dizziness and even as a stroke.

Subclavian steal syndrome occurs when the subclavian artery is occluded proximally to the origin of the vertebral artery. Blood is redirected from through the common carotid artery, internal carotid artery, and the circle of Willis, where it then passes in a retrograde direction through the vertebral artery to reach the subclavian artery, distal to the occlusion. In this way, circulation to the upper extremity distal to the occlusion is restored. Blood is "stolen" from the circle of Willis in order to supply the upper extremity; this can lead to the development of neurological symptoms resulting from brain ischaemia.

The internal thoracic artery arises from the caudal aspect of the subclavian artery and descends into the thorax. It supplies the anterior parts of the intercostal spaces, the sternum, the pericardium, and the diaphragm. **The thyrocervical trunk** supplies the thyroid gland, larynx, lateral cervical region, and cervical and scapular muscles. **The costocervical trunk** supplies the posterior parts of 1st and 2nd intercostal spaces and the deep nuchal muscles.

Internal thoracic artery (*arteria thoracica interna*)



Course

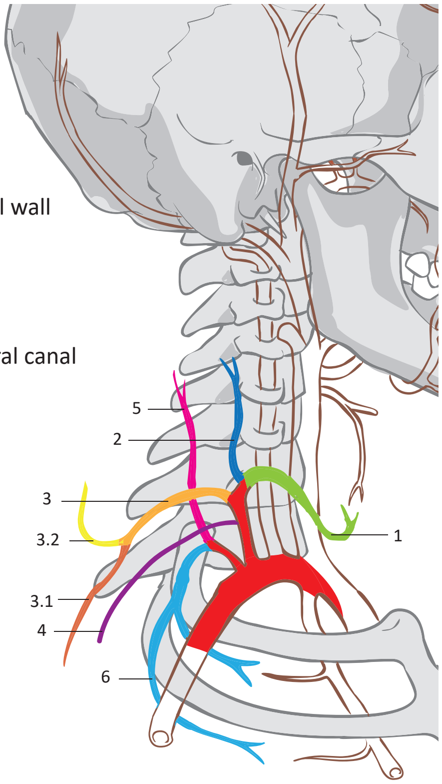
1. passes through the superior thoracic aperture to enter the thorax
2. runs approximately 1 cm laterally to the sternum through the anterior layer of the superior mediastinum and the anterior inferior mediastinum to the level of the 6th intercostal space
3. its terminal branch, the superior epigastric artery, passes through the diaphragm in front of the sternocostal triangle and terminates in the anterior abdominal wall

Branches and areas supplied

1 **Minor branches** – supply the thymus, sternum, trachea, and mediastinum

- 2 **Pericardiacophrenic artery** (*arteria pericardiacophrenica*) – travels with the phrenic nerve
  – supplies the pericardium and diaphragm
- 3 **Intercostal braches** (*rami intercostales*) – supply the intercostal muscles, pass within the superior part of the first six intercostal spaces and anastomose with the posterior intercostal arteries
  - 3.1 **Perforating branches** (*rami perforantes*)
    – perforate the intercostal spaces and supply the anterior thoracic wall
    - 3.1.1 **Medial mammary branches** (*rami mammarii mediales*) – supply the mammary gland
- 4 **Musculophrenic artery** (*arteria musculophrenica*) – the first terminal branch
  – passes from the 7th rib laterocaudally along the insertion of the diaphragm
  - 4.1 **Anterior intercostal branches** (*rami intercostales anteriores*)
    – supply the anterior part of the 7th to 9th intercostal spaces
- 5 **Superior epigastric artery** (*arteria epigastrica superior*) – the second terminal branch
  – passes through the diaphragm in front of the sternocostal triangle and descends on the posterior surface of the rectus abdominis (in the rectus sheath) where it anastomoses with the inferior epigastric artery

Thyrocervical trunk



Course – passes close to the medial margin of the anterior scalenus

Branches and areas supplied

- 1 **Inferior thyroid artery** (*arteria thyroidea inferior*) – supplies the thyroid glands
  – ascends and turns medially at the level of the 6th cervical vertebra
  – crosses the recurrent laryngeal nerve close to the thyroid gland
  - 1.1 **Inferior laryngeal artery** (*arteria laryngea inferior*) – supplies the posterior laryngeal wall
  - 1.2 **Pharyngeal and tracheal branches** (*rami pharyngeales et tracheales*)
- 2 **Ascending cervical artery** (*arteria cervicalis ascendens*)
  – a small branch supplying the deep cervical muscles and spinal cord
  – runs on the anterior surface of the anterior scalenus medially to the phrenic nerve
  - 2.1 **Spinal branches** (*rami spinales*) – pass via the intervertebral foramina to the vertebral canal
- 3 **Transverse cervical artery** (*arteria transversa colli*)
  – passes ventrally to the anterior scalenus to reach the lateral cervical triangle
  - 3.1 **Deep branch, dorsal scapular artery** (*ramus profundus, arteria dorsalis scapulae*)
    – courses together with the dorsal scapular nerve and supplies the rhomboidei and levator scapulae
  - 3.2 **Superficial branch** (*ramus superficialis*) – supplies the trapezius
- 4 **Suprascapular artery** (*arteria suprascapularis*)
  – supplies the muscles attached to the posterior surface of the scapula
  – runs ventrally to the anterior scalenus to the superior margin of the scapula
  – passes above the superior transverse scapular ligament to enter the supraspinous fossa and then travels to the infraspinous fossa through the sphenoglenoid notch

Costocervical trunk

Course – arises dorsally to the anterior scalenus

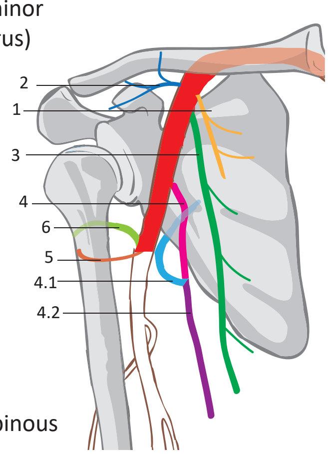
Branches and areas supplied

- 5 **Deep cervical artery** (*arteria cervicalis profunda*) – passes between the transverse processes of the 7th cervical vertebra and the 1st thoracic vertebra and between the semispinalis cervicis and capitis
- 6 **Supreme intercostal artery** (*arteria intercostalis suprema*) – passes in front of the heads of the first two ribs
  - 6.1 **Posterior intercostal arteries** (*arteriae intercostales posteriores*) – supply the first two intercostal spaces from the posterior side

The axillary artery is a direct continuation of the subclavian artery. It gives off branches in the axillary fossa and then continues as the brachial artery. It supplies the **muscles of the shoulder joint, muscles bordering the axillary fossa, deltoid, lateral thoracic wall including its muscles and the mammary gland**. The axillary artery can be divided into **three parts** according to its relationship with the pectoralis minor.

Course
1 **Suprapectoral part** (*pars suprapectoralis*)
– originates from the subclavian artery when it's crossing the lateral margin of the 1st rib
2 **Retropectoral part** (*pars retropectoralis*) – descends behind the tendon of the pectoralis minor
3 **Infrapectoral part** (*pars infrapectoralis*) – is located between the inferior margin of the pectoralis minor and the inferior part of the teres minor and latissimus dorsi (at the level of the surgical neck of the humerus)

Branches and areas supplied
- 1 **Superior thoracic artery** (*arteria thoracis superior*)
  – supplies the pectoral muscles and the mammary gland
- 2 **Thoraco-acromial artery** (*arteria thoracoacromialis*)
  – runs through the clavipectoral triangle
  - 2.1 **Acromial and deltoid branch, pectoral branches** (*ramus acromialis, deltoideus et rami pectorales*)
- 3 **Lateral thoracic artery** (*arteria thoracica lateralis*)
  – descends on the serratus anterior, which it supplies
  - 3.1 **Lateral mammary branches** (*rami mammarii laterales*)
- 4 **Subscapular artery** (*arteria subscapularis*)
  - 4.1 **Circumflex scapular artery** (*arteria circumflexa scapulae*)
    – passes through the omotricipital foramen to the infraspinous fossa where it supplies the posterior scapular muscles
  - 4.2 **Thoracodorsal artery** (*arteria thoracodorsalis*)
    – passes over the latissimus dorsi, which it supplies
- 5 **Anterior circumflex humeral artery** (*arteria circumflexa humeri anterior*)
  – a smaller artery located on the anterior surface of the surgical neck of the humerus
- 6 **Posterior circumflex humeral artery** (*arteria circumflexa humeri posterior*)
  – a larger artery that passes through the humerotricipital foramen to supply the deltoid



The lateral thoracic artery descends on the surface of the serratus anterior with the long thoracic nerve.

The thoracodorsal artery runs on the inner surface of the latissimus dorsi with the thoracodorsal nerve.

The posterior circumflex humeral artery can be damaged in a fracture of the surgical neck of the humerus.

The radial artery is palpable in the radial foveola (the "anatomical snuff-box").

The princeps pollicis artery is an inaccurate but widespread term, usually synonymous with the first palmar metacarpal artery.

The radialis indicis artery is synonymous with the radial proper palmar digital artery. It most commonly arises from the first palmar metacarpal artery.

The superficial palmar arch is completely formed only in 27 % of cases. It is unclosed in a majority of cases.

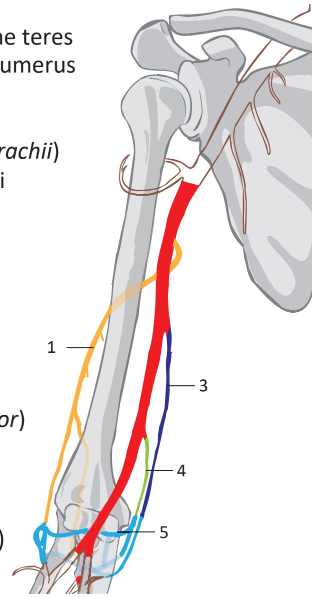
The arteria comitans nervi mediani (median artery) accompanies the median nerve and can be variably enlarged. The artery supports the arterial supply of the palm.

4.7 **Brachial artery** – Arteria brachialis

The brachial artery is the continuation of the axillary artery. It supplies **the whole arm and elbow joint**. It terminates in the cubital fossa where it **bifurcates into the ulnary and radial arteries**. Its collaterals terminate in the **cubital articular anastomosis**.

Course
1. originates from the axillary artery, close to the inferior margin of the teres major and latissimus dorsi, at the level of the surgical neck of the humerus
2. descends on the medial surface of the arm to the cubital fossa

Branches and areas supplied
- 1 **Profunda brachii artery / deep artery of arm** (*arteria profunda brachii*)
  – runs between the lateral and medial heads of the triceps brachii and then passes in the radial canal with the radial nerve
  - 1.1 **Deltoid branch** (*ramus deltoideus*) – supplies the deltoid
  - 1.2 **Middle collateral artery** (*arteria collateralis media*)
    – enters the medial head of the triceps brachii
  - 1.3 **Radial collateral artery** (*arteria collateralis radialis*)
    – terminates in the cubital articular anastomosis
- 2 **Humeral nutrient artery** (*arteria nutritia humeri*)
- 3 **Superior ulnar collateral artery** (*arteria collateralis ulnaris superior*)
  – descends on the anterior surface in the medial intermuscular septum of the arm with the ulnar nerve
  – terminates in the cubital articular anastomosis
- 4 **Inferior ulnar collateral artery** (*arteria collateralis ulnaris inferior*)
  – arises just above the cubital fossa
  – enters the cubital articular anastomosis
- 5 **Cubital articular anastomosis** (*rete articulare cubiti*) – a vascular plexus of the elbow joint
  – formed by the brachial, radial, and ulnar arteries



Clinical notes

The thoracodorsal artery serves as a nutritive vessel for muscle and myocutaneous flaps in plastic and reconstructive surgeries.

When measuring blood pressure, the stethoscope is placed over the brachial artery, medially to the biceps brachii tendon in the cubital fossa.

Pulse is most usually palpated by compressing the radial artery against the radius in the distal part of the forearm, a few centimetres proximal to the wrist.

The radial artery serves as an approach for diagnostic and therapeutic coronary catheterisation. The ulnar artery is less commonly used

The radial artery can be used to create an arterio-venous fistula (anastomosis) for haemodialysis. The most common type is the radiocephalic fistula.

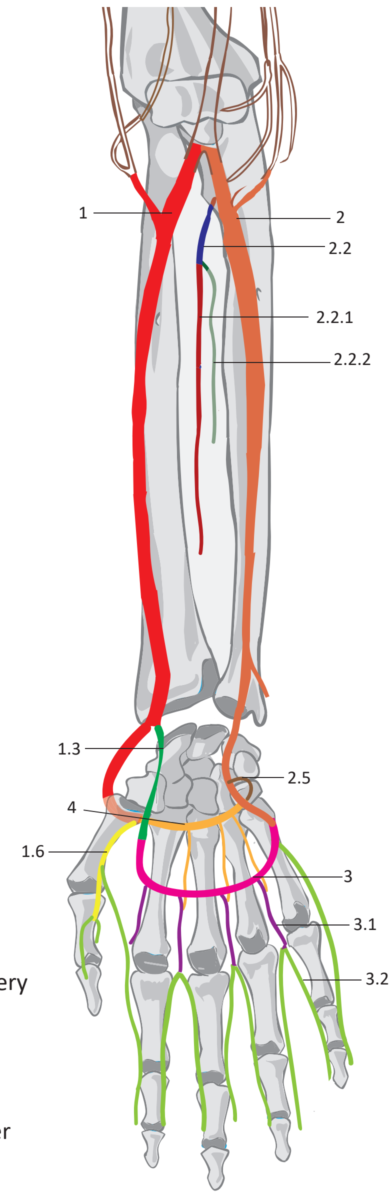
The radial artery can be also used as a graft in coronary artery bypass surgery. Its diameter is 2.0–3.2 mm.

The radial and ulnar arteries are the terminal branches of the brachial artery. The radial artery supplies the **lateral part of the forearm and hand** and the ulnar artery supplies the **medial part of the forearm and hand**. Both arteries continue to the palm, but neither of them pass in the carpal tunnel. They form vascular arches within the palm. **The superficial palmar arch** is supplied mainly by the **ulnar artery** and the **deep palmar arch** is supplied mainly by the **radial artery**.

Radial artery (*arteria radialis*)

Course
1. its initial segment is located between the pronator teres and the brachioradialis
2. passes between the brachioradialis and flexor carpi radialis and then continues between the radial styloid process and scaphoid
3. then it runs on the lateral surface of the wrist to the radial foveola and passes through the first interdigital space to reach the palm

- 1 **Radial artery**
  - 1.1 **Radial recurrent artery** (*arteria recurrens radialis*) – heads proximally to the cubital articular anastomosis
  - 1.2 **Palmar carpal branch** (*ramus carpalis palmaris*) – terminates in the palmar carpal articular anastomosis
  - 1.3 **Superficial palmar branch** (*ramus palmaris superficialis*)
    – runs between the thenar muscles and participates in the formation of the superficial palmar arch
  - 1.4 **Dorsal carpal branch** (*ramus carpalis dorsalis*)
    – arises at the radial foveola and gives branches supplying the dorsum of the hand
  - 1.5 **First dorsal metacarpal artery** (*arteria metacarpalis dorsalis prima*)
    – divides into the dorsal digital arteries supplying the thumb and lateral side of the forefinger
  - 1.6 **First palmar metacarpal artery** (*arteria metacarpalis palmaris prima*)
    – arises where the radial artery enters the palm and divides into the palmar digital arteries, which supply both margins of the thumb and the lateral margin of the forefinger



Ulnar artery (*arteria ulnaris*)

Course
1. after its origin in the cubital fossa it passes along the ulnar side of the forearm
2. lies between the flexor digitorum superficialis and flexor digitorum profundus
3. distally it runs between the flexor carpi ulnaris and the flexor digitorum superficialis
4. enters the palm through the ulnar canal (Guyon's canal) with the ulnar nerve

- 2 **Ulnar artery**
  - 2.1 **Ulnar recurrent artery** (*arteria recurrens ulnaris*)
    – heads proximally to the cubital articular anastomosis
  - 2.2 **Common interosseous artery** (*arteria interossea communis*)
    – heads to the interosseous membrane of the forearm
    - 2.2.1 **Anterior interosseous artery** (*arteria interossea anterior*)
      – runs on the anterior surface of the interosseous membrane of the forearm, perforates the membrane proximally to the pronator teres and terminates in the palmar carpal articular anastomosis
    - 2.2.2 **Posterior interosseous artery** (*arteria interossea posterior*)
      – perforates the interosseous membrane of the forearm (distally to the oblique cord) and runs on the posterior surface of the forearm between the superficial and deep layers of the extensors
  - 2.3 **Palmar carpal branch** (*ramus carpalis palmaris*)
    – heads to the palmar carpal articular anastomosis
  - 2.4 **Dorsal carpal branch** (*ramus carpalis dorsalis*)
    – heads to the dorsal carpal articular anastomosis
  - 2.5 **Deep palmar branch** (*ramus palmaris profundus*)
    – anastomoses with the radial artery and forms the deep palmar arch

Superficial and deep palmar arches (*arcus palmaris superficialis et profundus*)

- 3 **Superficial palmar arch** (*arcus palmaris superficialis*) – is formed by an anastomosis between the larger trunk of the ulnar artery and the smaller superficial palmar branch of the radial artery
  - 3.1 **Common palmar digital arteries** (*arteriae digitales palmares communes*)
    – descend distally between the metacarpals
  - 3.2 **Proper palmar digital arteries** (*arteriae digitales palmares propriae*) – paired branches arising at the level of the metacarpal heads that run along the sides of the fingers
- 4 **Deep palmar arch** (*arcus palmaris profundus*) – is formed by an anastomosis between the larger terminal branch of the radial artery and the smaller deep palmar branch of the ulnar artery
  - 4.1 **Palmar metacarpal arteries** (*arteriae metacarpales palmares*)
    – supply the fingers, head distally and anastomose with the common digital arteries

The thoracic aorta is located between the aortic arch and the abdominal aorta. It supplies the 3rd–11th intercostal spaces, lateral and posterior thoracic walls, the vertebral column with the spinal cord and meninges, part of the diaphragm, and the lungs.

Course

1. begins at the level of the 3rd thoracic vertebra as the continuation of the aortic arch
2. descends on the left side of the vertebral column and oesophagus
3. subsequently it gets between the vertebral column and oesophagus
4. terminates by passing via the aortic hiatus at the level of the 12th thoracic vertebra (continues as the abdominal aorta)

Parietal branches

- 1 **Posterior intercostal arteries** (*arteriae intercostales posteriores*)
  – nine pairs of arteries, which enter the 3rd to 11th intercostal spaces from the posterior side
  – pass between the internal and innermost intercostal muscles
  – anastomose with the anterior intercostal branches
  1.1 **Subcostal artery** (*arteria subcostalis*)
    – a pair of arteries running below the 12th rib
- 2 **Superior phrenic arteries** (*arteriae phrenicae superiores*)
  – a small paired artery supplying the adjacent part of the diaphragm

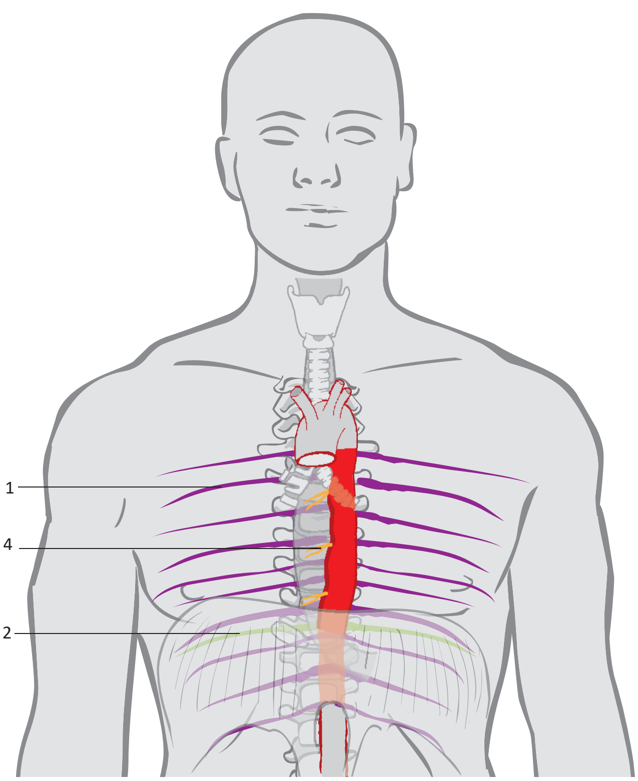
Visceral branches

3 **Bronchial branches** (*rami bronchiales*)
– supply for the nutritive circulation of the lungs
– usually the left one is doubled and the right one is single

4 **Oesophageal branches** (*rami oesophageales*)
– supply the thoracic part of the oesophagus

5 **Pericardiac branches** (*rami pericardiaci*)
– supply the posterior part of the pericardium

6 **Mediastinal branches** (*rami mediastinales*)
– supply the other mediastinal organs and connective tissue



The aortic isthmus (*isthmus aortae*) is an insignificant constriction of the aorta at the border between the aortic arch and the thoracic aorta. It is located distal to the origin of the left subclavian artery at the attachment of the ligamentum arteriosum (a remnant of a fetal shunt: the *ductus arteriosus*). The aorta is slightly enlarged immediately below the isthmus.

The right bronchial branch (*ramus bronchialis dexter*) usually arises from the beginning of the third posterior intercostal artery.

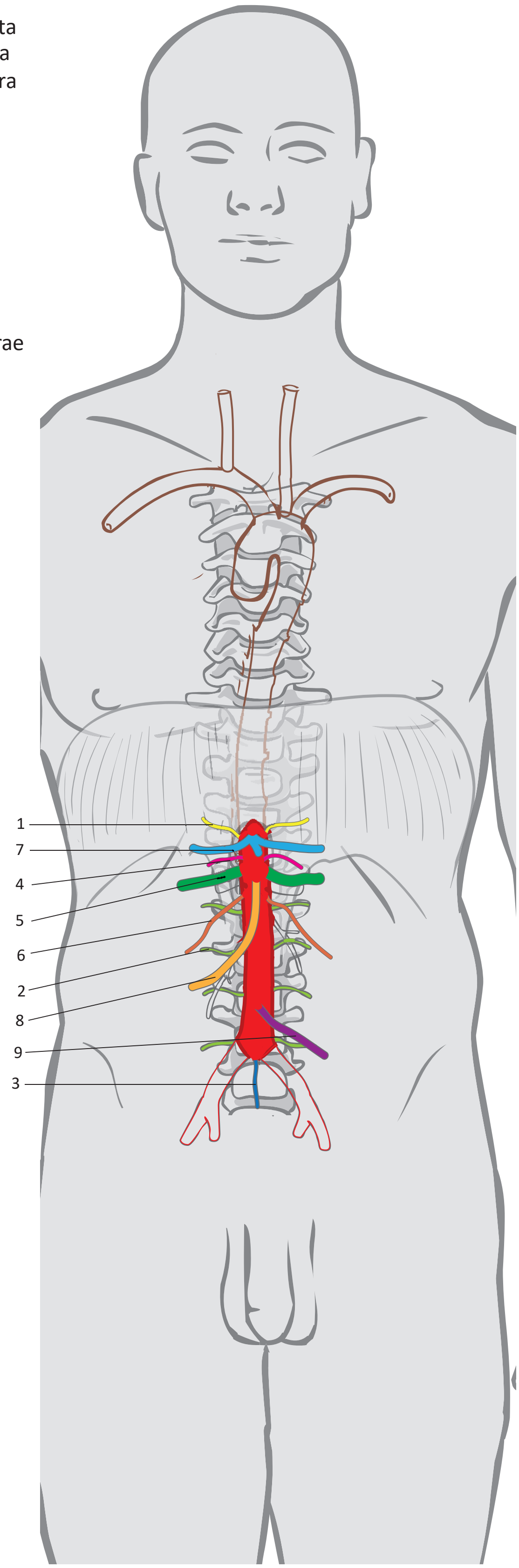
Clinical notes

Aortic coarctation (*coarctatio aortae*) is a congenital constriction of the aorta. In a majority of cases, it is located within the aortic isthmus. It can lead to a decrease in blood pressure distal to the coarctation and an increase in blood pressure proximal to the coarctation. If the coarctation is not severe, collateral blood circulation may develop within the thoracic walls.

An aortic aneurysm (*aneurysma aortae*) is most commonly located in the abdominal aorta, ascending aorta and aortic arch. Atherosclerosis is one of the major causes of aortic aneurysms.

The abdominal aorta is the last part of the aorta. It is the continuation of the thoracic aorta **after passing through the aortic hiatus**. The abdominal aorta **terminates at the level of the 4th lumbar vertebra** by bifurcating into the right and left common iliac arteries. It supplies **the abdominal and pelvic organs, muscles of the abdominal wall and back, part of the diaphragm, and external genital organs.** The abdominal aorta has **parietal** and **visceral branches**.

Course
1. begins at the level of the aortic hiatus as the continuation of the thoracic aorta
2. descends in front of the vertebral column, on the left of the inferior vena cava
3. bifurcates into the common iliac arteries at the level of the 4th lumbar vertebra



Parietal branches

- 1 **Inferior phrenic arteries** (*arteriae phrenicae inferiores*)
  - – a large paired artery supplying the inferior surface of the diaphragm
  - 1.1 **Right and left superior suprarenal arteries** (*arteriae suprarenales superiores dextrae et sinistrae*)
    - – several smaller arteries supplying the suprarenal gland from above
- 2 **Lumbar arteries** (*arteriae lumbales*)
  - – four paired arteries running on the vertebral bodies of the lumbar vertebrae
  - – terminate in the muscles of the abdominal wall
- 3 **Median sacral artery** (*arteria sacralis mediana*)
  - – an unpaired artery, the caudal axial continuation of the aorta
  - – arises from the aortic bifurcation and descends on the anterior surface of the sacrum to the coccyx
  - – **Coccygeal body** (*glomus coccygeum*) – a glomerulus of arteriovenous anastomoses located in front of the coccygeal apex

Paired visceral branches

- 4 **Middle suprarenal artery** (*arteria suprarenalis media*)
- 5 **Renal artery** (*arteria renalis*)
  - – arises almost at a right angle at the level of the 1st lumbar vertebra
  - – runs across the psoas major and enters the hilum of the kidney
  - 5.1 **Inferior suprarenal artery** (*arteria suprarenalis inferior*)
- 6 **Ovarian artery** (*arteria ovarica*) – in females
  - – supplies the ovary
  - – arises at the level of the 2nd lumbar vertebra and passes along the posterior abdominal wall
  - – enters the superior part of the broad ligamentum of the uterus via the suspensory ligament of the ovary
- 6 **Testicular artery** (*arteria testicularis*) – in males
  - – supplies the testis and epididymis
  - – arises at the level of the 2nd lumbar vertebra and passes along the posterior abdominal wall
  - – crosses ventrally the common iliac artery and vein, while passing the border between the pelvis and abdomen
  - – heads to the deep inguinal ring and runs inside the inguinal canal and spermatic cord to the scrotum

Unpaired visceral branches

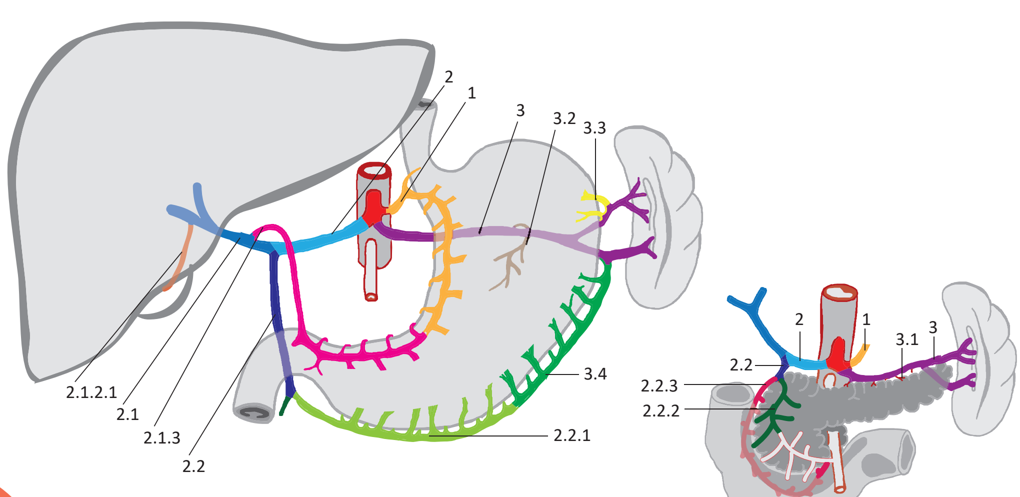
- 7 **Coeliac trunk** (*truncus coeliacus*)
  - – supplies the abdominal part of the oesophagus, stomach, spleen, liver, bile ducts, oral half of the duodenum, and the majority of the pancreas
- 8 **Superior mesenteric artery** (*arteria mesenterica superior*)
  - – supplies the aboral half of the duodenum, inferior part of the pancreatic head, jejunum, ileum, caecum, ascending colon, and 2/3 of the transverse colon
- 9 **Inferior mesenteric artery** (*arteria mesenterica inferior*)
  - – supplies the left 1/3 of the transverse colon, descending colon, sigmoid colon, and the majority of the rectum

The coeliac trunk is a one centimeter long trunk arising from the anterior side of the abdominal aorta at the level of T12 to L1. It trifurcates into **three main branches that supply the abdominal part of the oesophagus, stomach, spleen, liver, bile ducts, oral half of the duodenum, and the majority of the pancreas**.

- 1 **Left gastric artery** (*arteria gastrica sinistra*) – runs on the posterior wall of the omental bursa along the superior margin of the pancreas and the lesser curvature of the stomach, where it anastomoses with the right gastric artery forming the gastric arch

- 2 **Common hepatic artery** (*arteria hepatica communis*)
  – arises from the coeliac trunk, travels to the right and divides above the pylorus
  - 2.1 **Hepatic artery proper** (*arteria hepatica propria*)
    – passes inside the hepatoduodenal ligament with the main bile duct and the portal vein (which is positioned behind the artery) to the porta hepatis
    - 2.1.1 **Left branch** (*ramus sinister*) – supplies the left hepatic lobe
    - 2.1.2 **Right branch** (*ramus dexter*) – supplies the right hepatic lobe
      - 2.1.2.1 **Cystic artery** (*arteria cystica*)
        – arises from the right branch and supplies the gallbladder
    - 2.1.3 **Right gastric artery** (*arteria gastrica dextra*) – passes along the lesser curvature and anastomoses with the left gastric artery (forms the gastric arch)
  - 2.2 **Gastroduodenal artery** (*arteria gastroduodenalis*)
    – descends behind the pylorus
    - 2.2.1 **Right gastro-omental artery** (*arteria gastroomentalis, gastroepiploica dextra*)
      – anastomoses with the left gastro-omental artery along the greater curvature to form the gastro-omental arch supplying the adjacent part of the stomach and greater omentum
    - 2.2.2 **Anterior superior pancreaticoduodenal artery**
      (*arteria pancreaticoduodenalis superior anterior*)
      – descends behind the pancreatic head and descending part of the duodenum
      – anastomoses with the inferior pancreaticoduodenal artery to form the anterior pancreatic arch supplying the head of the pancreas
    - 2.2.3 **Posterior superior pancreaticoduodenal artery**
      (*arteria pancreaticoduodenalis superior posterior*)
      – anastomoses with the inferior pancreaticoduodenal artery to form the posterior pancreatic arch supplying the head of the pancreas

- 3 **Splenic artery** (*arteria splenica/lienalis*) – has a tortuous course as it runs to the left along the superior pancreatic margin to the splenic hilum
  - 3.1 **Pancreatic branches** (*rami pancreatici*) – supply the body and tail of the pancreas
  - 3.2 **Posterior gastric artery** (*arteria gastrica posterior*)
    – supplies the posterior wall of the stomach
  - 3.3 **Short gastric arteries** (*arteria gastricae breves*) – branches for the fundus of the stomach
  - 3.4 **Left gastro-omental artery** (*arteria gastroomentalis/gastroepiploica sinistra*)
    – descends along the greater curvature and anastomoses with the right gastro-omental artery



Tripus Halleri is an obsolete term for the coeliac trunk.

From the developmental point of view, the coeliac trunk is the artery supplying the foregut. The superior mesenteric artery supplies the midgut and the inferior mesenteric artery supplies the hindgut.

The marginal artery of the colon / marginal artery of Drummond (*arteria marginalis coli*) is formed by individual branches of the colic arteries. It has a variable caliber and runs along the wall of the large intestine.

The anastomosis of Haller / arc of Riolan *(anastomosis magna)* is an anastomosis between the middle and left colic arteries, located close to the left colic flexure. It interconnects the arterial beds of the superior and inferior mesenteric arteries.

The cystic artery passes through the cystohepatic triangle (triangle of Calot), where it can be ligated during cholecystecomy.

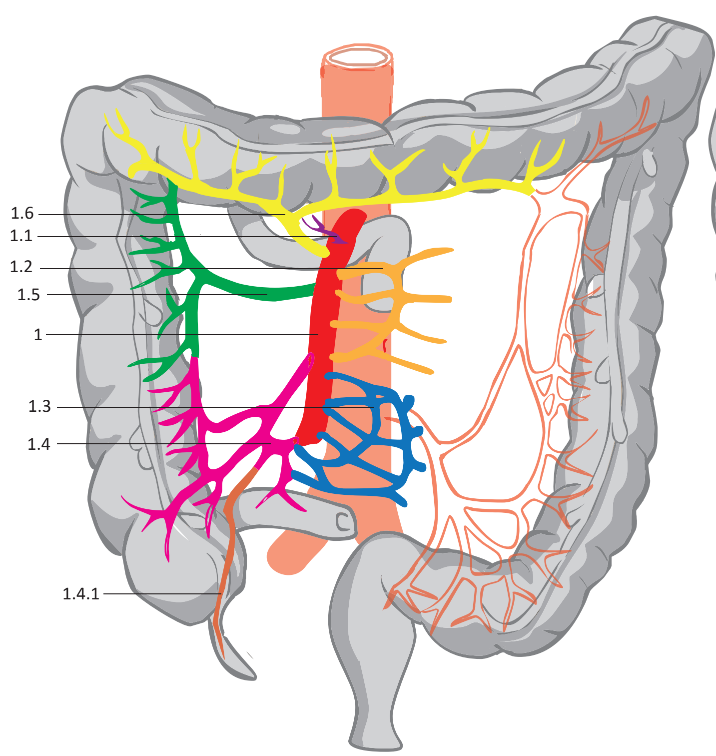
The right gastric artery arises from the hepatic artery proper in two thirds of cases. It is a branch of the common hepatic artery in the remaining cases.

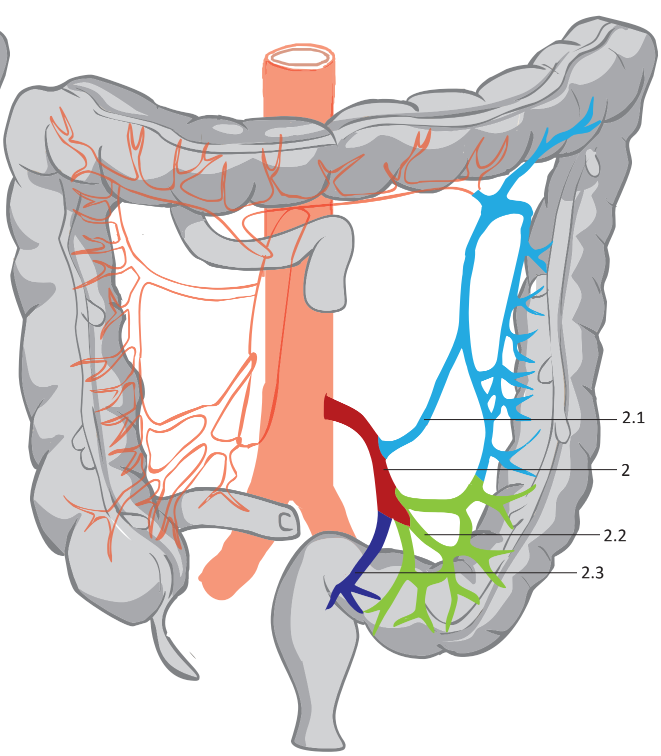
The pancreatic branches include individual smaller or larger branches supplying the pancreatic body and tail (dorsal, inferior and greater pancreatic artery, prepancreatic artery, and the artery to the tail of the pancreas).

The superior mesenteric artery is the second unpaired branch of the abdominal aorta. It supplies the **aboral half of the duodenum, inferior half of the head of the pancreas, jejunum, ileum, caecum, vermiform appendix, and ascending colon.** It also supplies the **right two thirds of the transverse colon** and anastomoses with a branch of the inferior mesenteric artery. **The inferior mesenteric artery** is the third unpaired branch of the abdominal aorta. It supplies the **left third of the transverse colon, descending colon, sigmoid colon**. It continues to the pelvis as the superior rectal artery supplying the majority of the rectum.

- 1 **Superior mesenteric artery** (*arteria mesenterica superior*)
  – arises from the aorta at the level of L1 and runs inside the root of the mesentery
  - 1.1 **Inferior pancreaticoduodenal artery** (*arteria pancreaticoduodenalis inferior*)
    – passes between the duodenum and pancreas, which it supplies
    - 1.1.1 **Anterior branch** (*ramus anterior*) – anastomoses with the corresponding branch from the superior pancreaticoduodenal artery forming the anterior pancreatic arch
    - 1.1.2 **Posterior branch** (*ramus posterior*) – anastomoses with the corresponding branch from the superior pancreaticoduodenal artery forming the posterior pancreatic arch
  - 1.2 **Jejunal arteries** (*arteriae jejunales*) – run inside the mesentery to the loops of the jejunum,
    – divide into one or two arcades, which give off long branches termed arteriae rectae (average of 5 cm long)
  - 1.3 **Ileal arteries** (*arteriae ileales*) – run inside the mesentery to the loops of the ileum
    divide into two or three arcades which give off short branches termed the arteriae rectae (average of 2 cm long)
  - 1.4 **Ileocolic artery** (*arteria ileocolica*) – a terminal branch passing inside the mesentery to the caecum, which it supplies
    - 1.4.1 **Appendicular artery** (*arteria appendicularis*) – passes in the mesoappendix and supplies the vermiform appendix
    - 1.4.2 **Anterior and posterior caecal artery** (*arteria caecalis anterior et posterior*)
      – supply the anterior and posterior aspects of the caecum
  - 1.5 **Right colic artery** (*arteria colica dextra*) – passes retroperitoneally and heads to the ascending colon, which it supplies
    – anastomoses with the ileocolic artery and middle colic artery
  - 1.6 **Middle colic artery** (*arteria colica media*) – passes in the transverse mesocolon and supplies the transverse colon
    – anastomoses with both the right colic artery and the left colic artery, which is a branch of the inferior mesenteric artery

- 2 **Inferior mesenteric artery** (*arteria mesenterica inferior*)
  – arises at the level of L3 and heads down and to the left in the retroperitoneum
  where its branches either ascend to the left colic flexure or descend to the lesser pelvis
  - 2.1 **Left colic artery** (*arteria colica sinistra*) – supplies the descending colon
    and anastomoses with the middle colic artery close to the left colic flexure
  - 2.2 **Sigmoid arteries** (*arteriae sigmoideae*) – two to four arteries supplying the sigmoid colon
  - 2.3 **Superior rectal artery** (*arteria rectalis superior*)
    – a terminal branch of the inferior mesenteric artery while crossing the linea terminalis
    – supplies the rectal ampulla



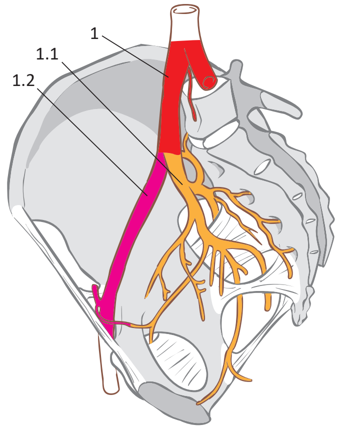


The common iliac artery is a paired continuation of the abdominal aorta from its bifurcation **at the level of L4**. The artery passes on the medial surface of the psoas major, and divides into the external and internal iliac arteries at the level of the sacroiliac joint. The common iliac artery **supplies the pelvis and lower extremities**.

Course

– arises from the bifurcation of the abdominal aorta

- 1 **Right common iliac artery** (*arteria iliaca communis dextra*)
  – crosses the left common iliac vein at a right angle and descemds ventrally to the right common iliac vein
  - 1.1 **Internal iliac artery** (*arteria iliaca interna*)
    – passes across the sacroiliac joint to the lesser pelvis and supplies both the pelvic wall and organs
  - 1.2 **External iliac artery** (*arteria iliaca externa*)
    – passes medially to the psoas major and through the vascular space (under the inguinal ligament)
    – continues as the femoral artery and continues to the lower extremity which it supplies



The iliolumbar artery gives off the iliac branch, which runs across the iliac fossa, and the lumbar branch, which ascends medially to the psoas major where it produces the spinal branch to the vertebral canal.

The internal iliac artery bifurcates into an anterior and a posterior trunk. The posterior trunk usually gives off the iliolumbar artery, lateral sacral arteries, and superior gluteal artery. All the other parietal and visceral branches originate from the anterior trunk.

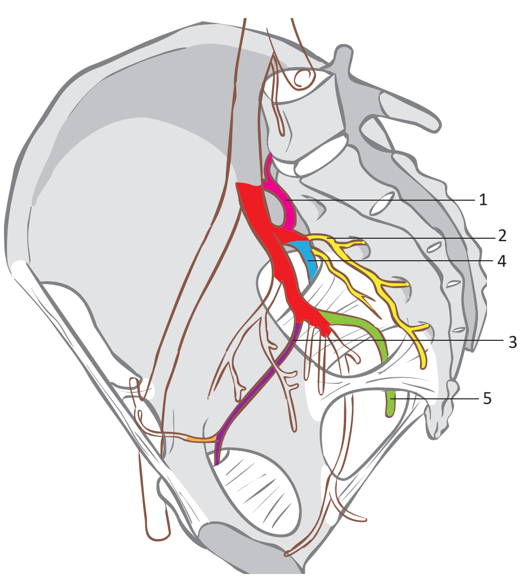
The obturator artery gives off the pubic branch within the lesser pelvis.

The artery to the sciatic nerve (*arteria comitans nervi ischiadici*) is a thin artery accompanying the sciatic nerve.

4.11.1 Internal iliac artery (parietal branches) – Arteria iliaca interna

The internal iliac artery is the medial branch of the common iliac artery. It descends behind the peritoneum to the lesser pelvis, where it passes in front of the sacroiliac joint and produces branches close to the sacral plexus. **The parietal branches supply the pelvic wall, gluteal muscles, and medial side of the thigh**.

- 1 **Iliolumbar artery** (*arteria iliolumbalis*) – supplies the iliopsoas, quadratus lumborum, ilium, and vertebral column including the cauda equina
- 2 **Lateral sacral arteries** (*arteriae sacrales laterales*)
  – two branches (each then bifurcate) enter the anterior sacral foramina and supply the sacrum, cauda equina, and adjacent muscles of the back
- 3 **Obturator artery** (*arteria obturatoria*) – supplies the muscles of the medial femoral group, sacroiliac joint, accompanies the obturator nerve and vein passing via the obturator canal
- 4 **Superior gluteal artery** (*arteria glutea superior*) – supplies the gluteus maximus, medius, and minimus, passes together with the superior gluteal vein and nerve via the suprapiriform foramen and issues branches which course between passing gluteal muscles
- 5 **Inferior gluteal artery** (*arteria glutea inferior*) – supplies the gluteus maximus
  – passes together with the inferior gluteal nerve and vein via the infrapiriform foramen



Pelvis with arteries, left anterior view to the right side of the pelvis

Clinical notes

The corona mortis of Hesselbach is a term for an aberrant obturator artery arising from the bed of the external iliac artery, usually from the inferior epigastric artery. It crosses the pecten pubis (linea terminalis) and heads to the lesser pelvis and enters the obturator canal. The corona mortis is considered to be an anatomic variant, which is present in 25 % of cases. In 5 % of cases the beds of both arteries are interconnected via an anastomosis (accessory aberrant obturator artery).

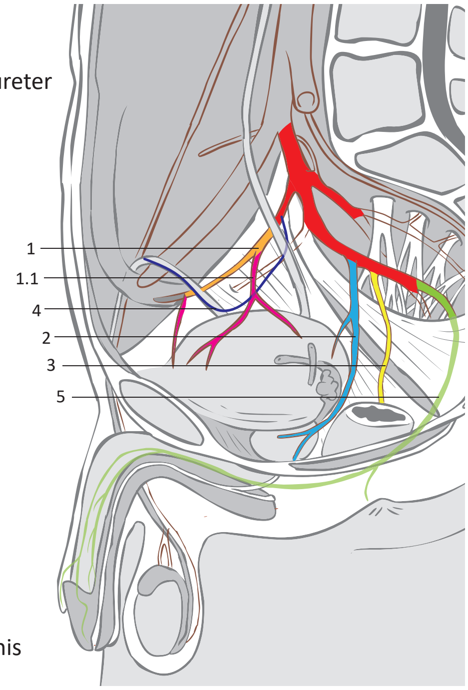
The inferior epigastric artery serves as a nutritive vessel for muscle and myocutaneous flaps (rectus abdominis) in reconstructive surgery.

The visceral branches of the internal iliac artery supply the perineum and the organs within the lesser pelvis except for the rectal ampulla and part of the ovary and uterine tube.

- 1 **Umbilical artery** (*arteria umbilicalis*) – the main foetal artery carrying deoxygenated blood from fetus to the placenta
  – soon after the delivery its distal part obliterates (occluded part, *pars occlusa*) and forms the lateral umbilical ligament (inside the medial umbilical fold) within the anterior abdominal wall
  - 1.1 **Superior vesical arteries** (*arteriae vesicales superiores*) – arise from the unoccluded part (patent part, *pars patens*) of the umbilical artery and supply the superior part of the urinary bladder, part of the ureter and the urethra
- 2 **Inferior vesical artery** (*arteria vesicalis inferior*) – supplies the fundus of the urinary bladder
  – in female, it participates in the blood supply of the vagina, in male supplies blood to the prostate and seminal glands
- 3 **Middle rectal artery** (*arteria rectalis media*)
  – supplies the levator ani and terminates within the outer layers of the rectum (it is not necessary for its nutrition)
  – in female, it participates in the blood supply of the vagina, in male in the blood supply of the prostate and seminal glands
- 4 **Artery to ductus deferens** (*arteria ductus deferentis*) – in male
  – supplies the ductus deferens, prostate, seminal glands, and part of the ureter

4 **Uterine artery** (*arteria uterina*) – in female, supplies the uterus and part of the vagina, uterine tube, ovary, and ureter
– passes in the inferior part of the broad ligament of the uterus to the cervix of the uterus, crosses the ureter ventrally, close to the vaginal fornix and then tortuously ascends along the border of the uterus
  - 4.1 **Ureteric branch** (*ramus uretericus*) – arises where the uterine artery crosses the ureter
  - 4.2 **Uterine branches** (*rami uterini*) – travel to the uterine wall
  - 4.3 **Vaginal branches** (*rami vaginales*) – travel to the vaginal fornices
  - 4.4 **Tubal branch** (*ramus tubarius*) – the first terminal branch is a continuation from the uterine horn and enters the mesosalpinx to supply part of the uterine tube
  - 4.5 **Ovarian branch** (*ramus ovaricus*) – the other terminal branch, enters the mesovary and forms the ovarian arcade with the ovarian artery

- 5 **Internal pudendal artery** (*arteria pudenda interna*)
  – accompanies the sciatic nerve, inferior gluteal artery and vein through the infrapiriform foramen, then winds around the ischial spine and passes through the lesser sciatic foramen to the ischioanal fossa, where it runs through the pudendal canal with the internal pudendal veins and pudendal nerve to the posterior margin of the urogenital diaphragm
  - 5.1 **Inferior rectal artery** (*arteria rectalis inferior*) – supplies the anal canal and anus
  - 5.2 **Perineal artery** (*arteria perinealis*) – runs forward along the perineum
    - 5.2.1 **Posterior scrotal/labial branches** (*rami scrotales/labiales posteriores*) – supply the posterior part of the scrotum / labia majora
  - 5.3 **Dorsal artery of penis/clitoris** (*arteria dorsalis penis/clitoridis*)
  - 5.4 **Deep artery of penis/clitoris** (*arteria profunda penis/clitoridis*)
  - 5.5 **Urethral artery** (*arteria urethralis*) – in male, heads to the corpus spongiosum penis
  - 5.6 **Artery of bulb of penis/vestibule** (*arteria bulbi penis / vestibuli*)
    – supplies the bulb of the penis / vestibule of the vagina and bulb of the vestibule



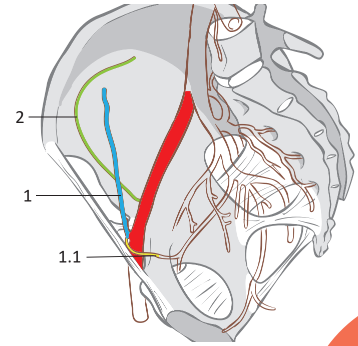
Sagittal section of the male pelvis

4.11.3 **External iliac artery** – arteria iliaca externa

The external iliac artery **travels to the vascular space (*lacuna vasorum*)** and passes laterally to the external iliac vein. After leaving the vascular space it **continues as the femoral artery and supplies the majority of the lower extremity, the anterior and lateral muscles of the abdominal wall, part of the wall of the greater pelvis and part of the scrotum** in males and the broad ligament of the uterus in females. The external iliac artery **issues two branches just medial to the inguinal ligament**.

Branches

- 1 **Inferior epigastric artery** (*arteria epigastrica inferior*) – ascends along the posterior surface of the anterior abdominal wall inside the lateral umbilical fold
  – anastomoses with the superior epigastric artery within the rectus abdominis, which it supplies
  - 1.1 **Pubic branch** (*ramus pubicus*) – runs to the pubic symphysis
  - 1.2 **Cremasteric artery** (*arteria cremasterica*) – in males
    – enters the inguinal canal and supplies the cremaster and spermatic cord
  - 1.3 **Artery of round ligament of uterus** (*arteria ligamenti teretis uteri*) – in females
    – enters the inguinal canal and travels to the round ligament of the uterus
- 2 **Deep circumflex iliac artery** (*arteria circumflexa ilium profunda*) – passes on the inner surface of the inferior margin of the anterior abdominal wall and travels to the anterior superior iliac spine



The femoral artery is the continuation of the external iliac artery after passing through the **vascular space** (*lacuna vasorum*)**.** It enters **the femoral triangle** on the anterior side of the thigh where it **gives off branches that supply the muscles of the thigh**. It passes between the two insertions of the adductor magnus (via the adductor hiatus) and **continues to the popliteal fossa as the popliteal artery**. The popliteal artery **supplies the knee joint and adjacent muscles and continues as the anterior and posterior tibial arteries to the leg**.

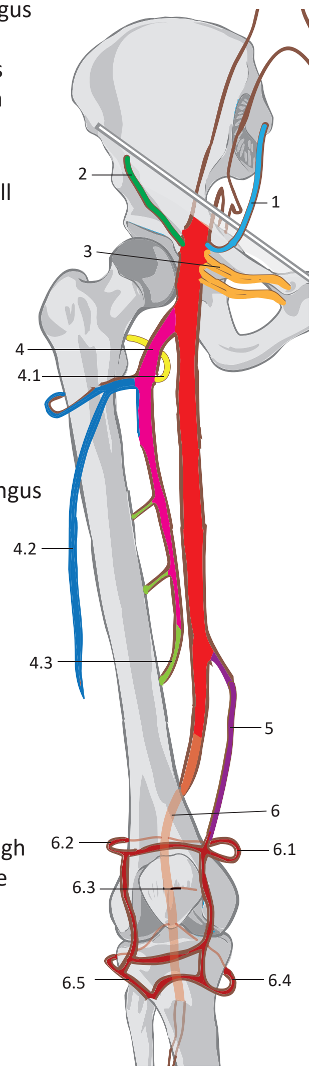
Femoral artery (*arteria femoralis*)

Course

1. runs through the vascular space and between the iliopsoas and pectineus in the iliopectineal fossa where it is positioned laterally to the common femoral vein
2. then it passes beneath the sartorius and between the adductor longus and the vastus medialis to reach the adductor canal of Hunter
3. then it courses between the two insertions of the adductor magnus and passes through the adductor hiatus to reach the popliteal fossa

Branches

- 1 **Superficial epigastric artery** (*arteria epigastrica superficialis*) – passes in the subcutaneous tissue of the anterior abdominal wall
- 2 **Superficial circumflex iliac artery** (*arteria circumflexa ilium superficialis*) – passes in subcutaneous tissue along the inguinal canal to the anterior superior iliac spine
- 3 **External pudendal arteries** (*arteriae pudendae externae*) – the deep and superficial external pudendal arteries supply the anterior part of the scrotum/labia majora as the anterior scrotal/labial branches (*rami scrotales/labiales anteriores*)
- 4 **Profunda femoris artery / deep artery of thigh** (*arteria profunda femoris*) – arises from the proximal part of the femoral atery and passes between the adductor magnus and longus – its branches supply the posterior side of the thigh
  - 4.1 **Medial circumflex femoral artery** (*arteria circumflexa femoris medialis*) – supplies the sacroiliac joint and the adductors and flexors of the thigh
  - 4.2 **Lateral circumflex femoral artery** (*arteria circumflexa femoris lateralis*) – supplies the quadriceps femoris
  - 4.3 **Perforating arteries** (*arteriae perforantes*) – three branches, which travel to the posterior side of the thigh to supply the adductors and flexors
- 5 **Descending genicular artery** (*arteria genus descendens*) – supplies the anterior thigh muscles and knee joint, passes through the adductor canal and perforates the vastoadductor membrane



Popliteal artery (*arteria poplitea*)

Course

1. passes through the adductor hiatus to reach the popliteal fossa
2. then runs medially and below the popliteal vein and tibial nerve
3. bifurcates into the anterior and posterior tibial arteries over the popliteus

Branches

- 6 **Popliteal artery**
  - 6.1 **Superior medial genicular artery** (*arteria superior medialis genus*) – passes below the insertion of the semitendinosus and semimembranosus
  - 6.2 **Superior lateral genicular artery** (*arteria superior lateralis genus*) – passes below the insertion of the biceps femoris
  - 6.3 **Middle genicular artery** (*arteria media genus*) – passes over the posterior surface of the articular capsule
  - 6.4 **Inferior medial genicular artery** (*arteria inferior medialis genus*) – passes below the medial head of the gastrocnemius
  - 6.5 **Inferior lateral genicular artery** (*arteria inferior lateralis genus*) – passes below the lateral head of the gastrocnemius
  - 6.6 **Sural arteries** (*arteriae surales*) – paired branches supplying both heads of the gastrocnemius

The common femoral artery (*arteria femoralis communis*) is a clinical term for the common trunk of the femoral artery in the proximal part of the thigh. The common femoral artery bifurcates into the femoral artery (clinically termed the superficial femoral artery) and the deep artery of the thigh (clinically termed the deep femoral artery).

The genicular (articular) anastomosis (*rete articulare genus*) is a vascular network of the knee joint, which is formed by the branches of the femoral artery, the deep artery of the thigh, the popliteal artery and anterior and posterior tibial arteries.

The malleolar articular network (*rete malleolare*) is a vascular network around the ankles formed by the branches of the anterior tibial, posterior tibial and fibular arteries.

The arcuate artery (*arteria arcuata*) is a variable arch interconnecting the dorsal pedis artery and the lateral tarsal artery on the dorsum of the foot. The arcuate artery is present in 12 % of cases.

Clinical notes

The arterial pulse on the lower extremity can be palpated in four places:
– **Femoral artery** in the femoral triangle, caudal to the inguinal canal
– **Popliteal artery** in the popliteal fossa
– **Posterior tibial artery** behind the medial ankle
– **Dorsal pedis artery** on the dorsum of the foot, laterally to the tendon of the extensor hallucis longus

The medial circumflex femoral artery is the main artery of the hip joint. Fractures of the femoral neck may lead to compression or disruption of the retinacula of Weibrecht, through which vessels of the femoral head pass, which impairs the healing process.

Arterial aneurysms have a predilection for elastic arteries, such as the popliteal artery – the most affected peripheral artery.

Chronic lower limb ischaemia is a pathological condition caused by atherosclerosis of the arteries of the lower extremity. Small arteries are damaged in the microangiopathy of chronic diabetes mellitus, which can also lead to peripheral ischaemia or necrosis. Chronic lower limb ischaemia manifests as claudication (pain in the leg that occurs when walking).

The anterior tibial artery passes perforates the **interosseous membrane of the leg** and then descends along the anterior side of the leg **to the dorsum of the foot**. The anterior tibial artery supplies the knee joint, anterior side of the leg, dorsum of the foot, and part of the sole. **The posterior tibial artery** is a direct continuation of the popliteal artery. It descends along the posterior side of the leg and passes behind the medial malleolus. It passes **through the malleolar canal to the sole**, where it bifurcates into its terminal branches. The posterior tibial artery supplies the posterior and lateral sides of the calf and sole.

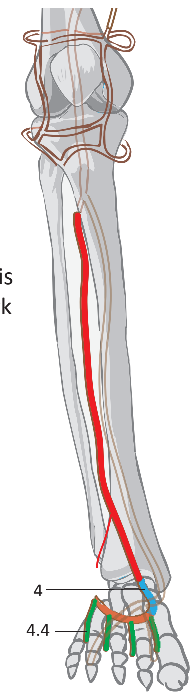
Anterior tibial artery (*arteria tibialis anterior*)

Course

1. perforates then travels along the anterior side of the interosseous membrane of the leg, to which it is attached by fibrous loops
2. is accompanied by the deep fibular nerve and anterior tibial veins
3. distally, it gets to the dorsum of the foot after passing under the superior and inferior extensor retinacula and continues as the dorsal pedis artery

Branches

1 **Anterior tibial recurrent artery** (*arteria recurrens tibialis anterior*) – terminates in the genicular articular anastomosis
2 **Anterior medial malleolar artery** (*arteria malleolaris anterior medialis*) – terminates in the medial malleolar network
3 **Anterior lateral malleolar artery** (*arteria malleolaris anterior lateralis*) – terminates in the lateral malleolar network
4 **Dorsalis pedis artery / dorsal artery of foot** (*arteria dorsalis pedis*)
– passes on the dorsum of the foot between the tendons of the extensor hallucis longus and the extensor digitorum longus and then continues as the deep plantar artery into the sole
4.1 **Lateral tarsal artery** (*arteria tarsalis lateralis*) – passes along the metatarsus laterally towards the toes
4.2 **Medial tarsal arteries** (*arteriaa tarsales mediales*) – small branches supplying the medial metatarsal margin
4.3 **Dorsal digital artery** (*arteria digitalis dorsales*) – supplies the lateral margin of the little toe
4.4 **Dorsal metatarsal arteries** (*arteriae metatarsales dorsales*)
– four arteries travelling to the toes, the medial ones are usually branches of the dorsalis pedis artery and the lateral ones are the branches of the lateral tarsal artery, but they can also be branches of the plantar arch from that pass from the sole of the foot
4.5 **Dorsal digital arteries** (*arteriae digitales dorsales*) – supply the adjacent surfaces of all toes
4.6 **Deep plantar artery** (*arteria plantaris profunda*) – the continuation of the dorsalis pedis artery, piercing through the first interdigital space to the sole where it anastomoses with the plantar arch



Right leg, anterior view

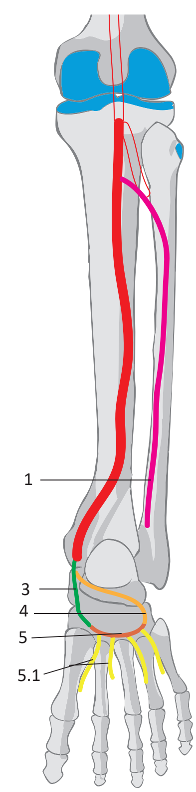
Posterior tibial artery (*arteria tibialis posterior*)

Course

1. passes under the tendinous arch of the soleus with the tibial nerve and posterior tibial veins
2. then descends under the soleus, between the two muscular layers of the calf
3. passes behind the medial malleolus and through the malleolar canal below the flexor retinaculum
4. while passing within the canal it bifurcates into two terminal branches: the medial and lateral plantar arteries

Branches

1 **Fibular/peroneal artery** (*arteria fibularis/peronea*) – descends in the musculofibular canal of Hyrtl to the ankle
1.1 **Calcaneal branches** (*rami calcanei*) – terminate in the calcaneal anastomosis and supply the calcaneus
1.2 **Lateral malleolar branches** (*rami malleolares laterales*) – terminate in the lateral malleolar network
1.3 **Tibial nutrient artery** (*arteria nutritia tibiae*) – supplies the tibia
1.4 **Perforating branch** (*ramus perforans*) – perforates the interosseous membrane of the leg and anastomoses with the anterior tibial artery above the ankle
2 **Medial malleolar branches** (*rami malleolares mediales*) – terminate in the medial malleolar network
3 **Medial plantar artery** (*arteria plantaris medialis*) – passes between the abductor hallucis and flexor digitorum brevis together with the medial plantar nerve and veins
3.1 **Superficial branch** (*ramus superficialis*) – supplies the subcutaneous tissue on the medial surface of the foot
3.2 **Deep branch** (*ramus profundus*) – supplies the muscles of the great toe
4 **Lateral plantar artery** (*arteria plantaris lateralis*) – passes between the flexor digitorum brevis and quadratus plantae proximally to the base of the fifth metatarsal bone
4.1 **Common plantar digital arteries and plantar digital arteries proper** (*arteriae digitales plantares communes et propriae*) – supply the adjacent parts of all toes
4.2 **Perforating branches** (*rami perforantes*) – interconnect the arteries of the sole and dorsum of the foot, occasionally individually supply the second to the fifth toe
5 **Plantar arch** (*arcus plantaris*) – both plantar arteries anastomose with each other and form an arterial arch within the sole, which gives off the plantar metatarsal arteries supplying the toes
5.1 **Plantar metatarsal arteries** (*arteriae metatarsales plantares*) – four arteries leading to the toes
5.1.1 **Plantar digital arteries** (*arteriae digitales plantares*) – supply the sides of the toes



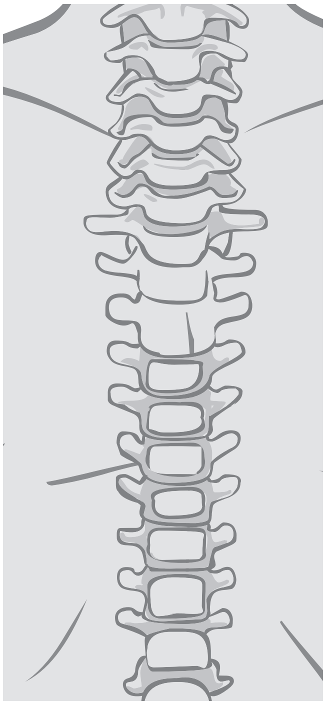
Right leg, posterior view

The superior vena cava is a **short vein** of a **large diameter (2–3 cm) without valves**, arising from the confluence of the right and left brachiocephalic veins at the level of the cartilage of the first right rib. It **descends vertically through the superior mediastinum** to **empty into the right atrium at the level of the third sternocostal joint**. It carries deoxygenated blood from the upper half of the body (head, neck, upper limbs, thorax and upper part of the back).

1 **Right brachiocephalic vein** (*vena brachiocephalica dextra*)
– arises from the confluence of the right internal jugular vein with the right subclavian vein, descends behind the right margin of the manubrium of the sternum
Right venous angle (*angulus venosus dexter*) – the angle formed by the confluence of the right internal jugular vein with the right subclavian vein
– the right lymphatic duct opens into the bloodstream here
1.1 **Vertebral vein** (*vena vertebralis*) – follows the course of the vertebral artery
1.2 Other tributaries – inferior laryngeal veins (*vv. laryngeae inferiores*), right internal thoracic vein (*vena thoracica interna dextra*), tracheal, thymic, bronchial, mediastinal, pericardiac veins (*vv. tracheales, thymicae, bronchiales, mediastinales, pericardiacae*) and supreme intercostal vein (*v. intercostalis suprema*)

2 **Left brachiocephalic vein** (*vena brachiocephalica sinistra*)
– arises from the confluence of the left internal jugular vein with the left subclavian vein
– is located anterior to the three arterial branches of the aortic arch
– lies dorsally to the thymus in childhood
Left venous angle (*angulus venosus sinister*) – the angle formed the confluence of the left internal jugular vein with the left subclavian vein
– the thoracic duct opens into the bloodstream here
2.1 **Unpaired thyroid plexus** (*plexus thyroideus impar*), or the separate inferior thyroid vein (*v. thyroidea inferior*)
– drains the inferior part of the thyroid gland
2.2 **Vertebral vein** (*vena vertebralis*) – follows the course of the vertebral artery
2.3 **Left internal thoracic vein** (*vena thoracica interna sinistra*)
– usually doubled, accompanies the internal thoracic artery
– tributaries: superior epigastric vein, musculophrenic vein, and anterior intercostal veins
2.4 **Accessory hemiazygos vein** (*vena hemiazygos accessoria*)
– cranially connected to the left superior intercostal vein
– caudally drains into the superior part of the hemiazygos vein
2.5 Other tributaries – inferior laryngeal veins (*vv. laryngeae inferiores*), tracheal, thymic, bronchial, mediastinal , pericardiac veins (*vv. tracheales, thymicae, bronchiales, mediastinales, pericardiacae*), supreme intercostal vein (*v. intercostalis suprema*) and the left superior intercostal vein (*v. intercostalis superior sinistra*)

3 **Azygos vein** (*vena azygos*)
–arises on the right side of the lumbar part of the vertebral column from the confluence of the right subcostal vein with the right ascending lumbar vein
– passes through the right crus of the diaphragm into the posterior inferior mediastinum and ascends behind the hilum of the right lung, then turns ventrally to drain into the superior vena cava
3.1 **Hemiazygos vein** (*vena hemiazygos*)
– arises on the left side of the lumbar part of the vertebral column from the confluence of the left subcostal vein with the left ascending lumbar vein
– passes through the left crus of the diaphragm into the posterior inferior mediastinum,
– diagonally crosses the vertebral column at level of T7–T9 and drains into the azygos vein
3.2 Other tributaries – posterior intercostal veins (*vv. intercostales posteriors*); oesophageal, bronchial, pericardial, mediastinal and superior phrenic veins (*vv. oesophageales, bronchiales, pericardiacae, mediastinales, phrenicae superiores*)



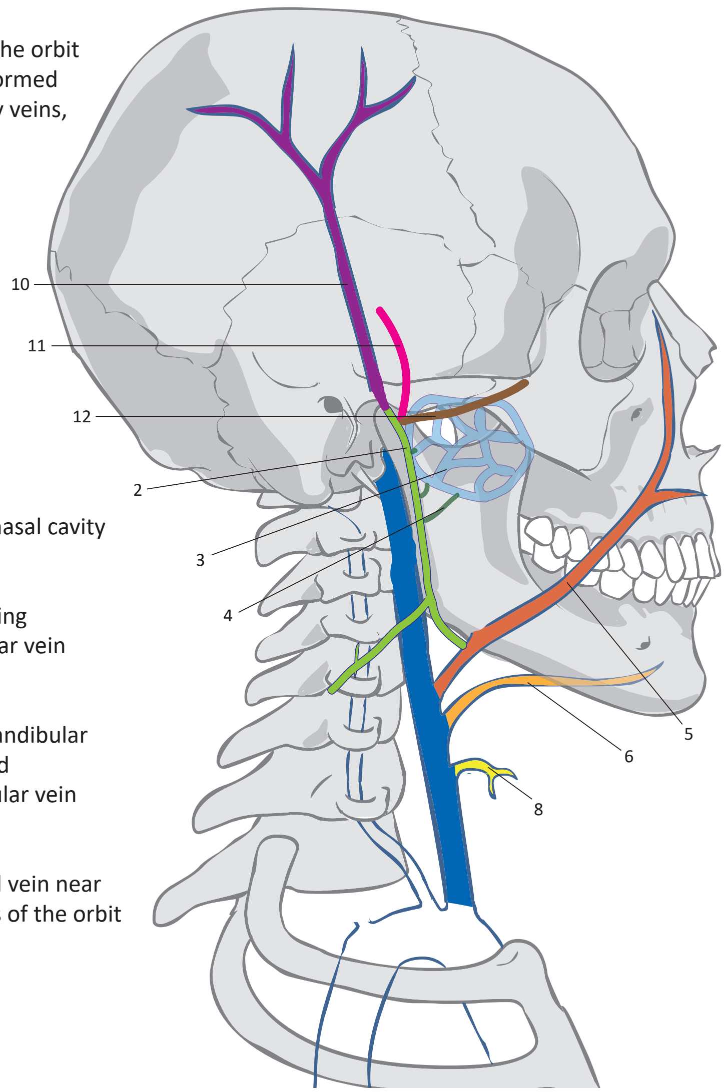
The internal jugular vein is formed from the confluence of two intracranial sinuses: the sigmoid sinus and the inferior petrosal sinus. Its dilated initial segment, located in the jugular foramen, is called **the superior bulb**. It runs **caudally through the parapharyngeal space with the internal carotid artery and vagus nerve**. It widens behind the sternoclavicular joint as the inferior bulb, before joining **the subclavian vein to form the brachiocephalic vein**. It has both intracranial and extracranial tributaries and collects blood from the head (including the eye and brain) and the ventral part of the neck. The internal jugular vein and its tributaries usually don't have valves.

Intracranial tributaries

1 **Cerebral veins** (*venae cerebri*) – superficial and deep cerebral veins
2 **Meningeal veins** (*venae meningeae*) – veins from the cranial dura mater
3 **Dural venous sinuses** (*sinus durae matris*) – intracranial venous sinuses
4 **Diploic veins** (*venae diploicae*) – veins from the bones of the neurocranium
5 **Labyrinthine veins** (*venae labyrinthi*) – veins from the inner ear
6 **Emissary veins** (*venae emissariae*) – connections between intracranial and extracranial veins

Extracranial tributaries

1 **Ophthalmic veins** (*venae ophthalmicae*) – drain the eye and the orbit
2 **Retromandibular vein** (*vena retromandibularis*) – is usually formed by the confluence of the superficial temporal vein, maxillary veins, middle temporal vein, and transverse facial vein
– descends in the retromandibular fossa within the parenchyma of the parotid gland
– bifurcates into anterior and posterior branches
– the anterior branch usually joins the facial vein
– the posterior branch joins the posterior auricular vein to form the external jugular vein
3 **Pterygoid plexus** (*plexus pterygoideus*)
– a venous plexus within the infratemporal fossa
3.1 **Superior alveolar veins** (*venae alveolares superiores*)
– drain the upper jaw
3.2 **Inferior alveolar vein** (*vena alveolaris inferior*)
– drain the lower jaw and mandibular region
3.3 **Sphenopalatine vein** (*vena sphenopalatina*) – drains the nasal cavity
3.4 **Palatine veins** (*venae palatinae*)
– drain the region of the soft and hard palate
4 **Maxillary veins** (*venae maxillares*) – very short veins connecting the pterygoid plexus to the initial part of the retromandibular vein
5 **Facial vein** (*vena facialis*) – drains the facial part of the head
– courses from the medial angle of the eye on the face, ventrally to the masseter, over the mandible into the submandibular region, where it lies superficially to the submandibular gland
– is usually joined by the anterior branch of the retromandibular vein
– alternatively, it may terminate alone in the retromandibular or external jugular vein
5.1 **Angular vein** (*vena angularis*) – the beginning of the facial vein near the medial angle of the eye, anastomoses with the veins of the orbit
5.2 **Deep facial vein** (*vena profunda faciei*)
– connects the pterygoid plexus to the facial vein, passing between the masseter and buccinator
5.3 **External palatine vein** (*vena palatina externa*)
– drains the soft palate and the pharyngeal tonsil

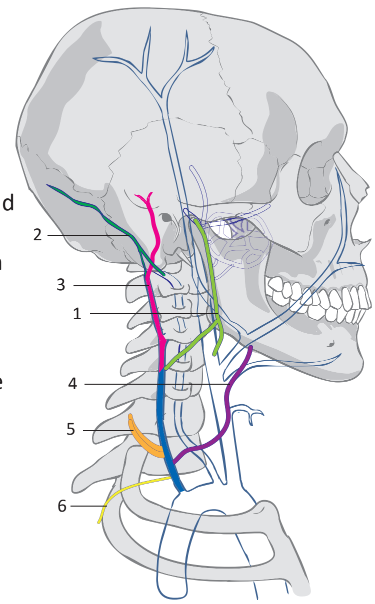


6 **Lingual vein** (*vena lingualis*) – collects blood from the tongue
6.1 **Sublingual vein** (*vena sublingualis*) – drains the deep parts of the tongue
6.2 **Vena comitans of hypoglossal nerve** (*vena comitans nervi hypoglossi*) – follows the hypoglossal nerve
7 **Pharyngeal veins** (*venae pharyngeae*) – drain the pharyngeal plexus
8 **Superior thyroid veins** (*venae thyroideae superiores*) – drain the cranial parts of the thyroid gland and larynx and contribute to one of two common venous trunks (thyrolingual or thyrolinguofacial trunk)
9 **Middle thyroid vein** (*vena thyroidea media*) – drains the middle parts of the thyroid gland
10 **Superficial temporal vein** (*vena temporalis superficialis*) – drains the frontal, parietal, and temporal regions
11 **Middle temporal vein** (*vena temporalis media*) – drains the temporal muscle
12 **Transverse facial vein** (*vena transversa faciei*) – follows the parotid duct and drains the facial region
13 **Posterior auricular vein** (*vena auricularis posterior*) – drains the region of the auricle

5.3 External jugular vein – Vena jugularis externa

The external jugular vein is a superficial vein of the neck that usually **arises from the confluence of the posterior auricular vein and the posterior branch of the retromandibular vein.** It descends **on the lateral side of the neck, crosses the sternocleidomastoid** and opens **into the venous angle** or in its vicinity. It usually has a valve at its termination. It drains the regions of the auricle and neck.

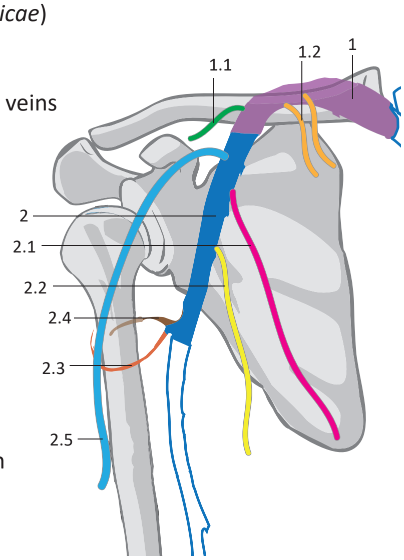
- 1 **Retromandibular vein** (*vena retromandibularis*)
  – runs within the parotid gland parenchyma
  – one of the few veins that bifurcates
  – the anterior branch usually opens into the facial vein
  – the posterior branch joins the posterior auricular vein to form the external jugular vein
- 2 **Occipital vein** (*vena occipitalis*) – drains the back of the head
  – usually opens into the vertebral vein
  – frequently has a connection with the external jugular vein
- 3 **Posterior auricular vein** (*vena auricularis posterior*)
  – drains the region of the auricle
- 4 **Anterior jugular vein** (*vena jugularis anterior*) – drains the anterior cervical region, descends close to the midline
  – the jugular venous arch (*arcus venosus jugularis*) is a short connection with the contralateral vein
- 5 **Transverse cervical veins** (*venae transversae colli*)
  – drain the lateral cervical region
- 6 **Suprascapular vein** (*vena suprascapularis*)
  – drains the area supplied by the suprascapular artery, doubled in its peripheral course



5.4 Subclavian and axillary veins – Vena subclavia et vena axillaris

The axillary vein is the continuation of the brachial veins in the axilla. It becomes **the subclavian vein at the outer border the 1st rib**. The subclavian vein **courses ventrally to the anterior scalene muscle** and the scalene fissure, and dorsally to the sternoclavicular joint. **It opens into the venous angle**, where it joins the internal jugular vein to **form the brachiocephalic vein**. The subclavian vein has only minor tributaries as most of the veins following branches of the subclavian artery open into either the brachiocephalic vein or the internal jugular vein. Both veins have at least one valve.

- 1 **Subclavian vein** (*vena subclavia*)
  - 1.1 **Dorsal scapular vein** (*vena scapularis dorsalis*)
    – drains the area supplied by the dorsal scapular artery
  - 1.2 **Pectoral veins** (*venae pectorales*) – drains the pectoral muscles
- 2 **Axillary vein** (*vena axillaris*)
  - 2.1 **Thoraco-epigastric veins** (*venae thoracoepigastricae*)
    – drain the subcutaneous tissue of the anterior and lateral aspects of the thoracic wall
    – anastomose with the subcutaneous abdominal veins
  - 2.2 **Lateral thoracic vein** (*vena thoracica lateralis*)
    – drains the lateral aspect of the thoracic wall
  - 2.3 **Anterior circumflex humeral vein** (*vena circumflexa humeri anterior*)
    – drains the area supplied by the anterior circumflex humeral artery
  - 2.4 **Posterior circumflex humeral vein** (*vena circumflexa humeri posterior*)
    – drains the area supplied by the posterior circumflex humeral artery
  - 2.5 **Cephalic vein** (*vena cephalica*)
    – a superficial vein of the upper limb arising from the dorsal venous plexus of the hand
    – passes through the infraclavicular (oval) fossa in the clavipectoral fascia



Venous blood flows from the palm to the dorsum of the hand due to higher pressure on the palm. This explains the location of the venous network on the dorsum of the hand.

The inferior vena cava into four parts: the infrarenal part, suprarenal part, hepatic part and thoracic part.
The ratio of the diameter of the suprarenal part to the hepatic part is useful for the evaluation of the venous blood flow through the liver.
The inferior vena cava has a diameter of 3.5 cm at its opening in the heart.

The median antebrachial vein (*vena mediana antebrachii*) courses on the anterior aspect of the forearm and opens into the median cubital vein.

The cubital perforating vein of Gracz (*vena cubitalis perforans*) is a connection between the superficial and deep venous systems in the cubital fossa.

Clinical notes

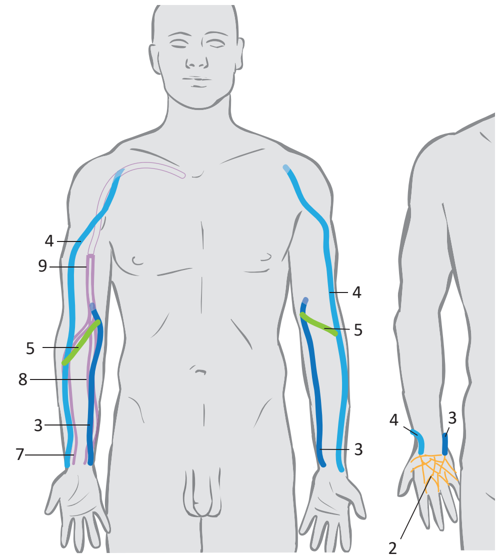
Central venous pressure can be evaluated by examination of the jugular vein filling pressure. In a patient sitting at 30° the jugular veins shouldn't fill more than 2cm above the sternoclavicular joint. Distension of the jugular vein beyond this level can occur in right ventricular insufficiency, tricuspidal valve disease and superior vena cava syndrome. In patients with right ventricular failure, applying sustained pressure for 10 s on the right hypochondrium causes the jugular venous pressure to increase significantly (> 4cm).

Central venous catheterisation is the insertion of a catheter into the internal jugular or subclavian vein. It is used for long-term intravenous administration of medication, fluids or nutrition, and for the measurement of central venous pressure (normal value 1–10 mmHg). Potential complications are: pneumothorax caused by puncturing the cervical pleura and injury to the brachial plexus and subclavian artery.

Patients with pulmonary emboli originating in the veins of the lower limbs or the pelvis, in which medical anticoagulation is ineffective or contraindicated, may benefit from the insertion of a metallic filter (of Greenfield) into the inferior vena cava, to prevent thrombi from reaching the pulmonary circulation.

The veins of the upper limb form two systems. **The superficial system** lies superficially to the fascia of the muscles and lacks corresponding arteries. **The veins of the deep system** follow the course of the arteries; **two veins accompany each artery** up to the armpit.

Superficial system



1 **Intercapitular veins** (*venae intercapitulares*) – connections draining venous blood from the palm to the dorsal venous network of the hand
- 2 **Dorsal venous network of hand** (*rete venosum dorsale manus*)
  – a venous plexus on the dorsum of the hand
- 3 **Basilic vein** (*vena basilica*) – arises at the dorsal venous network of the hand
  – courses on the medial side of the forearm, runs in the medial bicipital groove in the arm and pierces the brachial fascia at the hiatus basilicus to join the brachial veins
  – alternatively, may ascend deep to the fascia all the way up to the axilla
- 4 **Cephalic vein** (*vena cephalica*) –arises at the dorsal venous network of the hand
  – courses on the lateral side of the forearm, runs in the lateral bicipital groove in the arm and then in the deltopectoral groove (*sulcus deltopectoralis*) at the shoulder
  – pierces the clavipectoral fascia in the clavipectoral triangle (in the infraclavicular fossa) to join the axillary vein
- 5 **Median cubital vein** (*vena mediana cubiti*)
  – connects the cephalic vein with basilic vein across the cubital fossa

Deep system

6 **Deep venous palmar arch** (*arcus venosus palmaris profundus*) – drains the fingers and metacarpals
- 7 **Radial veins** (*venae radiales*) – thin veins draining the area supplied by the radial artery
- 8 **Ulnar veins** (*venae ulnares*) – thin veins draining the area supplied by the ulnar artery
- 9 **Brachial veins** (*venae brachiales*) – drain the area supplied by the brachial artery

5.6 Inferior vena cava – Vena cava inferior

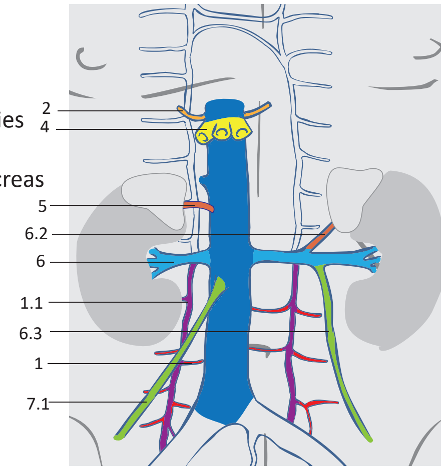
The inferior vena cava is the largest vein in the human body. It collects blood from the inferior half of the body (below the diaphragm) and possesses **no valves**. **It arises from the junction of the right and left common iliac veins** at the level of L4–L5. **It is runs on the right of the abdominal aorta**, **courses dorsally to the liver** in the groove for the inferior vena cava and passes **through the diaphragm** in the caval opening. It has both parietal and visceral tributaries.

Parietal tributaries

- 1 **Lumbar veins** (*venae lumbales*) – four pairs, follow the lumbar arteries
  - 1.1 **Ascending lumbar vein** (*vena lumbalis ascendens*)
    – a vertical connection between the common iliac vein and the azygos/hemiazygos vein, intersecting the lumbar veins
- 2 **Inferior phrenic veins** (*venae phrenicae inferiores*) – drain the inferior part of the diaphragm

3 **External vertebral venous plexuses** (*plexus venosi vertebrales externi*) – a venous networks surrounding the vertebral column

Visceral tributaries



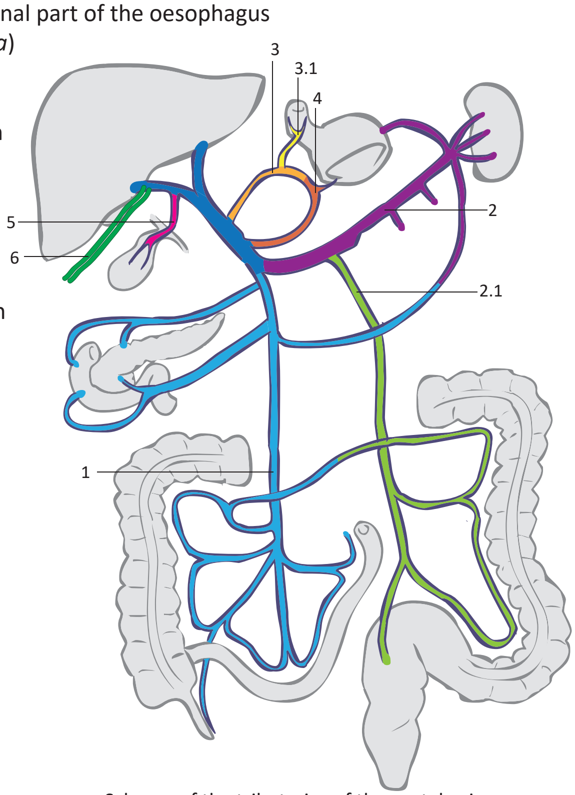
- 4 **Hepatic veins** (*venae hepaticae*) – three veins from the liver opening into the hepatic part of the inferior vena cava
- 5 **Right suprarenal vein** (*vena suprarenalis dextra*) – drains the right suprarenal gland
- 6 **Renal veins** (*venae renales*) – arise from the kidneys following the course of the renal arteries
  – the left one runs dorsally to the pancreas and ventrally to the abdominal aorta
  – the right one runs dorsally to the descending part of the duodenum and head of the pancreas
  - 6.1 **Capsular veins** (*venae capsulares*) – drain adipose tissue surrounding the kidney
  - 6.2 **Left suprarenal vein** (*vena suprarenalis sinistra*) – drains the left suprarenal gland
  - 6.3 **Left testicular/ovarian vein** (*vena testicularis/ovarica sinistra*)
    – opens into into the left renal vein

7 **Testicular vein** (*vena testicularis*) – in male, begins as the pampiniform plexus on the dorsal aspect of the testis and epididymis, courses through the inguinal canal, then runs on the anterior aspect of the psoas major and passes ventrally to the external iliac vessels
– doubled in most of its course and possesses valves
- 7.1 **Right testicular vein** – opens into the inferior vena cava

7 **Ovarian vein** (*vena ovarica*) – in female, begins in the ovary, courses ventrally to the external iliac vessels on the anterior aspect of the psoas major, doubled in most of its course, possesses valves
7.1 **Right ovarian vein** – opens into the inferior vena cava

The portal vein drains blood from the unpaired organs of the abdominal cavity into the liver. It is formed by the confluence of the superior mesenteric vein with the splenic vein. It passes inside the hepatoduodenal ligament (behind the bile duct and the hepatic artery proper) into the porta hepatis where it bifurcates into right and left branches supplying the hepatic lobes. The portal vein possesses no valves.

- 1 **Superior mesenteric vein** (*vena mesenterica superior*)
  – passes on the right side of the superior mesenteric artery, behind the pancreas within the pancreatic notch, and joins the splenic vein to form the portal vein
  - 1.1 **Pancreaticoduodenal veins** (*venae pancreaticoduodenales*)
    – collect blood from the head of the pancreas and from the duodenum
  - 1.2 **Pancreatic veins** (*venae pancreaticae*)
    – collect blood from the body and tail of the pancreas
  - 1.3 **Right gastro-omental vein** (*vena gastroomentalis dextra*)
    – collects blood from the greater curvature of the stomach and the greater omentum
  - 1.4 **Jejunal and ileal veins** (*venae jejunales et ileales*)
    – collect blood from the loops of the jejunum and ileum
  - 1.5 **Ileocolic vein** (*vena ileocolica*)
    – collects blood from the terminal part of the small intestine and from the caecum
    - 1.5.1 **Appendicular vein** (*vena appendicularis*)
      – collects blood from the vermiform appendix
  - 1.6 **Right colic vein** (*vena colica dextra*) – collects blood from the ascending colon
  - 1.7 **Middle colic vein** (*vena colica media*) – collects blood from the transverse colon
- 2 **Splenic vein** (*vena splenica*)
  – collects blood from the spleen, the fundus of the stomach and part of the pancreas
  – originates in the splenic hilum, passes above the tail of the pancreas and then behind the body of the pancreas to join the superior mesenteric vein forming the portal vein
  - 2.1 **Inferior mesenteric vein** (*vena mesenterica inferior*) – its caudal part accompanies the inferior mesenteric artery, its cranial part accompanies the left colic vein
    – usually opens into the splenic vein behind the head of the pancreas
    - 2.1.1 **Left colic vein** (*vena colica sinistra*) – collects blood from the descending colon
    - 2.1.2 **Sigmoid veins** (*venae sigmoideae*) – collect blood from the sigmoid colon
    - 2.1.3 **Superior rectal vein** (*vena rectalis superior*) – collects blood from the rectal ampulla
- 3 **Left gastric vein** (*vena gastrica sinistra*) – collects blood from the left part of the lesser curvature
  – anastomoses with the right gastric vein to form a venous arch along the lesser curvature
  - 3.1 **Oesophageal veins** (*venae oesophageae*)
    – collect blood from the abdominal part of the oesophagus
- 4 **Right gastric vein** (*vena gastrica sinistra*)
  – collects blood from the right part of the lesser curvature of the stomach, anastomoses with the left gastric vein to form a venous arch along the lesser curvature
  - 4.1 **Prepyloric vein** (*vena prepylorica*)
    – collects blood from the pyloric part of the stomach
- 5 **Cystic vein** (*vena cystica*)
  – collects blood from the gallbladder (doubled in some cases)
- 6 **Paraumbilical veins** (*venae paraumbilicales*)
  – two thin veins passing along the round ligament of the liver



Scheme of the tributaries of the portal vein

The inferior mesenteric vein empties into the splenic vein in 50 % of cases, into the superior mesenteric vein in 40 % of cases and directly into the beginning of the portal vein in 10 % of cases.

The gastropancreaticocolic trunk of Henle is a short trunk formed by the confluence of the right gastro-omental vein, right superior colic vein and anterior pancreaticoduodenal vein. It is situated behind the head of the pancreas.

Portal hypertension may cause the adventitial veins of the oesophagus to dilate, leading to the formation of paraoesophageal varices.

Interconnections between the vertebral venous plexuses and other extravertebral veins as known as the plexus of Batson. These interconnections do not possess valves and may serve as a route for the spread of metastasis or inflammation.

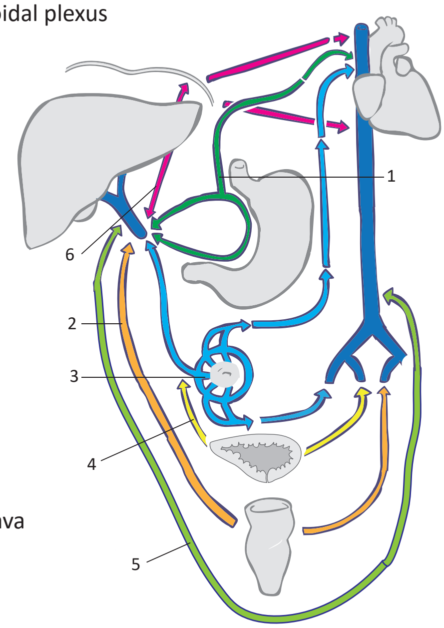
Cavocaval anastomoses may dilated to allow collateral circulation.

Clinical notes

Portal hypertension is a condition of increased pressure in the portal system. It is caused by obstruction to the portal venous blood flow, which may occur in the circulation before the liver (prehepatic portal hypertension – within the portal vein and its tributaries), inside the liver parenchyma (intrahepatic portal hypertension) or after the liver (posthepatic portal hypertension – between the liver and right heart). One of the most common causes of intrahepatic portal hypertension is liver cirrhosis. Increased pressure inside the liver leads to the redistribution of portal blood into the systemic circulation, circumventing the liver. This is achieved through portocaval anastomoses, connections between the portal and the caval systems.

The porto-caval anastomoses are **connections between the tributaries of the portal vein and the superior or inferior vena cava**. Under physiological conditions, blood flows **from the portal vein to the liver** and these anastomoses are small and closed. **If blood pressure in the portal system rises, these anastomoses enlarge and form varicose dilations, which have a high propensity to rupture and bleed.**

- 1 **Anastomoses between the gastric and oesophageal veins**
  – are located in the submucosa of the oesophagus and can enlarge into oesophageal varices
  1.1 **Oesophageal veins** – open into the azygos and hemi-azygos veins (to the inferior vena cava)
- 2 **Anastomoses in the rectum** – are located in the internal rectal venous plexus / haemorrhoidal plexus in the submucosa of the rectum and can enlarge, forming internal haemorrhoids
  2.1 **Superior rectal vein** – drains into the inferior mesenteric vein (to the portal vein)
  2.2 **Middle rectal veins** – drains into the internal iliac vein (to the inferior vena cava)
  2.3 **Inferior rectal veins** – drains into the internal pudendal vein (to the inferior vena cava)
- 3 **Anastomoses between the paraumbilical veins and the subcutaneous veins around the umbilicus** – are located within the subcutaneous tissue of the umbilicus
  – their enlargement is termed caput Medusae (“head of Medusa”)
  3.1 **Thoraco-epigastric veins** – open into the axillary vein (to the superior vena cava)
  3.2 **Superficial epigastric veins** – open into the great saphenous vein and then subsequently into the common femoral vein (to the inferior vena cava)
- 4 **Anastomoses between the paraumbilical veins and the vesical venous plexus**
  – are located around the umbilicus and are connected to the vesical venous plexus via the veins of Burrow, which pass along the median umbilical ligament
  4.1 **Vesical venous plexus** – opens into the internal iliac vein (to the inferior vena cava)
- 5 **Anastomoses within the retroperitoneal space** (veins of Retzius) – are located within the mesentery and mesocolons and in places where the unpaired abdominal organs are attached, between the superior and inferior mesenteric veins and the inferior vena cava
- 6 **Anastomoses between the veins of the liver and the phrenic veins**
  – are located within the posterior surface of the liver and on the bare area of the liver
  6.1 **Inferior phrenic veins** – open into the inferior vena cava
  6.2 **Pericardiacophrenic veins** – open into the brachiocephalic veins (to the superior vena cava)

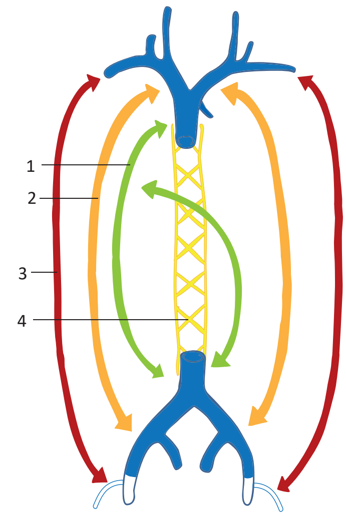


Scheme of the porto-caval anastomoses

5.9 Cavo-caval anastomoses

The cavo-caval anastomoses are **connections between the circulatory beds of the superior and inferior vena cava**. Under physiological conditions they are small. They **become clinically significant when one of the caval veins is occluded**.

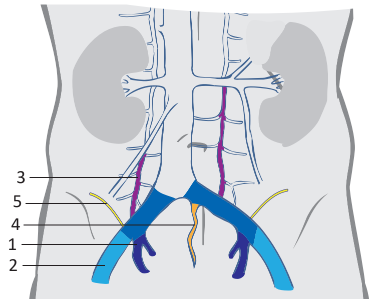
- 1 **Anastomoses between the lumbar veins and the azygos vein or hemi-azygos vein** – are located retroperitoneally in the lumbar region
  1.1 **Azygos vein** – arises on the right side of the lumbar vertebral column from the confluence of the right subcostal vein with the right ascending lumbar vein, passes through the right crus of the diaphragm into the posterior inferior mediastinum to drain into the superior vena cava at the level of T4
  1.2 **Hemi-azygos vein** – arises on the left side of the lumbar vertebral column from the confluence of the left subcostal with the left ascending lumbar vein, passes through the left crus of the diaphragm to the posterior inferior mediastinum and crosses the vertebral column at the level of T7–9 to empty into the azygos vein
  1.3 **Ascending lumbar vein** – intersects the lumbar veins (which drain into the inferior vena cava) as it ascends to join the subcostal vein to form the azygos/hemi-azygos vein, which empties into the superior vena cava
- 2 **Anastomoses between the inferior and superior epigastric veins**
  – are located inside the rectus sheath
  2.1 **Superior epigastric veins** – continue as the internal thoracic veins and open into the left brachiocephalic vein on the left and directly into the superior vena cava on the right
  2.2 **Inferior epigastric veins** – open into the external iliac vein (to inferior vena cava)
- 3 **Anastomoses between the superficial epigastric veins and the thoraco-epigastric veins**
  – are located within the subcutaneous layer of the abdominal wall
  3.1 **Superficial epigastric veins** – open into the great saphenous vein, which flows into the common femoral vein (to the inferior vena cava)
  3.2 **Thoraco-epigastric veins** – open into the axillary vein (to the superior vena cava)
- 4 **Anastomoses between the vertebral venous plexuses and other veins**
  4.1 **Internal vertebral venous plexus** – is located in the epidural space of the vertebral canal and is connected to the vertebral and occipital veins and the suboccipital venous plexus (to the superior vena cava)
  4.2 **Anterior and posterior external vertebral venous plexuses** – surround the vertebral column and are connected to the azygos/hemiazygos vein (to the superior vena cava) and the lumbar, iliolumbar, and sacral veins (to the inferior vena cava)



Scheme of the cavo-caval anastomoses

The common iliac vein originates at the confluence of the internal and external iliac veins ventrally to the sacro-iliac joint. The common iliac veins join to form the inferior vena cava at the level of the vertebra L4. These veins usually do not possess valves.

- 1 **Internal iliac vein** (*vena iliaca interna*)
  – collects blood from the lesser pelvis
- 2 **External iliac vein** (*vena iliaca externa*)
  – collects blood from the lower extremities and part of the anterior abdominal wall
- 3 **Ascending lumbar vein** (*vena lumbalis ascendens*)
  – crosses the lumbar veins within the lumbar region
- 4 **Median sacral vein** (*vena sacralis mediana*)
  – an unpaired vein opening into the left common iliac vein
- 5 **Iliolumbar vein** (*vena iliolumbalis*) – collects blood from the region supplied by the iliolumbar artery

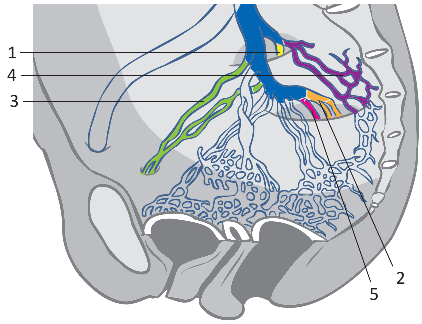


5.10.1 Internal iliac vein – Vena iliaca interna

The internal iliac vein is **formed by the confluence of the inferior gluteal veins with the internal pudendal veins.** It has both parietal and visceral tributaries. The parietal tributaries collect blood from regions supplied by homonymous arteries; usually two parietal veins accompany each of the arteries outside the lesser pelvis.

Parietal tributaries

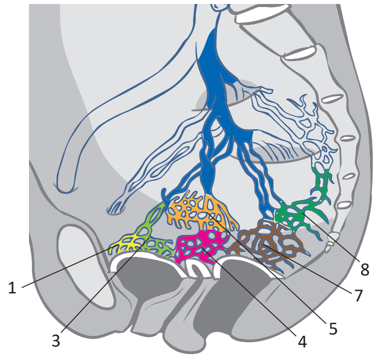
- 1 **Superior gluteal veins** (*venae gluteae superiores*)
- 2 **Inferior gluteal veins** (*venae gluteae inferiores*)
- 3 **Obturator veins** (*venae obturatoriae*)
- 4 **Lateral sacral veins** (*venae sacrales laterales*)
- 5 **Internal pudendal veins** (*venae pudendae internae*)



Veins of the right half of the female pelvis

Visceral tributaries

- 1 **Pudendal plexus of Santorini** (*plexus pudendus*)
  – is located in the retropubic space
  - 1.1 **Deep dorsal vein of penis/clitoris** (*vena dorsalis profunda penis/clitoridis*)
    – passes between the pubis and perineal muscles and enters the lesser pelvis
- 2 **Prostatic venous plexus** (*plexus venosus prostaticus*)
  – is located behind the symphysis
  – surrounds the prostate and
  – is connected to the vesical venous plexus
- 3 **Vesical venous plexus** (*plexus venosus vesicalis*)
  – is located below the base of the urinary bladder
- 4 **Vaginal venous plexus** (*plexus venosus vaginalis*)
  – surrounds the vagina
  – is connected to the vesical venous plexus
- 5 **Uterine venous plexus** (*plexus venosus uterinus*)
  – is located close to the uterine margins within the broad ligament of the uterus and is connected to the vaginal venous plexus, ovarian veins and via the veins of the round ligament of the uterus to the external pudendal veins
  - 5.1 **Uterine veins** (*venae uterinae*)
  - 5.2 **Vaginal veins** (*venae vaginales*)
- 6 **External rectal venous plexus** (*plexus venosus rectalis externus*) – is located around the rectum
- 7 **Internal rectal venous plexus / haemorrhoidal plexus** (*plexus venosus rectalis internus / plexus haemorrhoidalis*) – helps to retain stool
  - 7.1 **Inferior rectal veins** (*venae rectales inferiores*)
    – open into the internal pudendal veins within the area surrounding the anal canal
  - 7.2 **Middle rectal vein** (*vena rectalis media*) – a vein connected to the outer veins of the rectum
- 8 **Sacral venous plexus** (*plexus venosus sacralis*)
  – is connected to the median sacral vein and the lateral sacral veins



Venous plexuses of the right half of the female pelvis

The internal rectal venous plexus (haemorrhoidal plexus) is situated in the submucosa of the anal canal. Its distension contributes to stool continence. Its dilation in portal hypertension causes internal haemorrhoids.

The term perforator is in clinic used for both venous and arterial perforators. A venous perforator interconnects the superficial (subcutaneous) and deep venous system. An arterial perforator is an artery which penetrates through a muscle and fascia into subcutaneous tissue.

The so-called third system of the lower limb venous circulation is formed by a small subcutaneous system (*systema venosum laterale of Albanese*). It is situated on the lateral side of the knee region and is interconnected via perforators with the deep venous system.

The venous corona mortis (crown of death) is a variable interconnection between the inferior epigastric vein and obturator vein in the vicinity of the linea terminalis seen in 75 % of people.

The main trunk of the great/small saphenous vein is enclosed between the subcutaneous fascia (fascia saphena) and superficial muscular fascia. This organization resembles "Egyptian eye" on ultrasound examination.

Venous perforators page 311.

Clinical notes

Fractures and other trauma in the pelvic region are associated with severe bleeding (both arterial bleeding and profuse venous bleeding from the venous plexuses).

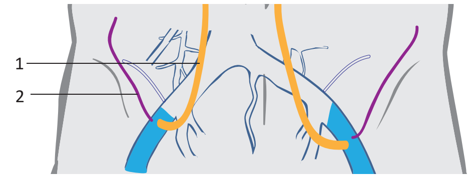
Slow blood flow in the pelvic venous plexus is associated with formation of blood clots (thrombosis), which carry a risk of embolisation to the lungs.

The venous systems of the lower limbs are connected to the pelvic plexuses by the perineal veins and the veins in the inguinal canal, obturator canal and infrapiriform foramen. The direction of blood flow depends on pressure gradients (two-way circulation), and great varices may develop inside the lesser pelvis.

The internal iliac vein and its tributaries are associated with profuse venous bleeding if harmed. The pelvic veins must be spared very carefully during surgical procedures because it is very difficult to suture the walls of the veins or to ligate bleeding veins in this region.

The external iliac vein is **the continuation of the common femoral vein**. It originates below the medial part of the inguinal ligament within the vascular space. It passes along the psoas major, medially to the external iliac artery and enters the lesser pelvis to join the **internal iliac vein to form the common iliac vein**.

- 1 **Inferior epigastric vein** (*vena epigastric inferior*)
  – collects blood from the anterior abdominal wall, its peripheral part is doubled
- 2 **Deep circumflex iliac vein** (*vena circumflexa ilium profunda*)
  – collects blood from the iliac region and the adjacent abdominal wall
  – its peripheral part is doubled



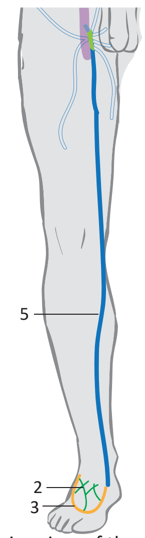
5.11 **Veins of the lower limb** – Venae membri inferioris

The veins of the lower extremity are divided into **2 systems. The superficial system** consists of the veins in the subcutaneous tissue, which do not accompany arteries. **The deep system** consists of the veins accompanying the arteries; two veins usually accompanying each artery. **Perforators** (perforating veins, *venae perforantes*) **interconnect the two systems**.

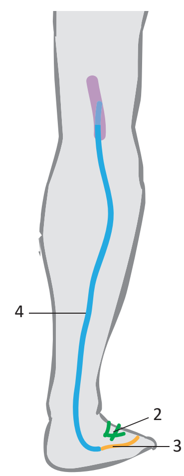
Superficial system

1 **Superficial dorsal digital and metatarsal veins** (*venae digitales et metatarsales dorsales superficiales*)
– collect blood from the dorsal aspect of the toes and metatarsals

- 2 **Dorsal venous network of foot** (*rete venosum dorsale pedis*) – a network of superficial veins on the dorsum of the foot
- 3 **Dorsal venous arch of foot** (*arcus venosus dorsalis pedis*) – a venous arch on the dorsum of the foot
  - 3.1 **Medial marginal vein** (*vena marginalis medialis*) – arises from the dorsal venous arch and passes along the medial margin of the dorsum of the foot and then continues as the great saphenous vein
  - 3.2 **Lateral marginal vein** (*vena marginalis lateralis*) – arises from the dorsal venous arch and passes along the lateral margin of the dorsum of the foot and then continues as the small saphenous vein
- 4 **Small saphenous vein** (*vena saphena parva*) – a continuation of the lateral marginal vein
  – passes on the lateral surface of the foot, behind the lateral malleolus, and on the posterior surface of the calf with the sural nerve
  – passes between the two heads of the gastrocnemius and penetrates the crural fascia to drain into the popliteal vein within the popliteal fossa
- 5 **Great saphenous vein** (*vena saphena magna*) – a continuation of the medial marginal vein
  – passes on the medial side of the foot, in front of the medial malleolus, then continues on the medial side of the calf accompanying the saphenous nerve
  – passes through the saphenous hiatus in the fascia lata and empties into the femoral vein
  - **Subinguinal venous confluence** (*confluens venosus subinguinalis*):
    - 5.1 **Superficialis, anterior, and posterior accessory saphenous vein**
    - 5.2 **Anterior and posterior circumflex femoral vein**
    - 5.3 **Superficial epigastric vein** – anastomoses with the thoraco-epigastric veins
    - 5.4 **Superficial circumflex iliac vein** – passes along the inguinal ligament
    - 5.5 **External pudendal veins** – drains the pubic region



Anterior view of the superficial veins of the right leg



Posterior view of the superficial veins of the right leg

Deep system

Dorsum of the foot

1 **Deep dorsal digital and metatarsal veins** (*venae digitales et metatarsales dorsales profundae*)
– open into the dorsal veins of the foot
2 **Dorsal veins of foot** (*venae dorsales pedis*) – open into the anterior tibial veins

Sole of the foot

1 **Deep plantar digital veins** (*venae digitales plantares profundae*) – open into the deep plantar metatarsal veins
2 **Deep plantar metatarsal veins** (*venae matatarsales profundae*) – open into the medial and lateral plantar veins
3 **Medial and lateral plantar veins** (*venae plantares mediales et laterales*) – open into the posterior tibial veins

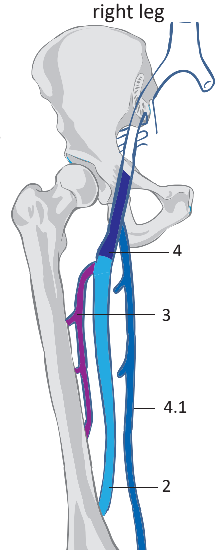
Veins of the leg

Anterior and posterior tibial veins *(vv. tibiales anteriores et posteriores)* – drain into the popliteal vein
Fibular veins *(vv. fibulares)* – drain into the posterior tibial veins

Veins of the thigh

1 **Popliteal vein** (*vena poplitea*) – arises from the confluence of the anterior and posterior tibial veins
– is located in the popliteal fossa and passes through the adductor hiatus to the adductor canal

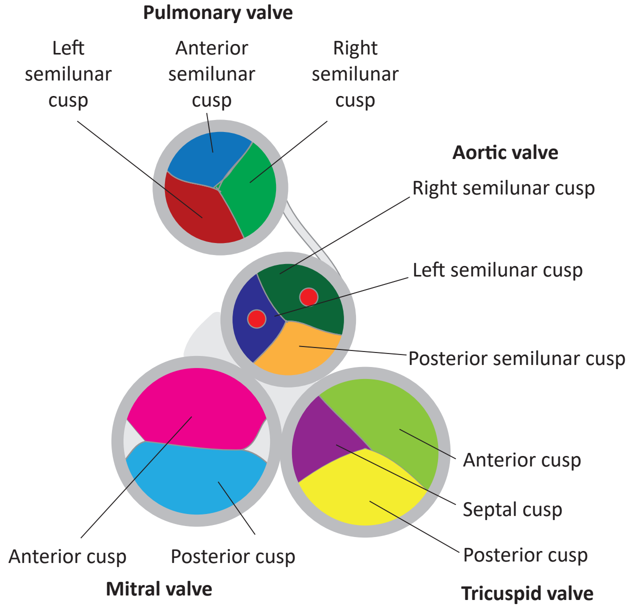
- 2 **Femoral vein** (*vena femoralis*) – courses medially to the femoral artery and empties into the common femoral vein at the femoral triangle
- 3 **Deep vein of thigh** (*vena profunda femoris*) – collects blood from the interior and posterior surface of the thigh
- 4 **Common femoral vein** (*vena femoralis communis*) – arises at the junction of the femoral vein with the deep vein of the thigh
  - 4.1 **Great saphenous vein** (*vena saphena magna*)



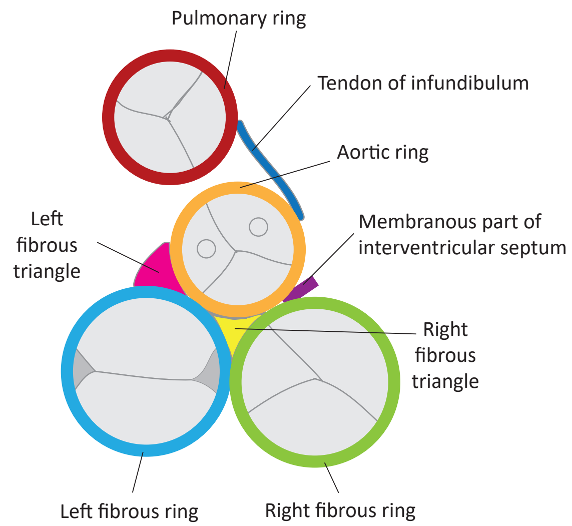
Deep and superficial veins of the right lower limb

6.1.1 Cardiac valves and fibrous skeleton of the heart

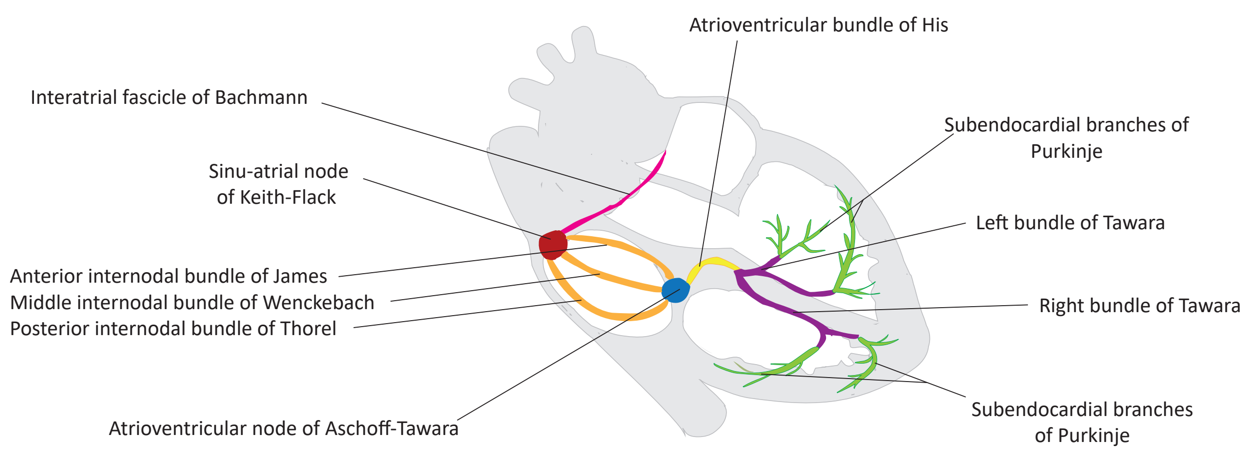
Cardiac valves



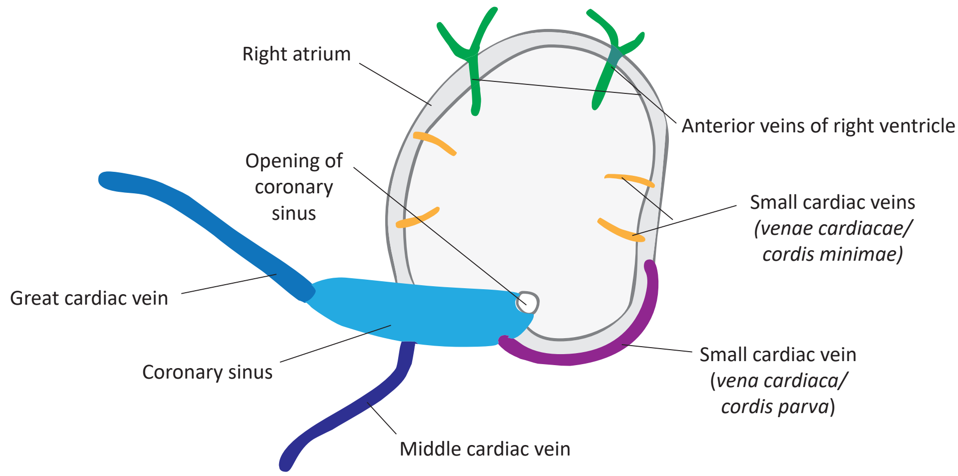
Fibrous skeleton of the heart

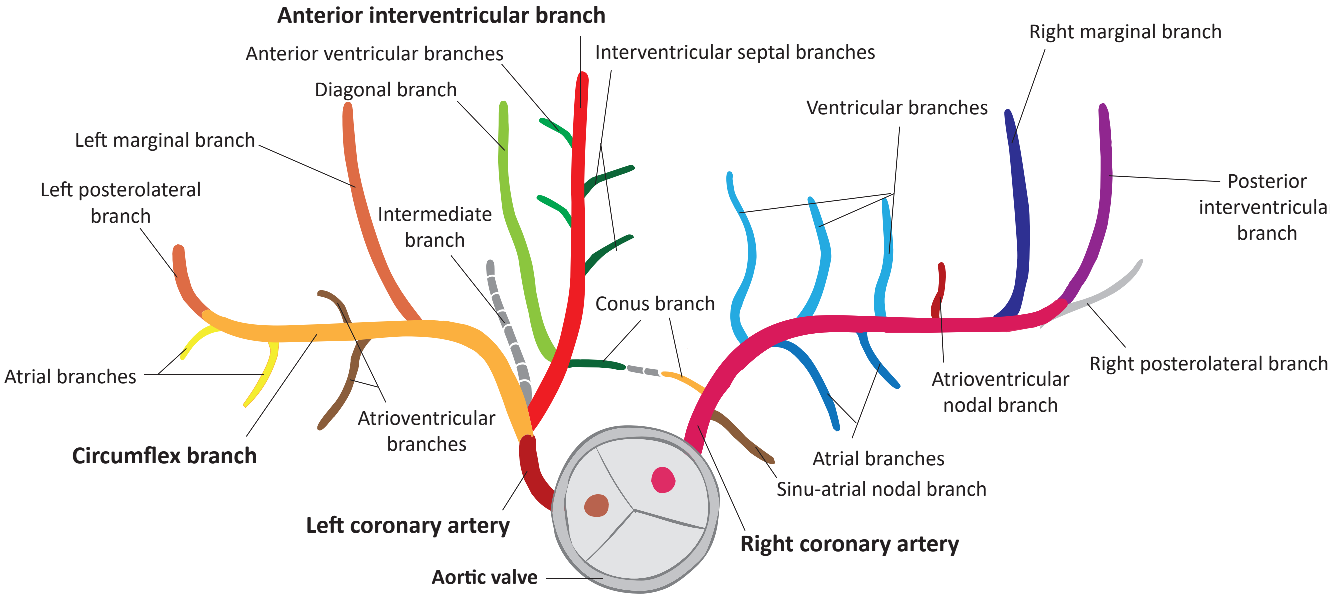


6.1.2 Conducting system of the heart



6.1.3 Venous drainage of the heart

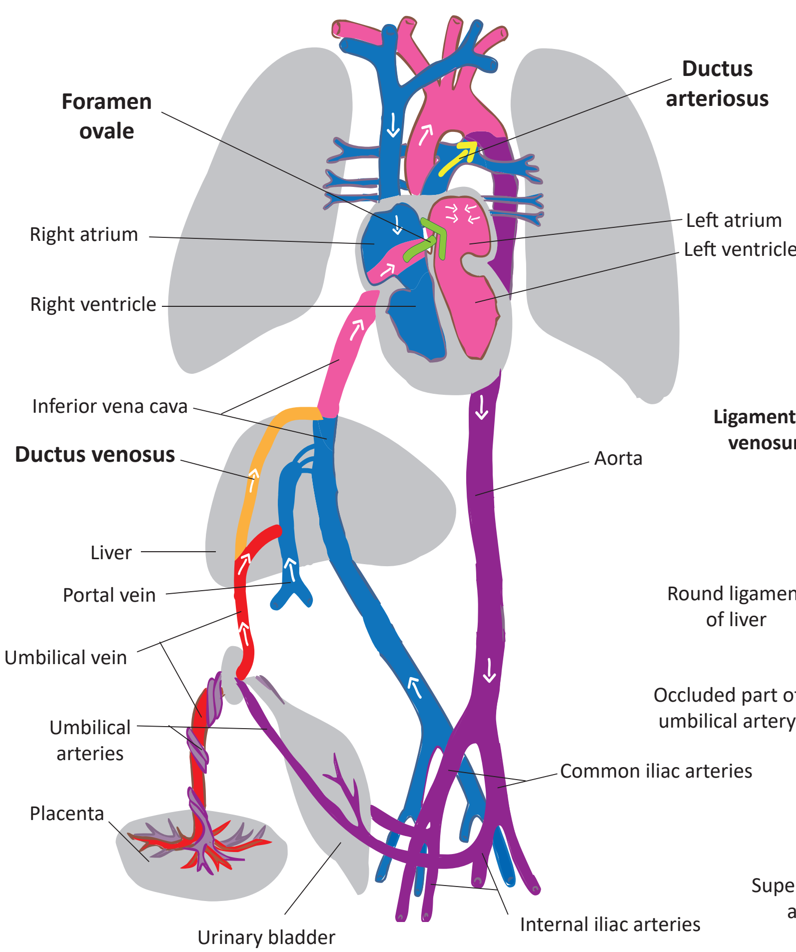


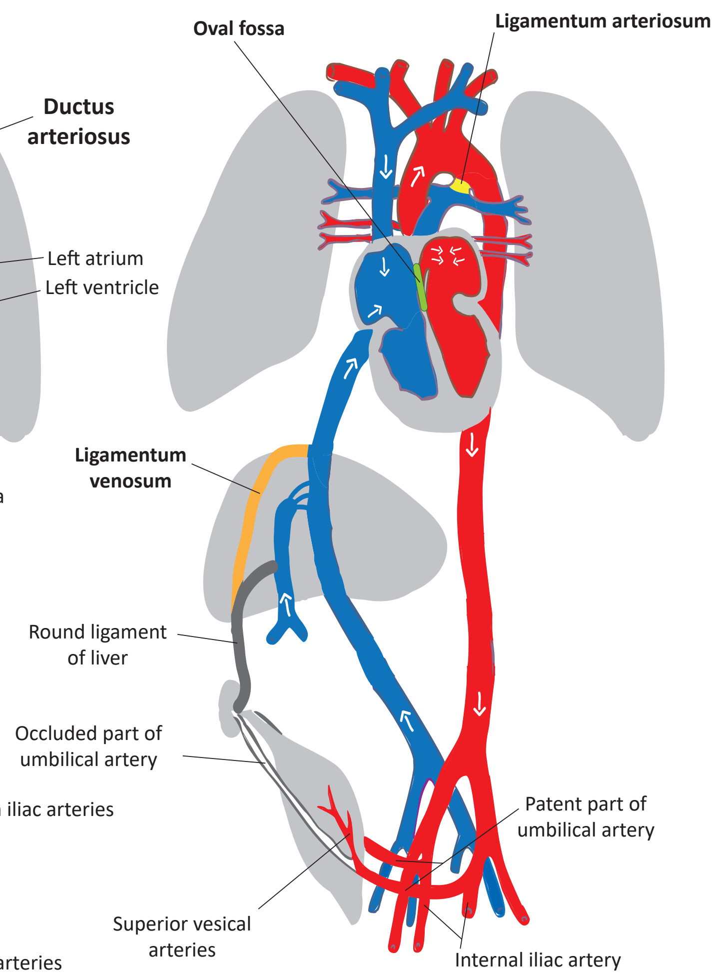


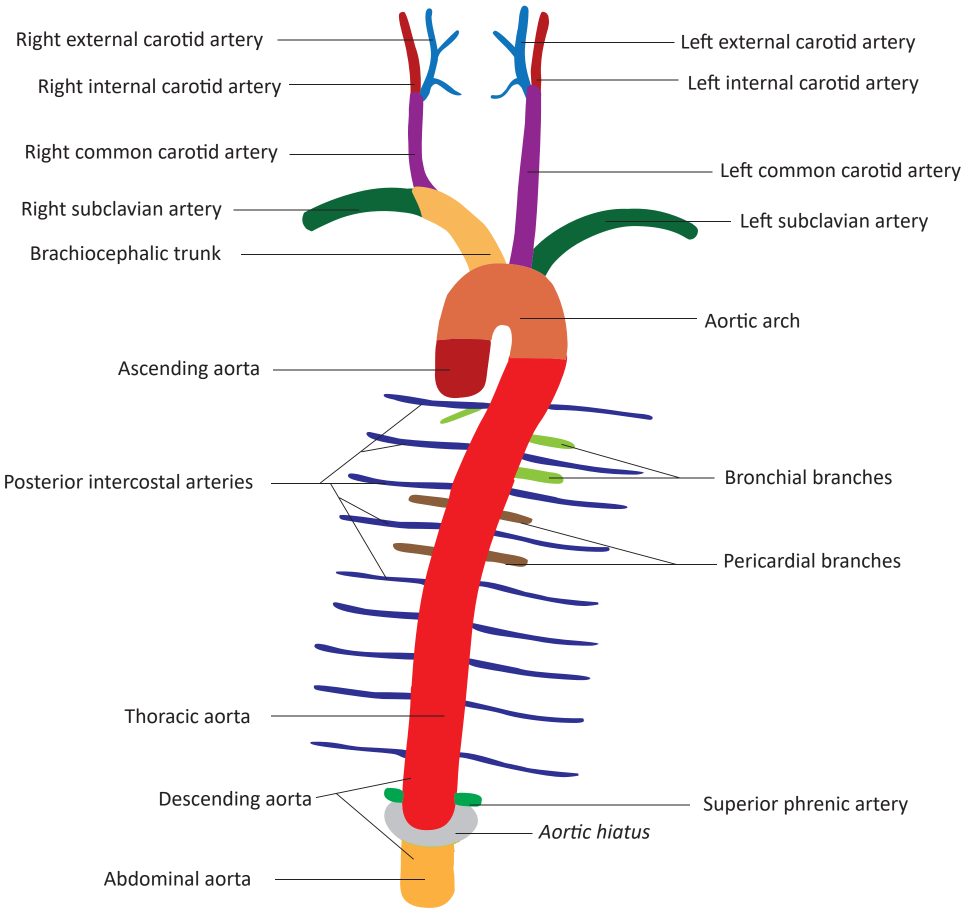
6.1.5 Foetal circulation

Basic pathway of the foetal circulation

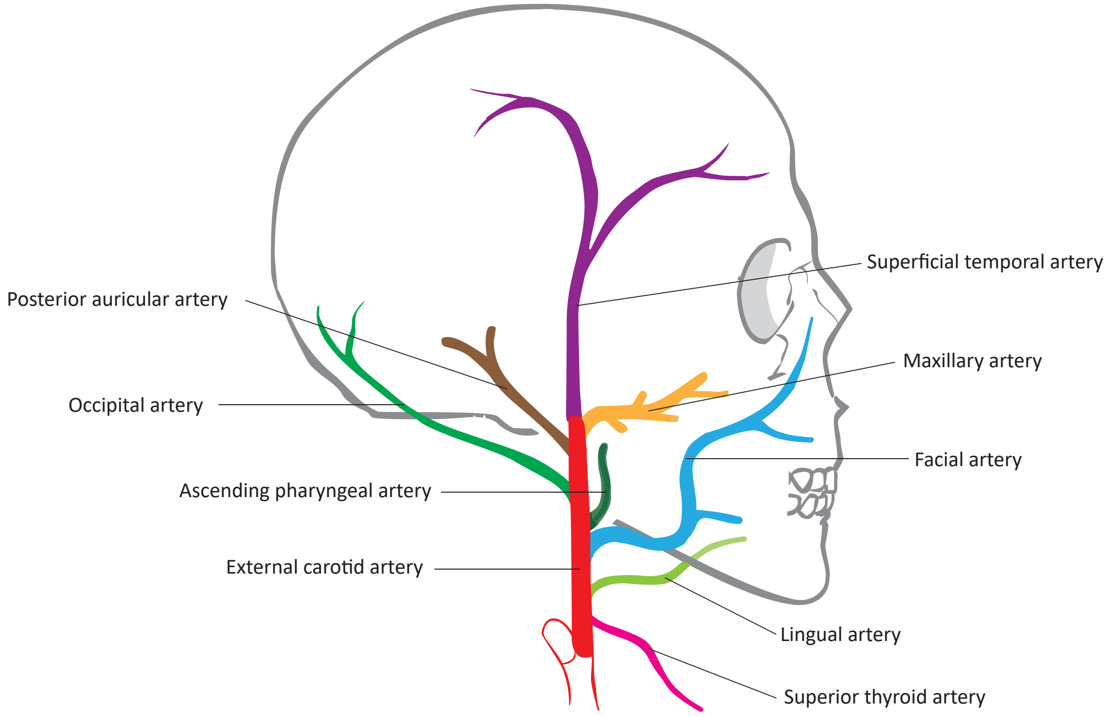
- **Deoxygenated blood**
- **Oxygenated blood:** placenta → umbilical vein → **ductus venosus** → inferior vena cava →
- **Mixed blood (high oxygen concentration):** inferior vena cava → right atrium → **foramen ovale** → left atrium → left ventricle →
- **Mixed blood (low oxygen concentration):** descending aorta → common iliac arteries → internal iliac arteries → umbilical arteries → placenta



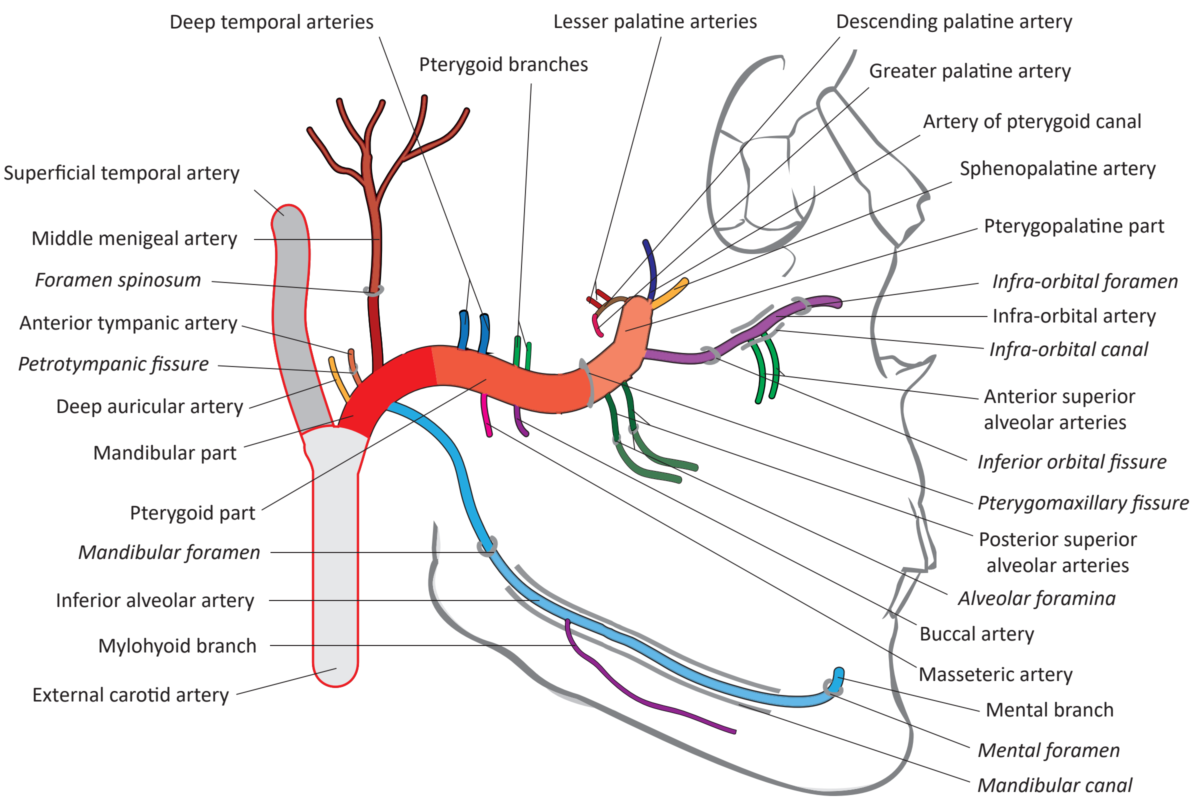




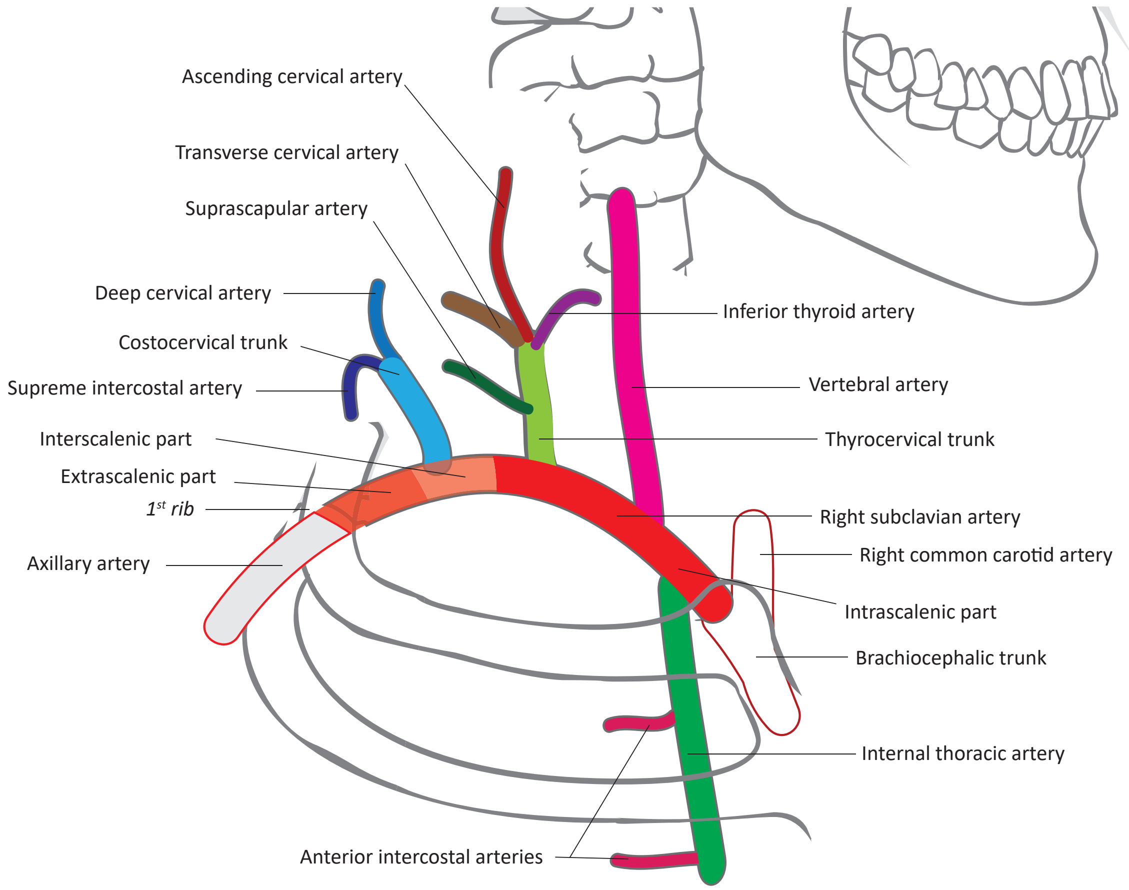
6.2.2 External carotid artery



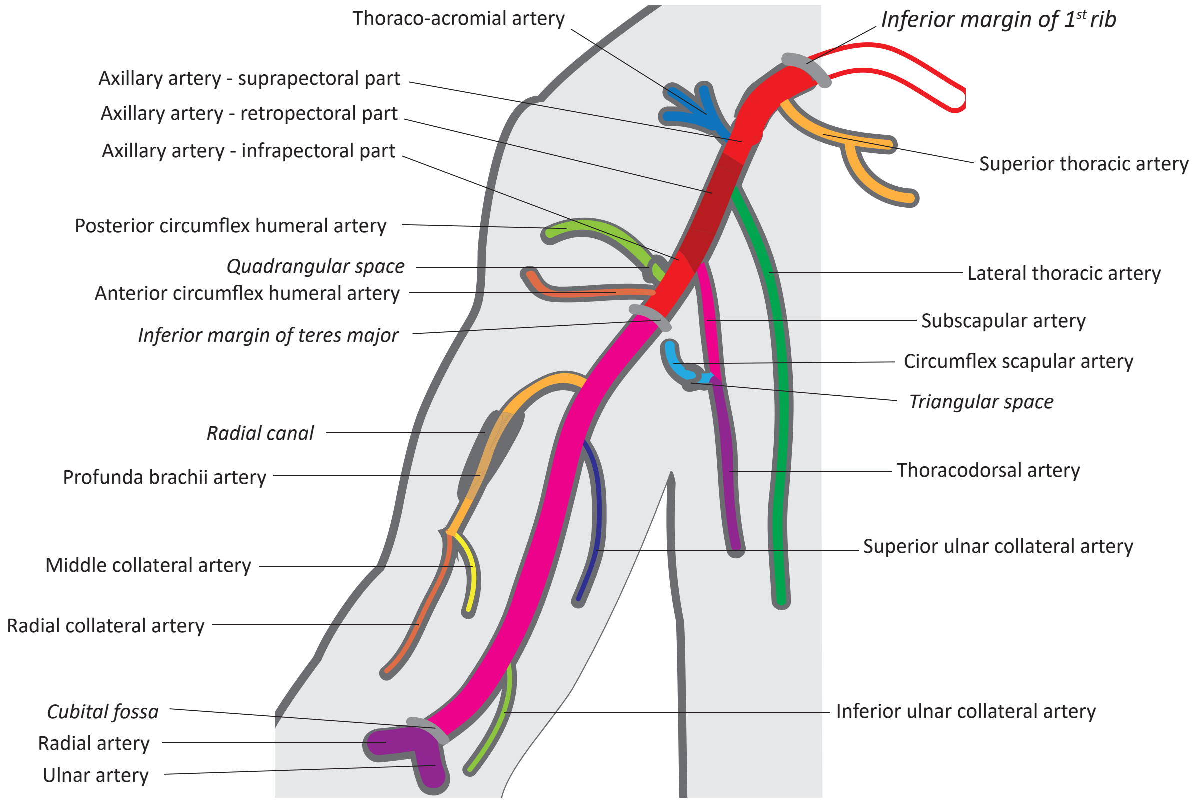
6.2.3 Maxillary artery



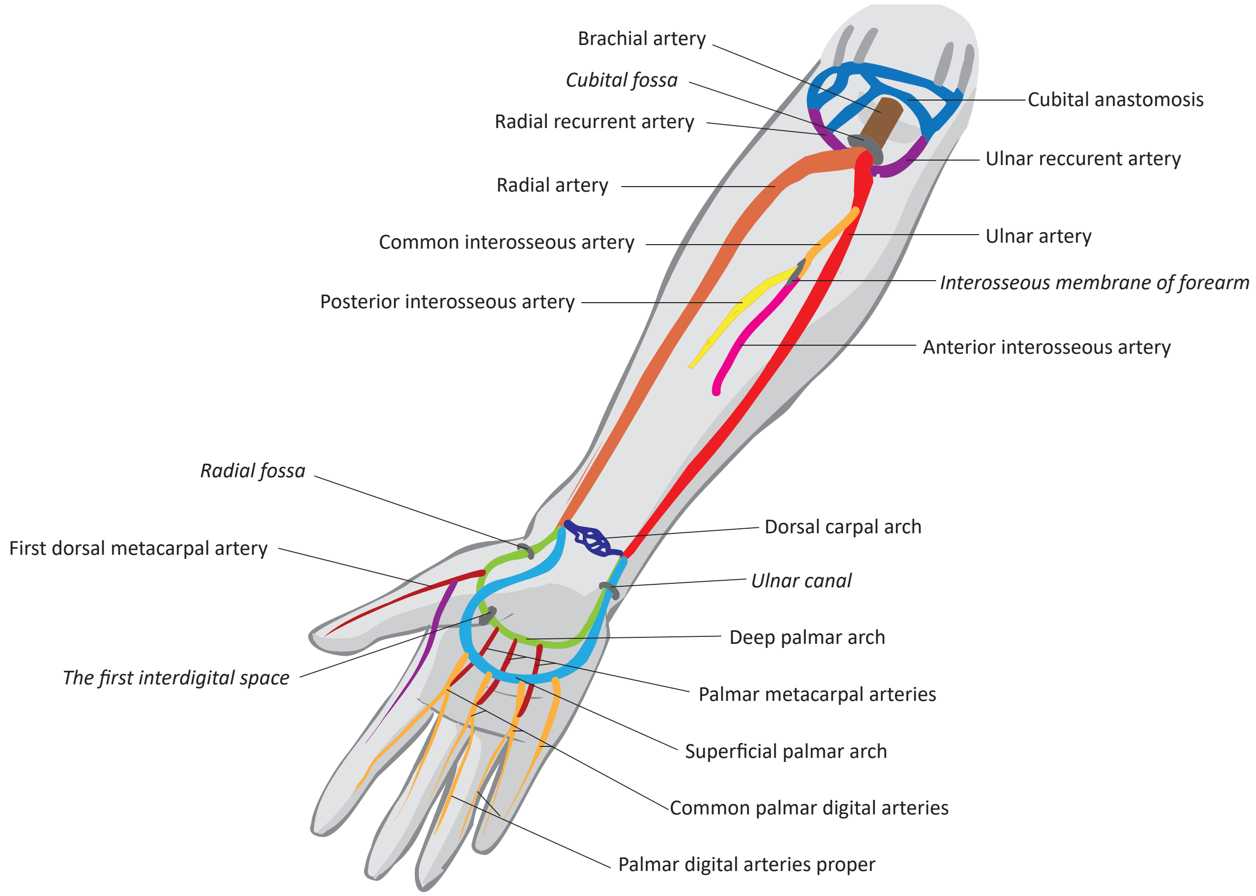
6.2.4 Subclavian artery

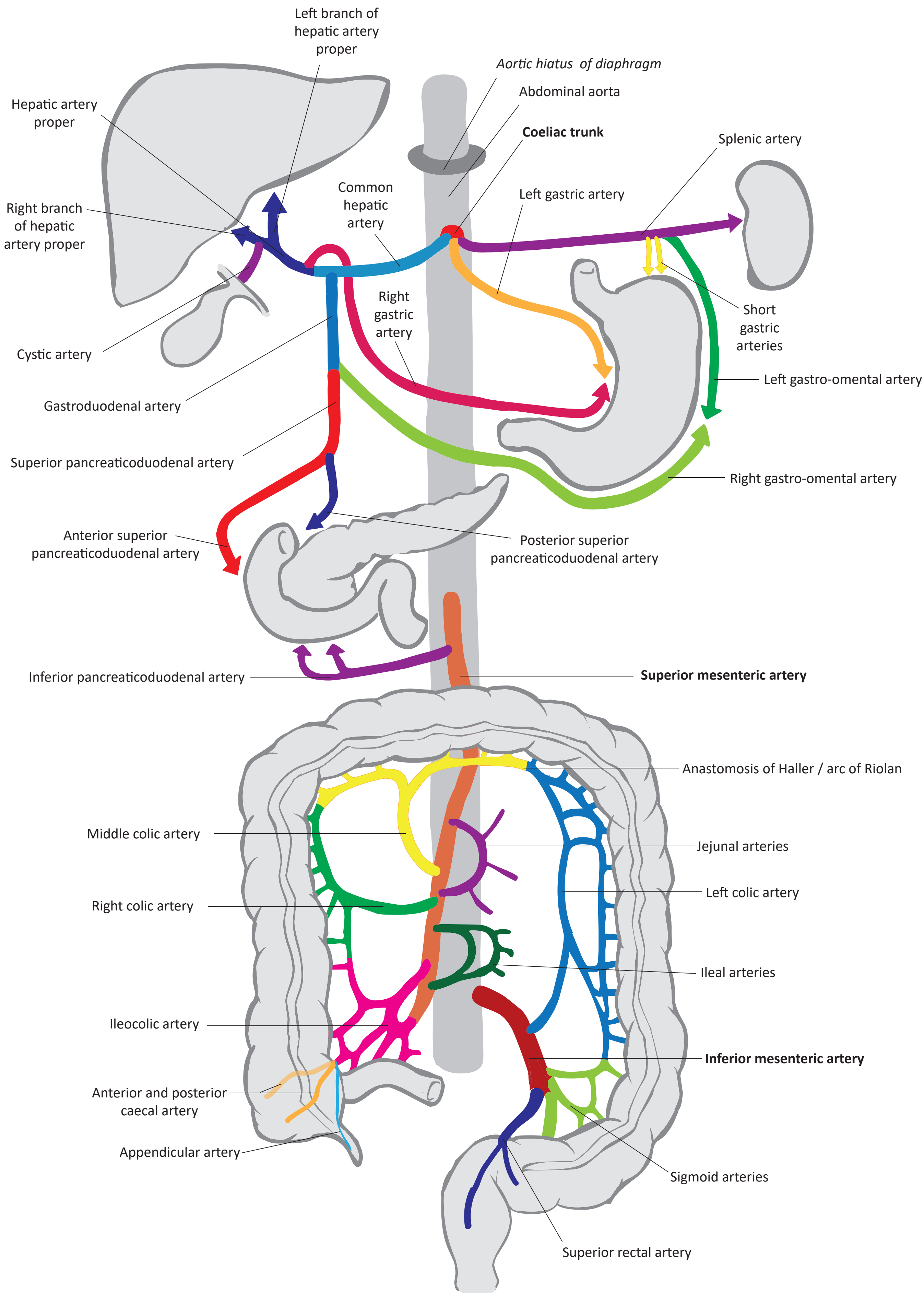


6.2.5 Axillary and brachial arteries

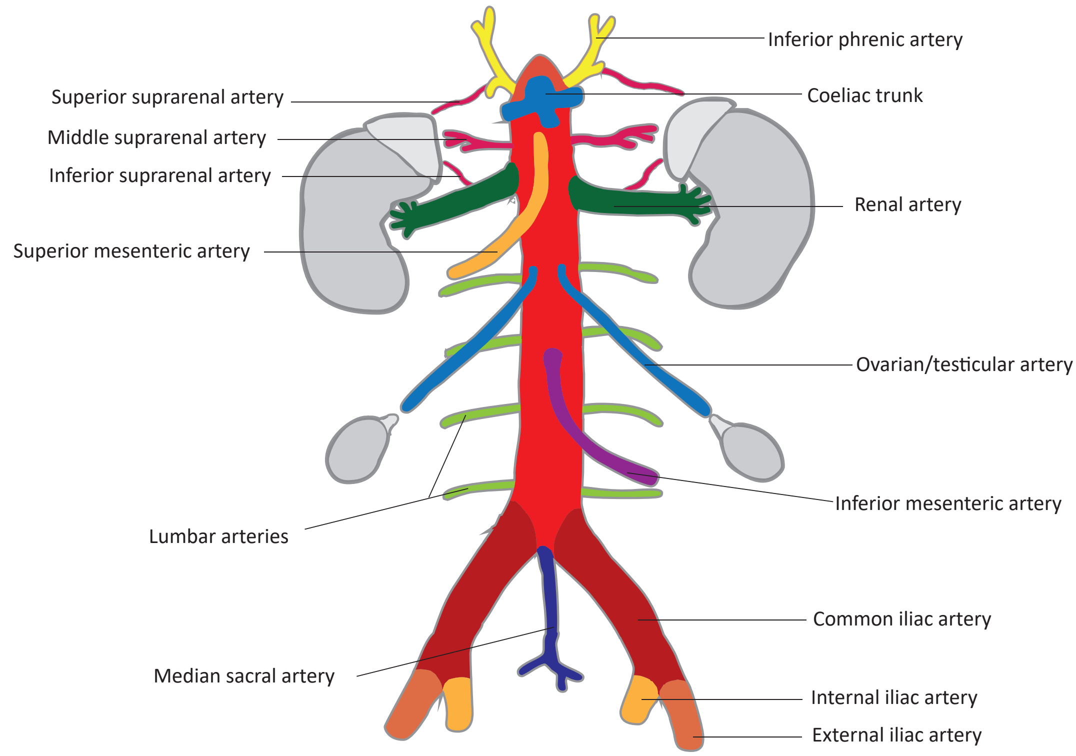


6.2.6 Radial and ulnar arteries

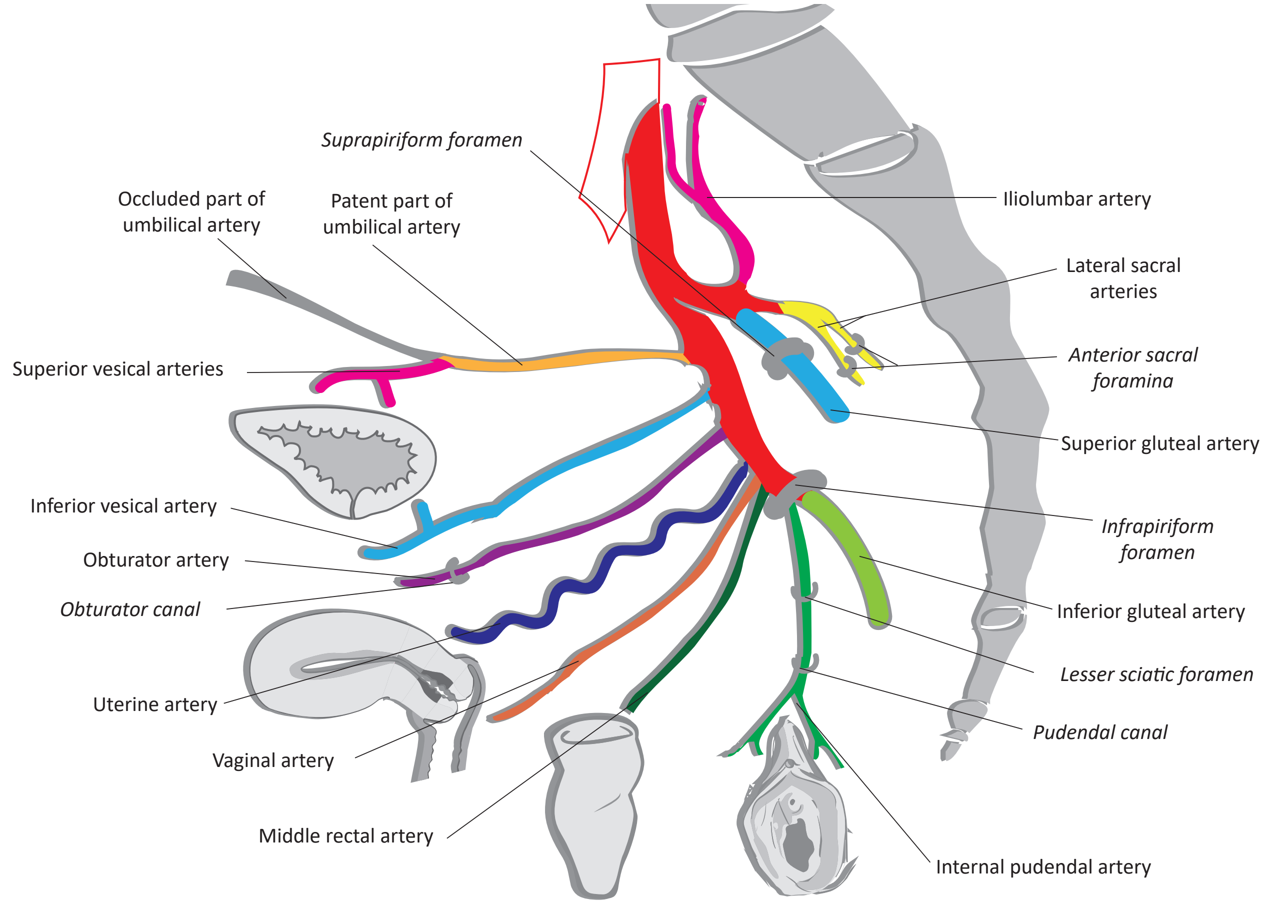


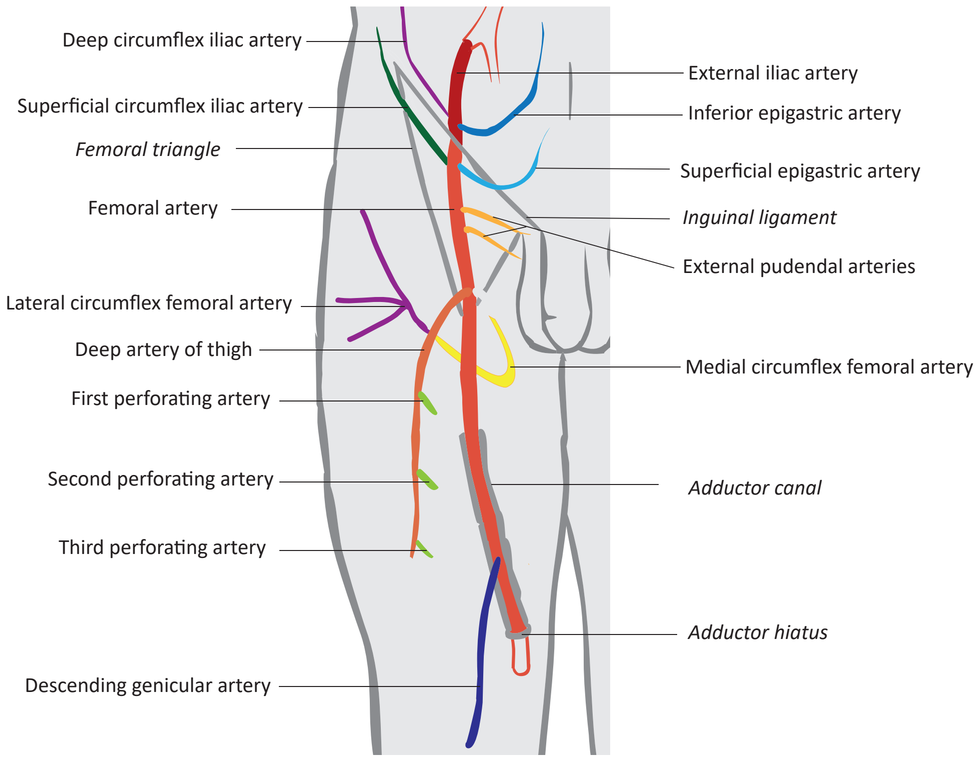

Left branch of hepatic artery proper
Aortic hiatus of diaphragm
Abdominal aorta
Coeliac trunk
Splenic artery
Hepatic artery proper
Common hepatic artery
Left gastric artery
Right branch of hepatic artery proper
Right gastric artery
Short gastric arteries
Cystic artery
Left gastro-omental artery
Gastroduodenal artery
Superior pancreaticoduodenal artery
Right gastro-omental artery
Anterior superior pancreaticoduodenal artery
Posterior superior pancreaticoduodenal artery
Inferior pancreaticoduodenal artery
Superior mesenteric artery
Anastomosis of Haller / arc of Riolan
Middle colic artery
Jejunal arteries
Left colic artery
Right colic artery
Ileal arteries
Ileocolic artery
Inferior mesenteric artery
Anterior and posterior caecal artery
Appendicular artery
Sigmoid arteries
Superior rectal artery

6.2.8 Abdominal aorta

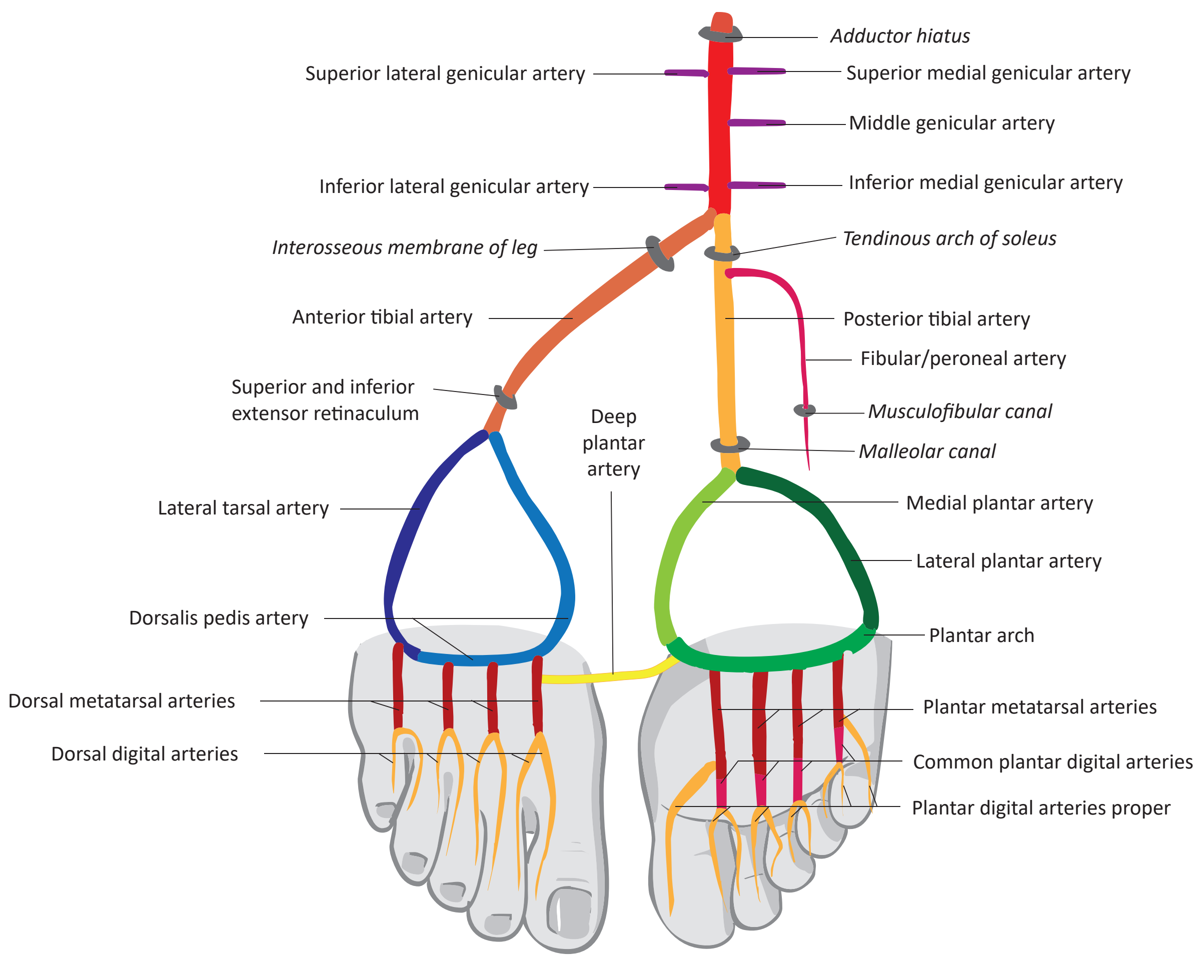


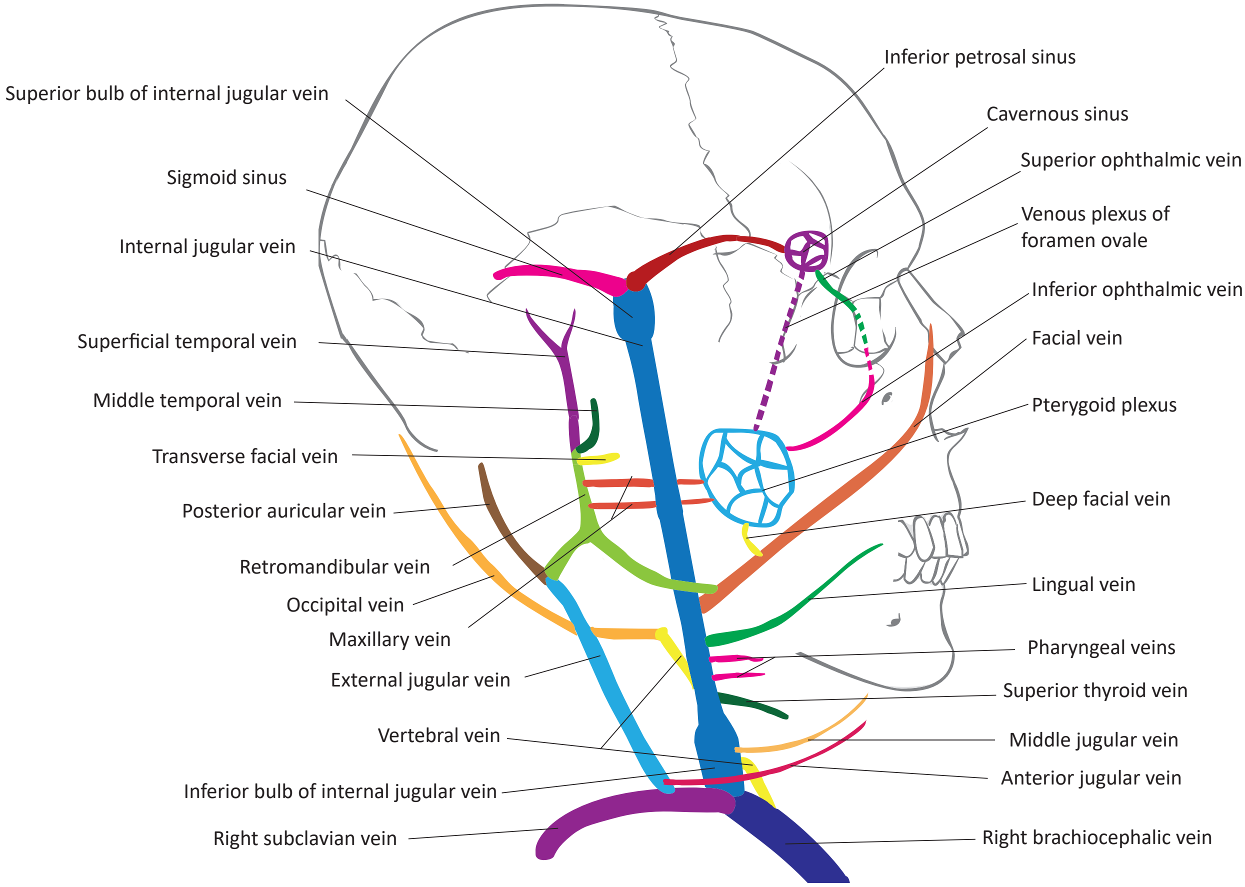
6.2.9 Internal iliac artery



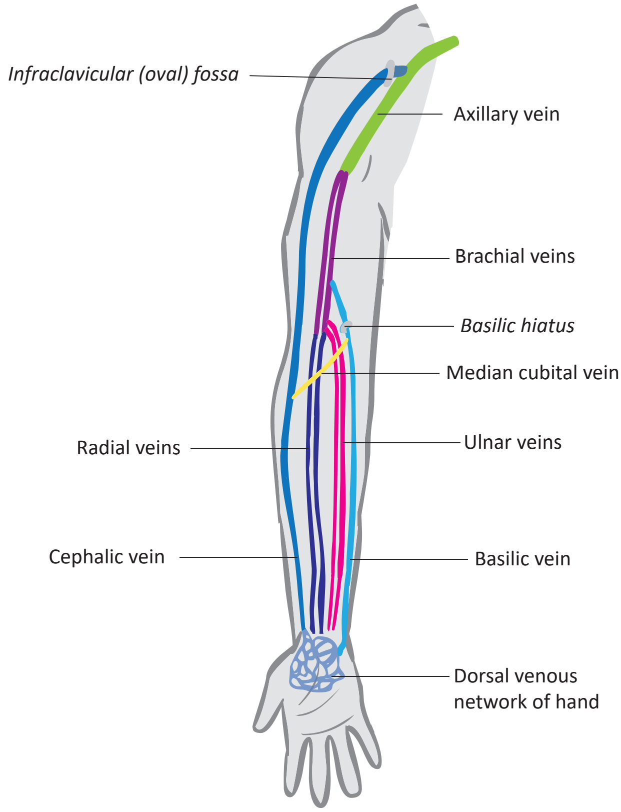


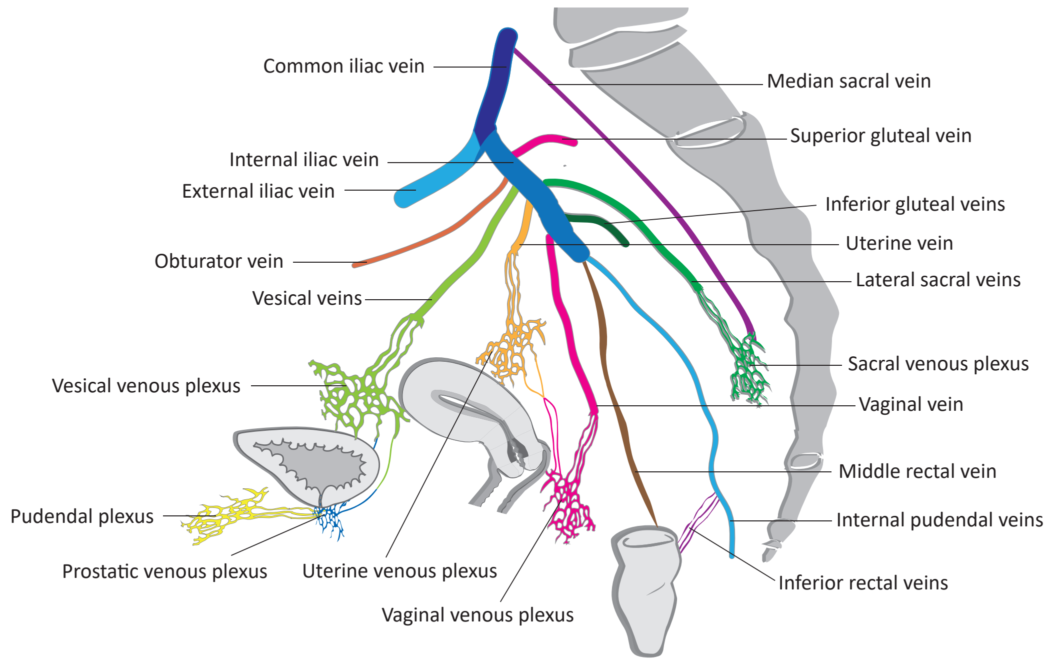
6.2.11 Popliteal artery and anterior and posterior tibial arteries



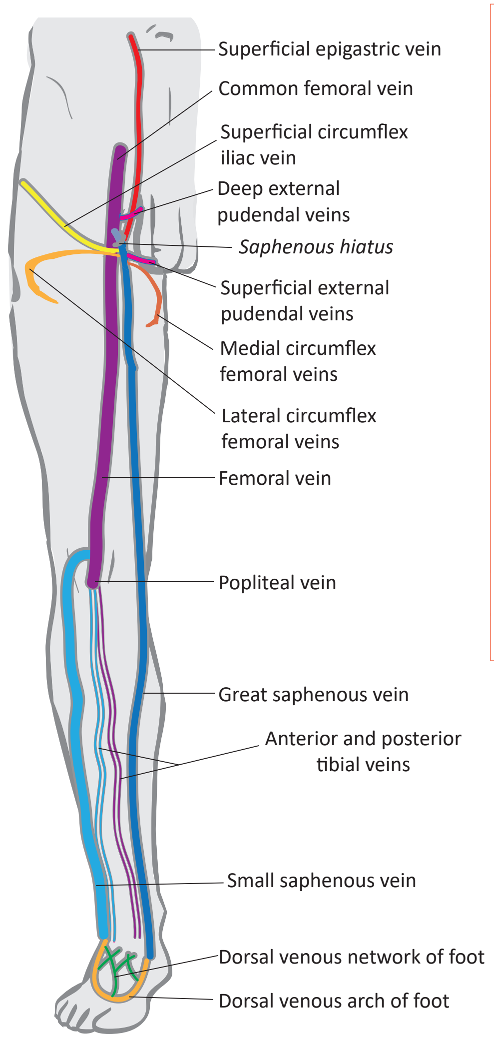


6.3 Venous drainage of the upper limb

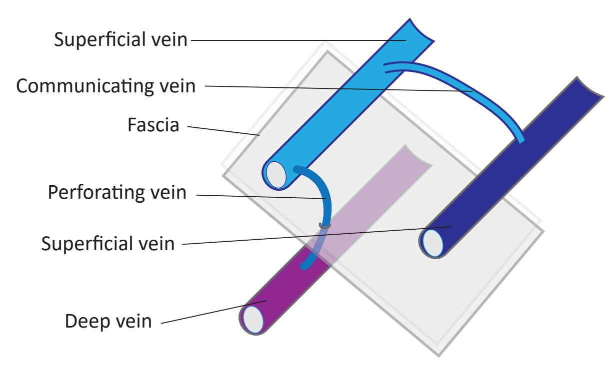


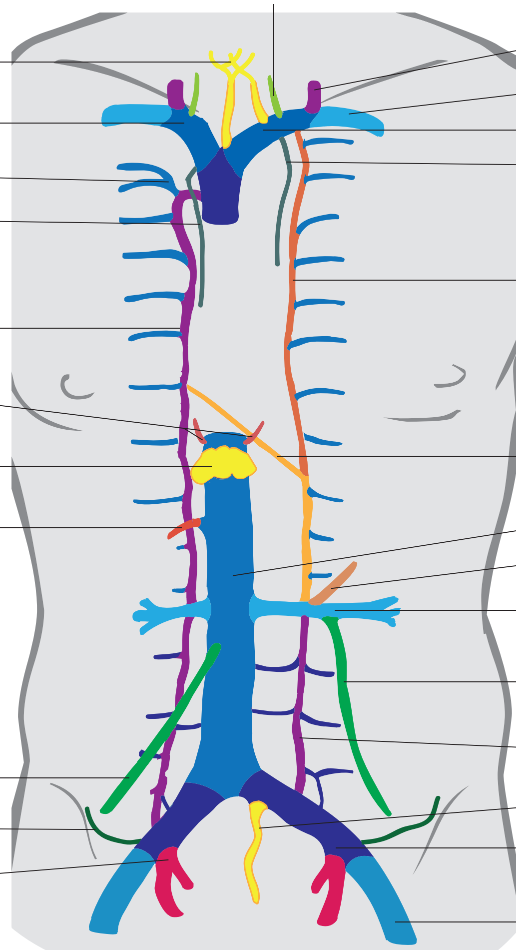


6.3 Venous drainage of the lower limb



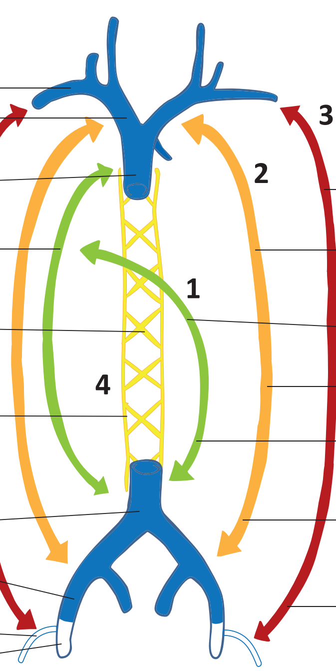
Perforators – perforating veins (*venae perforantes*)	
General structure and function	**Clinical importance**
Transfascial anastomoses that interconnect the deep (80% of blood) and superficial venous system of the lower extremity	Venous hypertension or valvular insufficiency can lead to a change of blood flow direction and subsequently cause ischemic changes
Contain valves	Varices can develop when insufficiency of valves occurs
Can be divided into 6 groups according to position (foot, ankle, leg, knee, thigh, buttocks)	Approximately 40 clinically important perforating veins (examination by ultrasound)
Simple, double or multiple	Their damage can cause venous ulcer of the leg
Accompanied by a small artery and cutaneous nerve (triad of Staubesand)	Are one of the causes of deep vein thrombosis which can lead to pulmonary embolism
The most important perforators	
Cockett's perforators – three veins called the venae perforantes cruris posteriores tibiales – interconnect the posterior accessory great saphenous vein and posterior tibial veins – 7cm, 13.5, and 18.5 proximally to the sole	
Boyd's perforator – vena perforans cruris paratibialis superior	
Dodd's perforator (adductor canal perforator) – vena perforans canalis adductorii	

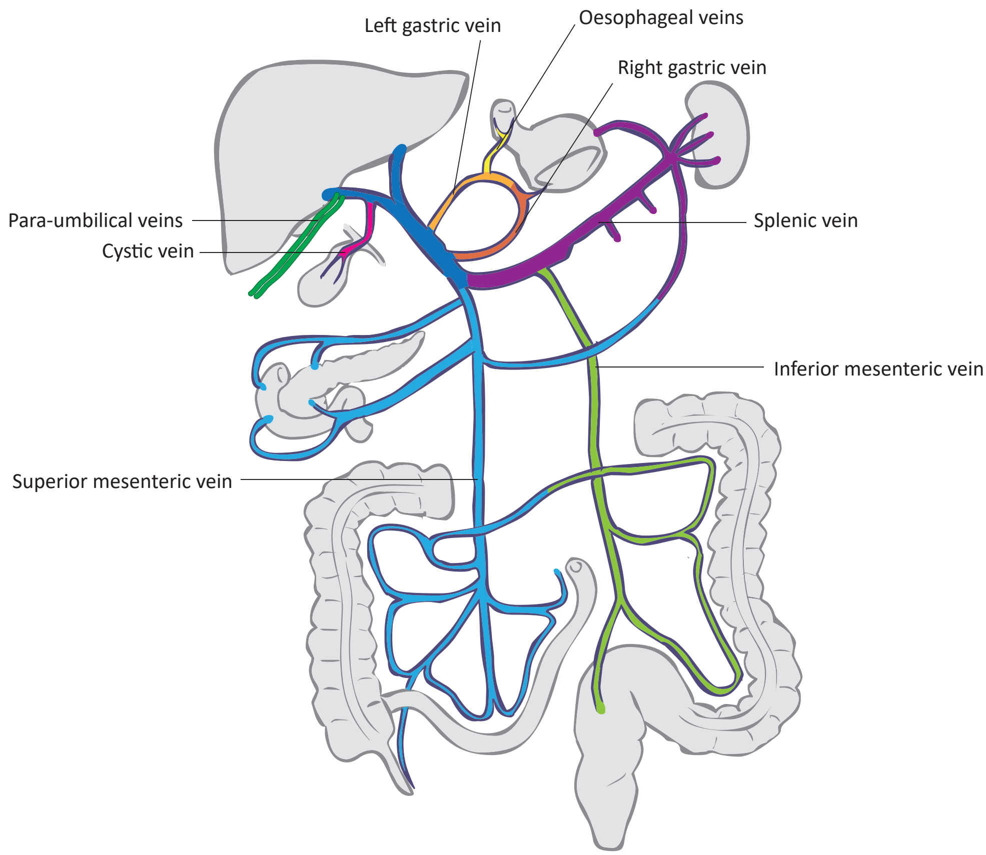




6.3 Cavo-caval anastomoses

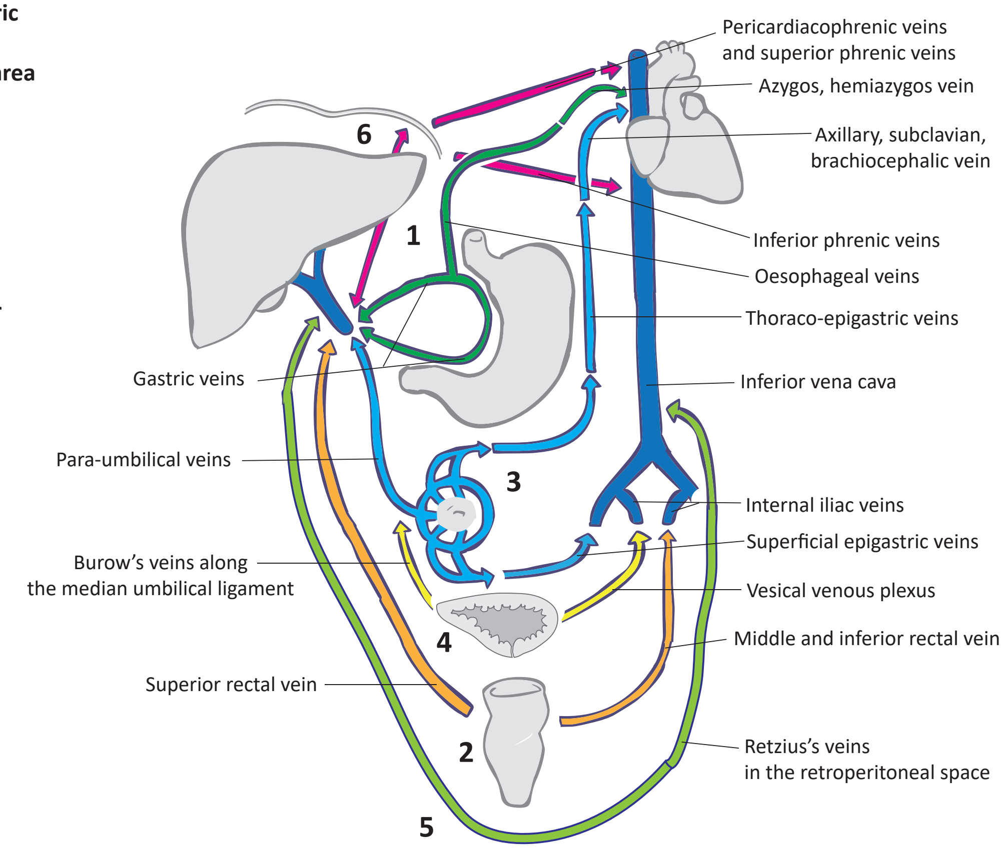
1 **Anastomoses between the lumbar veins and azygos (hemiazygos) vein**
2 **Anastomoses between the inferior epigastric veins and superior epigastric veins**
3 **Anastomoses between the superficial epigastric veins and thoraco-epigastric veins**
4 **Anastomoses between the vertebral plexuses and other veins**





6.3 Porto-caval anastomoses

1 **Anastomoses between the gastric and oesophageal veins**
2 **Anastomoses within the rectal area**
3 **Anastomoses between the para-umbilical veins and subcutaneous veins within the umbilical region**
4 **Anastomoses between the paraumbilical veins and venous plexus surrounding the urinary bladder**
5 **Anastomoses within the retroperitoneal space**
6 **Anastomoses between the hepatic and phrenic veins**

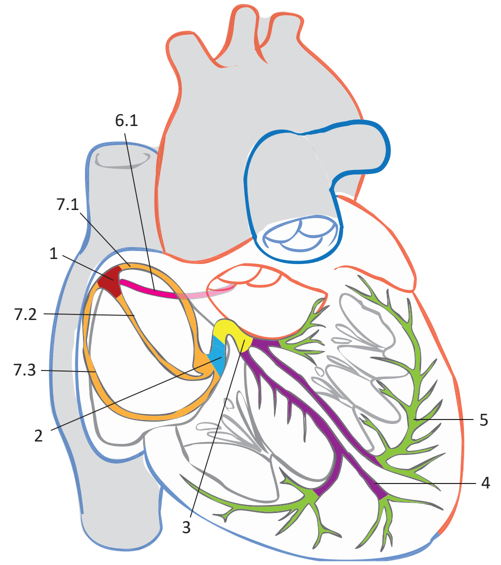


I. General structure of vessels

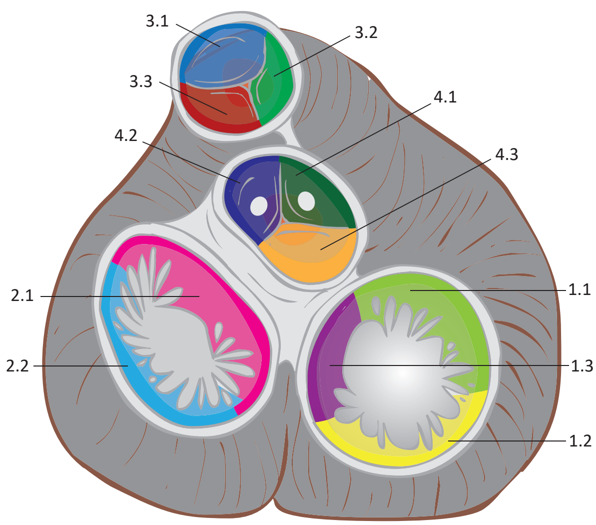
1. Explain the main difference between arteries and veins. (p. 264)
2. State the three types of anastomoses found in the circulatory system. (p. 264)

II. Heart

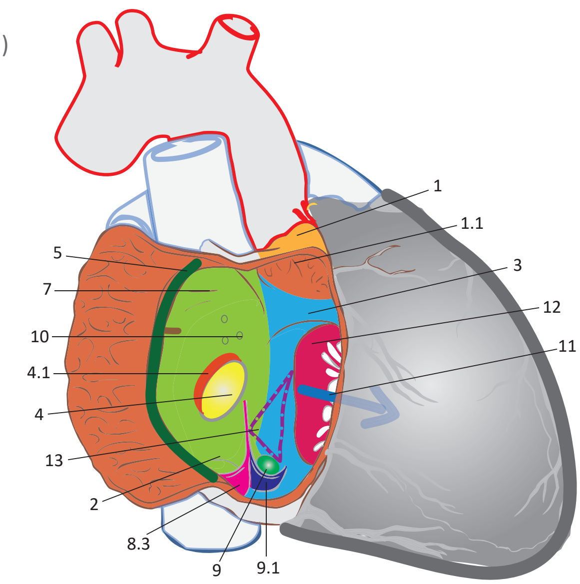
3. Describe the four layers of the heart wall. (p. 265)
4. Describe the places where the parietal and visceral serous pericardium meet. (p. 266)
5. State the surfaces of the heart and describe their features. (p. 267)
6. State the four heart grooves (sulci) and the structures passing through them. (p. 267)
7. State the venous openings in the right atrium. (p. 268)
8. State the structure positioned in the triangle of Koch. (p. 268)
9. State the parts of the interventricular septum. (p. 269)
10. Describe the course of the septomarginal trabecula. (p. 269)
11. Describe the structure that separates the right atrium from the right ventricle. (p. 269)
12. State the structure that separates the left atrium from the left ventricle. (p. 270)
13. Explain both the anatomical and functional differences between the atrioventricular and semilunar valves. (p. 272)
14. Explain the function of the papillary muscles with regard to their insertion. (p. 272)
15. Explain the three functions of the cardiac skeleton. (p. 273)
16. State the structure of the conducting system of the heart, which is also known as the pacemaker of the heart and explain what the term pacemaker means. (p. 273)
17. State the main branches of the left coronary artery. (p. 274)
18. Explain the function of the foetal shunts. (p. 275)
19. State the three foetal shunts and the vessels which they interconnect. (p. 275)



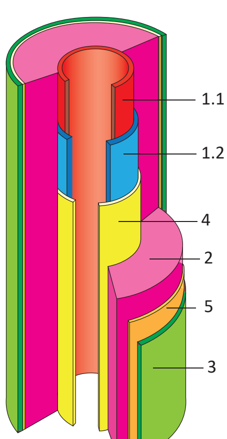
Describe the parts of the conducting system of the heart



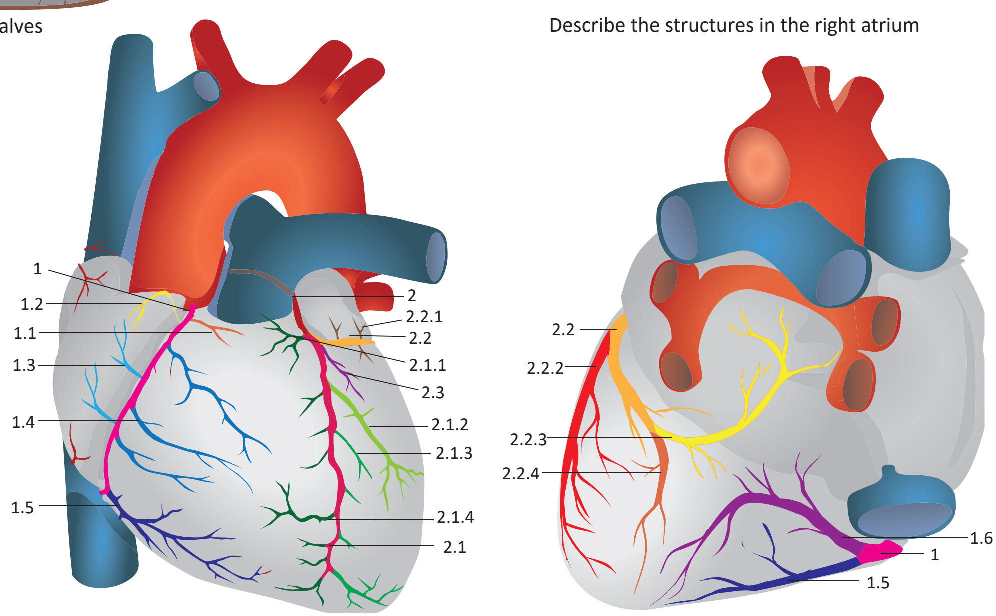
Describe the heart valves



Describe the structures in the right atrium



Describe the structure of the wall of a muscular artery



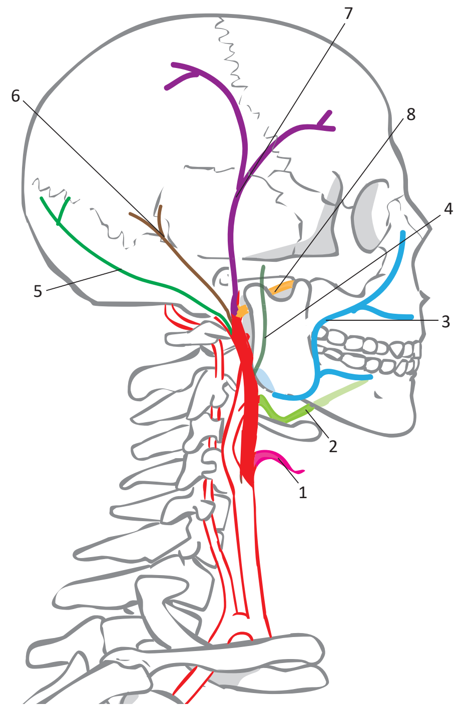
Describe the branches of the coronary arteries

III. Arteries

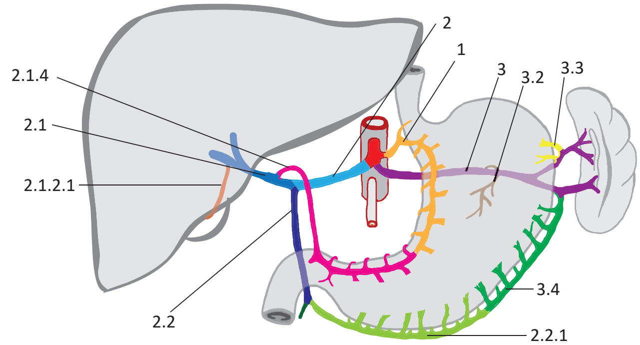
20. State the four parts of the aorta. (p. 277)
21. State the vessels that extend from the aortic arch and describe where they arise. (p. 277)
22. List the eight branches of the external carotid artery. (p. 278)
23. Explain the function of the carotid body. (p. 278)
24. Describe the course of the lingual artery. (p. 279)
25. Describe the parts of the maxillary artery and its course. (p. 281)
26. State the four main branches of the subclavian artery and where they arise. (p. 282)
27. Describe the topographical relationship between the suprascapular artery and the transverse scapular ligaments. (p. 283)
28. Describe the course of the radial artery. (p. 285)
29. Describe the course of the ulnar artery. (p. 285)
30. Explain the difference between the superficia and the deep palmar arch. (p. 285)
31. Describe the course of the thoracic aorta. (p. 286)
32. Describe the pattern of ramification of the abdominal aorta. (p. 287)
33. Describe the course of the abdominal aorta. (p. 287)
34. State the three branches of the coeliac trunk. (p. 288)
35. Explain the term anastomosis of Haller. (p. 288)
36. List the organs supplied by the superior mesenteric artery. (p. 289)
37. List the five parietal branches of the internal iliac artery. (p. 290)
38. State the four places where pulse on the lower extremity can be palpated. (p. 292)

IV. Veins

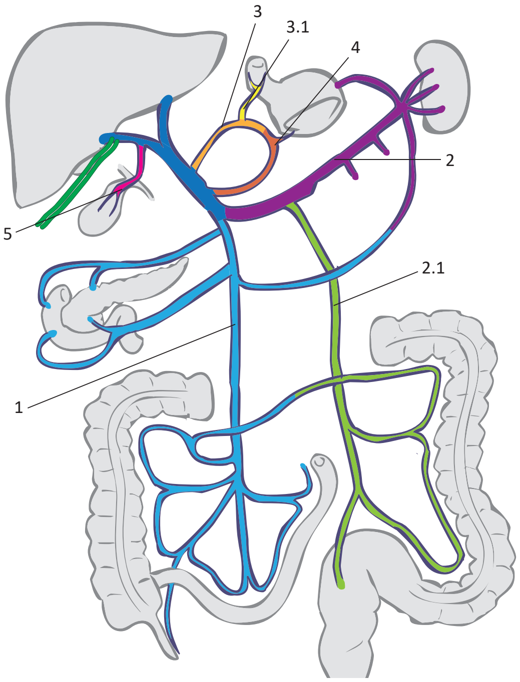
39. Describe the course of the superior vena cava. (p. 294)
40. List the thirteen extracranial tributaries of the internal jugular vein. (p. 295)
41. Describe the superficial venous system of the upper extremity. (p. 297)
42. State the two veins from which the portal vein is constituted. (p. 298)
43. Describe the place of origin of the portal vein. (p. 298)
44. List and describe the six portocaval anastomoses. (p. 299)
45. List and describe the four cavocaval anastomoses. (p. 299)
46. List the eight visceral tributaries of the internal iliac vein. (p. 300)
47. Describe the superficial venous system of the lower extremity. (p. 301)



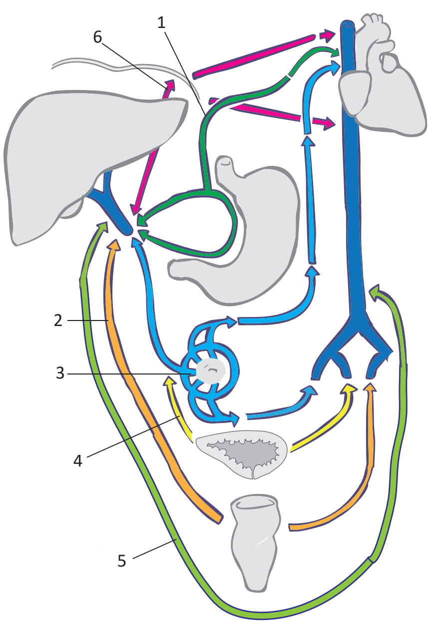
Describe the branches of the external carotid artery



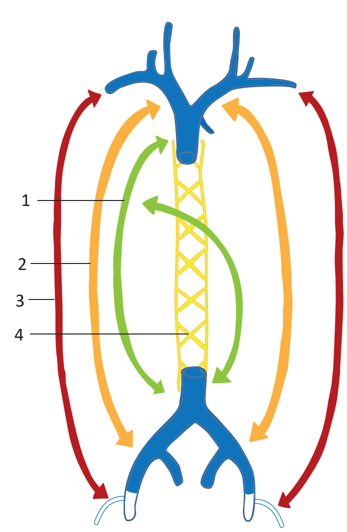
Describe the branches of the coeliac trunk



Describe the tributaries of the portal vein



Describe the scheme of the porto-caval anastomoses



Describe the scheme of the cavo-caval anastomoses

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